Linux Commands Summary
based on Embedded Linux course
(Real Time Colledge)

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Contents

1 Miscellaneous commands

1.1 py : evaluate python command from command line

```
-c : python command run

  Example :
  $ py -c "print(2**32-1)"
   4294967295
```

1.2 shutdown : shutdown the system in a safe way

```
  Shutdown the system in a safe way. You can shutdown
  the machine immediately, or schedule a shutdown using 24
  hour format.It brings the system down in a secure way. When
  the shutdown is initiated, all logged-in users and processes
  are notified that the system is going down, and no further
  logins are allowed.
  Only root user can execute shutdown command
     -r : Requests that the system be rebooted after it has
          been brought down.
     -P : Requests that the system be powered off after it has
           been brought down.
     -k : Only send out the warning messages and disable logins,
          do not actually bring the system down.

  Shut down :
    $ sudo shutdown

  How to shutdown the system at a specified time
    $ sudo shutdown 05:00

  System shutdown in 20 minutes from now:
    $ sudo shutdown +20

  How to shutdown the system immediately
    $ sudo shutdown now

  How to make shutdown power-off machine
  Although this is by default, you can still use the -P option
  to explicitly specify that you want shutdown to power off the system.
    $ shutdown -P

  How to broadcast a custom message
    $ sudo shutdown +10 "System upgrade"
```

1.3 minicom : RS232 communication on serial devices

```
minicom is a communication on serial devices, supports script language
interpreting, capturing to file, multiple users with individual
configurations, and more.

$ sudo apt install minicom
$ minicom -s # for settings, will edit /etc/minirc.dfl file
  serial port settings setup:
```

```
+---------------------------------------------------------------------+
| A -    Serial Device      : /dev/modem   # device name               |
| B - Lockfile Location     : /var/lock                                |
| C -   Callin Program      :                                          |
| D -  Callout Program      :                                          |
| E -    Bps/Par/Bits       : 115200 8N1                               |
| F - Hardware Flow Control : Yes                                      |
| G - Software Flow Control : No                                       |
| H -     RS485 Enable      : No                                       |
| I -   RS485 Rts On Send   : No                                       |
| J -  RS485 Rts After Send : No                                       |
| K -  RS485 Rx During Tx   : No                                       |
| L -  RS485 Terminate Bus  : No                                       |
| M - RS485 Delay Rts Before: 0                                        |
| N - RS485 Delay Rts After : 0                                        |
|                                                                      |
|    Change which setting?                                             |
+---------------------------------------------------------------------+
```

```
   after end of settings setuo we can choose
   "save as dflt"

minicom will control the terminal.
to choose options : ctrl+a+o : options
                    ctrl+a+q : exit
```

1.4 google-chrome : the web browser from Google

```
   $ google-chrome --new-window https://rt-ed.co.il/

--new-window
         If PATH or URL is given, open it in a new window.
```

1.5 mktemp : Create a temporary file or directory

```
    - Create an empty temporary directory and print its absolute path:
          $ mktemp -d
            /tmp/tmp.Ubw3bJBvR2
```

1.6 clamscan : antivirus

```
Scan home directory ("~"):
    $ clamscan -r --bell -i -v -o -a ~

-v: Be verbose.
-a: Show filenames while scan
-r: Recursive
-i: Only print infected files.
-o: Skip printing OK files

To view the version of the scan engine
    $ clamscan -V # Capital V
      ClamAV 0.103.11/27191/Tue Feb 20 11:25:13 2024

# Base Format
    $ clamscan [options] [directory or file to scan]
# To scan the current directory
    $ clamscan
# To view the version of the scan engine
    $ clamscan -V # Capital V
# To scan a specific folder or file
    $ clamscan /path or file
# To scan a specific path and its sub-folders
    $ clamscan -r /path
# To scan and display only the infected files
    $ clamscan -r -i /path
# To save the scan report
    $ clamscan -l /path/log-file-name /path
# Move the infected files to a different folder
    $ clamscan --move /directory /folder-or-file-to-scan
# Delete the infected files
    $ clamscan --remove yes /folder-or-file-to-scan
# Notify the user with a bell sound on identifying infected files
    $ clamscan --bell /folder-or-file-to-scan
# Different options in single go
    $ clamscan  -r --bell --remove yes -l /path/log-file-name  \
    /folder-or-file-to-scan

Freshclam runs as a background service and updates the virus signature
database in regular intervals. If you would like to manually perform an update,
execute the below commands in the terminal.
```

```
This would stop the Freshclam service and updates the db and finally starts
the service back.

   $ sudo systemctl stop clamav-freshclam.service
   $ sudo freshclam
   $ sudo systemctl start clamav-freshclam.service
See : man clamscan
```

1.7 reboot : Restart or reboot the system.

```
   Using 'reboot' command to restart our Linux system
     $ sudo reboot

   Force Immediate reboot in Linux system:
     $ sudo reboot -f
```

1.8 Zeal : Offline documentation browser

```
Zeal : Offline documentation browser
  for C,C++,Python,BASH etc.
```

1.9 wine : run microsoft software in Linux.

```
Wine is a free and open-source compatibility
layer that aims to allow application software and computer
games developed for Microsoft Windows to run on Unix-like
operating systems.
Wine also provides a software library, named Winelib,
against which developers can compile Windows applications
to help port them to Unix-like systems
```

1.10 yes : accept everything...

```
 - Repeatedly output "message":
      yes {{message}}

 - Repeatedly output "y":
      yes

 - Accept everything prompted by the apt-get command:
      yes | sudo apt-get install {{program}}
```

1.11 uptime : tells how long the system is running

```
Tell how long the system has been running
- Print current time, uptime, number of logged-in users and other information:
   $ uptime

- Show only the amount of time the system has been booted for:
   $ uptime --pretty

- Print the date and time the system booted up at:
   $ uptime --since
```

1.12 sleep : delay for a specified amount of time

```
sleep NUMBER[SUFFIX]...
Pause for NUMBER seconds.
SUFFIX may be
   's' for seconds (the default),
   'm' for minutes,
   'h' for hours or
   'd' for days.
 NUMBER need not be an integer.
 Given two or more arguments, pause for the amount of time
 specified by the sum of their values.
   - Delay in seconds:
       sleep {{seconds}}
       Example: sleep 3
   - Execute a specific command after 20 seconds delay:
       sleep 20 && {{command}}
```

1.13 Linux..O.S..version..check

- `lsb_release -a` `#lsb is for LinuxStandardBase:`
- `cat /etc/os-release`
- `hostnamectl`
- `cat /proc/version`

1.14 bc : "Basic Calculator" - simplified man pages

```
https://en.wikipedia.org/wiki/Bc_(programming_language)

bc, for basic calculator, is "an arbitrary-precision calculator
language" with syntax similar to the C programming language.
bc is typically used as either a mathematical scripting language or as
an interactive mathematical shell.

     scale = 5 : set precision to 5 numbers after the ".".

bc can be used non-interactively, with input through a pipe.
```

```
This is useful inside shell scripts:
     $ result=$(echo "scale=2; 5 * 7 /3;" | bc)
     $ echo $result
```

1.15 cal : "CALendar" - basic calendar

```
https://en.wikipedia.org/wiki/Cal_(command)

see ncal command with sligtly advanced options

 Prints calendar information, with the current day highlighted
   - Display a calendar for the current month:
       cal
   - Display previous, current and next month:
       cal -3
   - Display a calendar for a specific year (4 digits):
       cal 2024
   - Display a calendar for a specific month and year:
       cal 4 2024
   - Display calendar with working list numbers (example):
       cal -W 2024
```

1.16 date : show current date with format

```
Return current date

$ date          #return current date
   Wed 13 Sep 2023 15:31:52 IDT

$ date "+%Y"    # year long format
  2023
$ date "+%y"    # year short format
  23
$ date "+%m"    # month
  03
$ date "+%b"    #
  Mar
$ date "+%d"    # day
  13
$ date "+%S"    # seconds ?
  33
$ date "+%H"    # hours
  15
$ date "+%M"    # minutes
  35
$ date "+%Y-%m-%d %H:%M"
  2023-09-13 15:36

Example:
$ x=‘date +%Y‘ # command substitution
```

```
$ echo $x
  2024
$ x=$(date +%Y)
$ x=$(date +%Y)
$ echo $x
  2024
```

1.17 hdate : Hebrew DATE

```
Display Hebrew date.

-r --parasha      print weekly reading for Shabbat.
-h --holidays     print holidays.
-c                print Shabbat start/end times.



Location of Haifa : Haifa       32, 34, 2
   hdate -sRH 02 2024 -l32 -L34 -z2  :  sun rize+sunset (Haifa)+
                                         weekly reading+holidays
```

2 System administration basics

2.1 lspci : List all PCI devices

```
display information about PCI buses in the system and devices connected
to them.

-v - Be verbose, Include the Kernel Drivers for each device and Modules
     Modules - drivers in Linux.
-vv - Be more verbose. Show a more detailed version of the last command.
-vvv - Be as verbose as possible. Show all the details.

-t - Show a tree-like diagram containing all buses, bridges, devices
     and connections between them.

-b - Bus-centric view. Show all IRQ numbers and addresses as seen by the
     cards on the PCI bus instead of as seen by the kernel

  Example :
  $ lspci -t -v
-+-[10000:e0]-+-17.0  Intel Corporation Alder Lake-P SATA AHCI Controller
 |            +-1c.0  Intel Corporation Device 09ab
 |            \-1c.4-[e1]----00.0  Intel Corporation Device f1aa
 \-[0000:00]-+-00.0  Intel Corporation Device 4601
             +-02.0  Intel Corporation Device 46a8
```

```
+-04.0  Intel Corporation Alder Lake Innovation Platform
 Framework Processor Participant
+-08.0  Intel Corporation 12th Gen Core Processor Gaussian &
 Neural Accelerator
+-0e.0  Intel Corporation Volume Management Device NVMe RAID
 Controller
+-14.0  Intel Corporation Alder Lake PCH USB 3.2 xHCI Host
 Controller
+-14.2  Intel Corporation Alder Lake PCH Shared SRAM
+-14.3  Intel Corporation Alder Lake-P PCH CNVi WiFi
+-15.0  Intel Corporation Alder Lake PCH Serial IO I2C Controller
 #0
+-15.1  Intel Corporation Alder Lake PCH Serial IO I2C Controller
 #1
+-16.0  Intel Corporation Alder Lake PCH HECI Controller
+-17.0  Intel Corporation Device 09ab
+-1d.0-[01]----00.0  Realtek Semiconductor Co., Ltd.
 RTL8111/8168/8411 PCI Express Gigabit Ethernet Controller
```

2.2 kernel..modules : Linux drivers

```
module - Linux drivers
drivers are in files named *.ko (Kernel Object)
Add Code \Functionality to the Linux kernel while it is running.
  $ lsmod                 # list kernel modules (drivers)
  $ insmod module_name.ko # load kernel modules
  $ rmmod module_name     # unload kernel modules
```

Example

```
$ lsmod
 Module                  Size  Used by
 rfcomm                 98304  4
 ccm                    20480  6
 cmac                   12288  3
 algif_hash             12288  1
 algif_skcipher         12288  1
 spi_intel_pci          12288  0
 i2c_hid                40960  1 i2c_hid_acpi
 realtek                36864  1
 i2c_smbus              16384  1 i2c_i801
 idma64                 20480  0
 xhci_pci_renesas       20480  1 xhci_pci
 vmd                    24576  0
 spi_intel              32768  1 spi_intel_pci
```

2.3 dmesg..modules : Show kernel messages

```
 dmesg : Show kernel drivers messages
```

dmesg is used to examine or control the kernel ring buffer.

$ dmesg [-c] [-n level] [-s bufsize]

$ sudo dmsg -c   # -c is for clear after previous messages.
  after USB flash/ USB developement connection we will see
  USB messages.

 - Show kernel messages:
   $ dmesg

- Show kernel error messages:
    $ dmesg --level err

- Show kernel messages and keep reading new ones, similar to tail -f
  (available in kernels 3.5.0 and newer):
    $ dmesg -w

- Show how much physical memory is available on this system:
    $ dmesg | grep -i memory

 - Show kernel messages 1 page at a time:
    $ dmesg | less

 - Show kernel messages with a timestamp (available in kernels 3.5.0
   and newer):
    $ dmesg -T

 - Show kernel messages in human-readable form (available in kernels
   3.5.0 and newer):
    $ dmesg -H

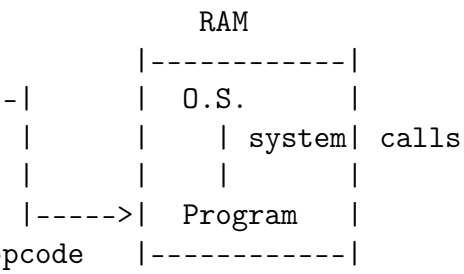
 - Colorize output (available in kernels 3.5.0 and newer):
    $ dmesg -L

The program helps users to print out their bootup/device
driver messages.

 - c: Clear the ring buffer contents after printing.
-sbufsize: Use a buffer of size bufsize to query the kernel ring
           buffer. This is 16392 by default.
-nlevel : Set the level at which logging of messages is done
           to the console.
           For example, -n 1 prevents all messages, expect
           panic messages, from appearing on the console.

 All levels of messages are still written to /proc/kmsg

2.4 strace : trace system calls of a program



Program communicates with CPU trough opcode and with O.S. with system calls , within program there are opcodes and system calls.

Withis assembly code (file *.s) there are opcodes and system calls, for example program hello_world.c:

```
# include <stdio.h>

  int main(void) {
     printf ("Hello world");
  }
```

present assembly source file:

```
$gcc -S hello_world.c
```

```
$ cat hello_world.s
 .file "hello_world.c"
 .text
 .section .rodata
.LC0:
 .string "Hello world"
 .text
 .globl main
 ...
 movl $0, %eax
 call printf@PLT  # here call is SYSTE CALL for Linux Kernell
 movl $0, %eax
 popq %rbp
```

```
$ gcc hello_world.c -o hello
$ ./ hello
```

```
$ strace ./hello
 execve("./hello", ["./hello"], 0x7ffd79c58fc0 /* 46 vars */) = 0
 brk(NULL)                                  = 0x64c5259e8000
 arch_prctl(0x3001 /* ARCH_??? */, 0x7ffe171e4d90) = -1 EINVAL
 (Invalid argument)
 mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0)
 = 0x743fb8c0c000
 access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or
```

```
 directory)
 openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
 newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=93767, ...},
 AT_EMPTY_PATH) = 0
 mmap(NULL, 93767, PROT_READ, MAP_PRIVATE, 3, 0) = 0x743fb8bf5000
 close(3)                                = 0
```

Used for tracing the system calls of a program.
when it is run in conjunction with a program,
it outputs all the calls made to the kernel by the program.

In many cases, a program may fail because it is
unable to open a file or because of insufficient memory.
And tracing the output of the program will clearly show the
cause of either problem.

Usage:
$ strace <name of the program>

Example:
$ strace ls

this will output a great amount of data on to the screen.
If it is hard to keep track of the scrolling mass of data, then there is
an option to write the output of strace to a file instead which is done
using the -o option:

```
$ strace -o strace_ls_output.txt ls
```

write all the tracing output of 'ls' to the 'strace_ls_output.txt' file.

```
$ strace echo "Hello World"
  execve("/usr/bin/echo", ["echo", "Hello World"], 0x7ffc1265a308
  /* 45 vars */) = 0
  brk(NULL)                               = 0x643239cd2000
  arch_prctl(0x3001 /* ARCH_??? */, 0x7ffc5285b970) = -1 EINVAL
  (Invalid argument)
  mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS,
  -1, 0) = 0x75c5d1538000
  access("/etc/ld.so.preload", R_OK)      = -1 ENOENT
  (No such file or directory)
  openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
  newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=93767, ...},
  AT_EMPTY_PATH) = 0
  mmap(NULL, 93767, PROT_READ, MAP_PRIVATE, 3, 0) =
  0x75c5d1521000
  close(3)                                = 0
  ....
```

....

2.5 uname : print system information

```
print all information:
$ uname -a
  Linux amits-vostro-3520 6.5.0-25-generic #25~22.04.1-Ubuntu SMP
  PREEMPT_DYNAMIC Tue Feb 20 16:09:15 UTC 2 x86_64 x86_64 x86_64
  GNU/Linux

-a | all info
-s | kernel name
-n | network node hostname
-r | kernel release
-v | kernel version
-m | machine hw name
-p | processor type
-i | hardware platform
-o | operating system



print the kernel release
$ uname -r
  6.5.0-25-generic
```

2.6 service..systemctl : service / deamon

```
A "service" is a program that starts automatically when you start your
computer, and runs in the background.

For example, the "network" service sets up your connection to
the Internet and keeps it running correctly.

service <service-name> status - Check if a service is running:
service <service-name> start  - Starting a service:
service <service-name> stop   - Stopping a service:

For example:
$ service network [start |stop| restart]
or
$ sytemctl [start |stop| restart] network)

   $ sudo systemctl status ssh  #check the status
   $ sudo systemctl stop ssh    #stop ssh for this session
```

```
$ sudo systemctl start ssh   #start ssh for this session
$ sudo systemctl disable ssh #stop ssh from boot to computer
$ sudo systemctl enable ssh  #start ssh from boot to computer
```

you can also modify services by /etc/init.d/service-name action_taken
(status, stop and start too).

look in /etc/init.d/ for the services available.

3 Bash Features and Scripting

3.1 only..bash..features

```
# ((  ))  $((  ))  $[   ]
above are features of BASH only
not other terminals as ZASH ... ("bashizm")
```

3.2 run..BASH..script../bin/bash

```
#!  ----> named "hash bang" or in short form "shebang"
 which bash
    /usr/bin/bash
```

By default, every text file can be run as BASH script
(file permissions must be executable):

```
$ cat hi555
   echo hi
$ ./hi555
   hi
$ ls -l hi555
  -rwxrw-r-- 1 amit amit 8 Oct  8 22:01 hi555
```

What happens if we add our own command "cat" that prints ‘‘miay’’ at the top of the PATH: our "cat" now is the first in search order.
Prepare following script named "cat" :

```
      $ nano cat
      --------------
      #! /bin/bash
      echo miay !!!
      --------------
 $ PATH=$PWD:$PATH
 $ cat
   miay!!!!
```

3.3 "hashbang"..."shebang"

```
https://en.wikipedia.org/wiki/Shebang_(Unix)

For bash script:
----------------
$ which bash
  /usr/bin/bash

#! symbol in named "hash bang" or shortly "shebang"
#! /usr/bin/bash


For Python:
-----------
https://en.wikipedia.org/wiki/Env

env may also be used in the hashbang line of a script to allow
the interpreter to be looked up via the $PATH.
For example, here is the code of a Python script:

      #!/usr/bin/env python3
      print("Hello, World!")

In this example, /usr/bin/env is the full path of the env command.
The environment is not altered, i.e. the original $PATH will be used.
Note that it is possible to specify the interpreter without using env, by giving
the full path of the python interpreter. A problem with that approach is that on
different computer systems, the exact path may be different.
By instead using env as in the example, the interpreter is searched for
and located at the time the script is run
```

3.4 stop..bash..script..at...error

```
https://ioflood.com/blog/bash-exit-on-error/

Using ‘set -e’
The simplest way to make a bash script exit when an error occurs is
by using the ’set -e’ command at the beginning of your bash script.
This command instructs the bash script to stop execution if any
command it runs returns a non-zero exit status, which is the
convention for indicating an error in bash.
   ----------------
   #!/bin/bash
   set -e
   ----------------

Using ‘set -o pipefail’
The ‘set -o pipefail’ command ensures that your script exits
```

```
if any command in a pipeline fails, not just the last one.
This can be useful in scripts where you're chaining together
multiple commands using pipes.
   ----------------
   #!/bin/bash
   set -o pipefail
   ---------------

Using 'set -u'
The 'set -u' command causes your script to exit if it tries
to use an uninitialized variable. This can help catch typos
and other bugs related to variables.
  -------------
  #!/bin/bash
  set -u
 --------------

- Execute a specific script and stop at the first [e]rror:
   $ bash -e {{path/to/script.sh}}
```

3.5 bash..debug

```
  Execute a specific script while printing each command before executing it:
  Usefull for debug !
     $ bash -x {{path/to/script.sh}}


  Print verbose information for each step (For Debug).
     $ env -v echo hello world
       executing: echo
       arg[0]= 'echo'
       arg[1]= 'hello'
       arg[2]= 'world'
       hello world

  Get warning on undefined variable use :
     $ set -u  # To
     $ echo [$b]
       []
     $ set -u
     $ echo [$b]
       bash: b: unbound variable     # warning on undefined var.
     $ set +u
     $ echo [$b]
       []
```

3.6 shopt : SHell OPTions manage

```
List shell definitions.
set/unset options

Options:
 -p print each shell option with an indication of its status
 -s enable (set) each OPTNAME
 -u disable (unset) each OPTNAME
```

- s : set
- u : unset

Examples :

```
$ shopt -s globstar
```

to return failure if glob did not get result::

```
$ shopt -s failglob
$ echo [j-z].txt
     bash: no match: [j-z].txt
$ shopt -u failglob
$ echo [j-z].txt
     [j-z].txt
```

to enable ** recursive glob :

```
$ shopt -s globstar  // set recursive glob on
$ shopt globstar
   globstar        on
$ echo **/*.txt   # recursive glob:
                  # find in the current dir (!)
                  # and all sub dirs under it
  1.txt 2.txt 3.txt 4.txt 5.txt   # files from current dir also
  temp1/a.txt temp1/b.txt temp1/c.txt temp1/d.txt
  temp2/10.txt temp2/1.txt temp2/2.txt temp2/3.txt
  temp2/4.txt temp2/5.txt temp2/6.txt temp2/7.txt
  temp2/8.txt temp2/9.txt
$ shopt -u globstar
$ shopt globstar
   globstar        off
$ echo **/*.*    # now it will search only in sub dirs
  temp1/a.txt temp1/b.txt temp1/c.txt temp1/d.txt
  temp2/10.txt temp2/1.txt temp2/2.txt temp2/3.txt
  temp2/4.txt temp2/5.txt temp2/6.txt temp2/7.txt
  temp2/8.txt temp2/9.txt
```

3.7 set : Change shell attributes / display shell vars

https://phoenixnap.com/kb/linux-set

```
$ type set
  set is a shell builtin

Change the value of shell attributes and positional parameters, or
display the names and values of shell variables.

set command: warnings for undefined variables.
set command is similar to shopt.

The set command is a built-in Linux shell command that displays
and sets the names and values of shell and Linux environment
variables.
On Unix-like operating systems, the set command functions within the
Bourne shell (sh), C shell (csh), and Korn shell (ksh).

If we type set without any additional parameters, we will get a list of
all shell variables, environmental variables, local variables,
and shell functions:
  $ set
This is usually a huge list. You probably want to pipe it into a pager
program to more easily deal with the amount of output:
  $ set | less
-----------------------------------------------------------------
Options to set/unset warning on undefined variable.
-u  Treat unset variables as an error when substituting.
+u  Return warning on undefined variable.
-----------------------------------------------------------------
set -e to exit immediately when there is an error in script
       and don't try to continue run
-----------------------------------------------------------------

$ set -u  # To get warning on undefined variable use
$ echo [$b]
  []
$ set -u
$ echo [$b]
  bash: b: unbound variable     # warning on undefined var.
$ set +u
$ echo [$b]
  []

$ help set
 set: set [-abefhkmnptuvxBCHP] [-o option-name] [--] [arg ...]
     Set or unset values of shell options and positional parameters.
     Change the value of shell attributes and positional parameters, or
     display the names and values of shell variables.
      -u  Treat unset variables as an error when substituting.
      +u  Return warning on undefined variable.
```

Split Strings
Use the set command to split strings based on spaces into separate variables. For example, split the strings in a variable called myvar, which says "This is a test".

```
$ myvar="This is a test"
$ set -- $myvar
$ echo $1
$ echo $2
$ echo $3
$ echo $4
 This
 is
 a
 test
```

3.8 chaining..operators explanation

https://www.geeksforgeeks.org/chaining-commands-in-linux/

Chaining commands in Linux allows us to execute multiple commands at the same time and directly through the terminal. It's like short shell scripts that can be executed through the terminal directly. Linux command chaining is a technique of merging several commands such that each of them can execute in sequence depending on the operator that separates them and these operators decide how the commands will get executed. It allows us to run multiple commands in succession or simultaneously.

Some commonly used chaining operators are as follows:

Operators	Function
& (Ampersand)	This command sends a process/script/command to the background.
&& (Logical AND)	The command following this operator will only execute if the command preceding this operator has been successfully executed.
; (Semi-colon)	The command following this operator will execute even if the command preceding this operator is not successfully executed.
|| (Logical OR)	The command succeeding this operator is only executed if the command preceding it has failed.
| (Pipe)	The output of the first command acts as input to the second command.
! (NOT)	Negates an expression within a command. It is

	much like an ‘‘except’’ statement.
>,>>,<(Redirection)	Redirects the output of a command or a group of commands to a file or stream.
&&-|| (AND-OR)	It is a combination of AND OR operator and is similar to the if-else statement.
\ (Concatenation)	Used to concatenate large commands over several lines in the shell.
() (Precedence)	Allows command to execute in precedence order.
{} (Combination)	The execution of the command succeeding this operator depends on the execution of the first command.

3.9 | : pipe - redirect stdout of one cmd to stdin of another cmd.

- ‘‘Pipe’’ redirects standard output of one command to standard input of another command
- t1, t2, tL are ‘‘temporary files’’ in this explanation.
- ‘‘|’’ acts as temporary file.
- Pipe redirects only REGULAR input/output, not error

```
keyboard --> command1 ---> | ---> command2 ---> |--> command3---> display
command1 > t1
command2 <t1 >t2
command3 > tL
```

- Example 1:
  Redirect output of (print of all files *.log)
  to (grep -i (insensitive) that search lines with ‘‘error’’)
  and sort all this lines:
  ```
  $ cat *log|grep -i error|sort
  ```

- Example 2:
  Redirect (result of recursive insensitive search ,
  starting with current directory) to (search of lines that not
  include word ‘‘ignored’’), the (sort the results with ‘‘unic’’)
  and send the result to file serious_errors.log:
  ```
  $ grep -ri error .|grep -v ‘‘ignored’’|sort u> serious_errors.log
  ```

- Example 3 (YES command usage):
  ```
  $ cat dir1_rm_inputs.txt
    y
    y
    n
    n
    y
    y
  ```

```
  y
$ cat dir1_rm_inputs.txt | rm ./dir1/*.* -i

$ rm -i dir2/*
   rm: remove regular empty file &apos;dir2/6.txt? n
   rm: remove regular empty file &apos;dir2/7.txt? n
   rm: remove regular file &apos;dir2/errors2.log? n
   rm: remove regular file &apos;dir2/errors3.log? n
   rm: remove regular file &apos;dir2/errors.log? n
$ yes | rm -i dir2/*
   rm: remove regular empty file ‘dir2/6.txt’? rm: remove regular
   empty file ‘dir2/7.txt’? rm: remove regular file ‘dir2/
   errors2.log’? rm: remove regular file ‘dir2/errors3.log’? rm:
   remove regular file ‘dir2/errors.log’?
```

- Example: automation to enter “yes” when install:

```
$ yes | sudo apt install some_programs
```

- apt has -flag to allow install without interrupts.

3.10 ; : send several commands in BASH

```
run several commands one after another:

$ echo hello ; sleep 1;echo world
   hello
   world

$ echo A; echo B; echo C
   A
   B
   C
This will output A, wait 3 sec., clear line and output B:
$ printf "A"; sleep 3; printf "\rB\n"
   B
```

3.11 || : logical OR

```
The command succeeding this operator will only execute if the
execution of the command preceding it fails.

Exit status: success: 0, others - fail

Exit status of grep command:
   Normally the exit status is
     0 if a line is selected,
```

```
        1 if no lines were selected,
        2 if an  error occurred.
      However, if the -q or --quiet or --silent is used and a line is
      selected, the exit status is 0 even if an error occurred.

Short circuiting:
      x || x || executed    :    stops on first success;
      Example: Do the same task with several alternatives automatically.
      Stop in case of first success in one of tasks.
   Example:
       $ ls
          latex-document-prototype.tex  try.docx  try.html  try.pdf  try.tex
       $ ll try.pdf || echo "2-nd" || echo "3 - rd"
          -rw-rw-r-- 1 amits amits 256176 Feb 28 12:11 try.pdf
       $ ll 222.txt || echo "2-nd" || echo "3 -rd"
          ls: cannot access ‘222.txt’: No such file or directory
          2-nd

   Example of "turnery if":
        $ ls
          latex-document-prototype.tex  try.docx  try.html  try.pdf  try.tex
     $ ll try.pdf && echo "OK !" || echo "FAILED !"
          -rw-rw-r-- 1 amits amits 256176 Feb 28 12:11 try.pdf
          OK !
        $ ll 222.txt && echo "OK !" || echo "FAILED !"
          ls: cannot access '222.txt': No such file or directory
          FAILED !
```

3.12 && : logical AND

```
Exit status: success: 0, others - fail.

The command succeeding this operator will only execute if the
command preceding it executed successfully

v && v && executed :   stops on first fail
Example: run chain of commands automatically, where we interested that
all the commands in the chain will run successfully.
Stop at first fail.

Example:
$ echo "1-st" && ll 222.txt && echo "3-rd"
  1-st
  ls: cannot access ‘222.txt’: No such file or directory
```

3.13 backslash : Split commands in BASH

```
To split the command in command line into
number of lines we use back slash  symbol \ :
```

```
$ ls\
>  -l
```

3.14 & symbol : start process in the background immediately

```
& Symbol
Another method to run a process in the background is by appending the
ampersand symbol & to the command you execute.
This symbol tells the shell to start the process
in the background immediately.
To just start the command in the background at the beginning:
all you need to do is put an ampersand at the end of any Linux command.
This will allow to use terminal along with running the command:

Example:

  $ gzip largefile.txt &


In this example, the gzip command starts running in the background
as soon as you execute it.
The process ID (PID) will be displayed on the terminal,
allowing you to continue using the shell while the process completes silently.

It's important to note that when using the & symbol,
the process may still produce output, which can interfere with
the prompt on the terminal.

Run several commands in background :
  $ xed & nemo & geany &
    [1] 19420
    [2] 19421
    [3] 19422



Note:
-----
To avoid this, it's recommended to redirect the output
to a file or to /dev/null, which discards the output.
You can do this by appending > filename or > /dev/null
to the command, respectively.
```

3.15 bash : Start an interactive shell session

```
 ------------------------------------------------
 when the variable is defined in following form
```

```
it is used in internal bash !
$ x=123 bash # x will be defined in internal bash

$ echo $x
  123
$ exit
  exit
$ echo $x
  2024
-------------------------------------------------
- Execute a specific script:
  Allows to run script without execution permissions !

      $ bash {{path/to/script.sh}}

bash command allows to run additional bash session within parent bash window.
To exit it use exit command.

- Start an interactive shell session:
  $ bash

- Start an interactive shell session without loading startup
  configs:
  $ bash --norc


- Execute a specific script while printing each command before executing it:
  Usefull for debug !

  $ bash -x {{path/to/script.sh}}

- Execute a specific script and stop at the first [e]rror:
  $ bash -e {{path/to/script.sh}}

  Exit bash session :
  $ exit

-----------------------------------------

$SHLVL  presents shell level:

$ bash
$ echo $SHLVL
  2
$ bash
$ echo $SHLVL
  3
$ exit
  exit
$ echo $SHLVL
```

```
  2
$ exit
  exit
$ echo $SHLVL
  1
```

3.16 SHELL.variables..vs..Env.varables

https://www.digitalocean.com/community/tutorials/how-to-read-and-set-environmental-

https://www.digitalocean.com/community/tutorials/how-to-read-and-set-environmental-and-shell-variables-on-linux

```
By convention, Env. and Shell variables are
usually defined using all capital
letters. This helps users distinguish
environmental variables within other contexts

for example command grep uses environmental variables:
 $ man grep
 ...
 ENVIRONMENT
 The behavior of grep is affected by the following environment
 variables.... LC_ALL ... LANG ... GREP_COLOR ...
 ...


Shell variables :
-----------------
variables that are contained exclusively within
the shell in which they were set or defined. They
are often used to keep track of ephemeral data,
like the current working directory. Shell variable
shouldn't be passed on to any child processes

Environmental variables :
-------------------------
variables that are defined for the current shell
and are inherited by any child shells or
processes. Environmental variables are used to
pass information into processes that are spawned
from the shell.

Env variables are local to the program/process that use
by them and they children and are deleted after the parent
program run finish.
For example, environ  for C programs allows to use env. variables.
Java uses CLASS PATH and JAVA HOME variables that define where
```

```
to find executable files to run them.

       Creating shell variable.
       ------------------------
$ TEST_VAR='Hello World!'
$ echo $TEST_VAR
     Hello World!

We can see this by grepping for our new variable within
the set output:
$ set | grep TEST_VAR
     TEST_VAR='Hello World!'

We can verify that this is not an environmental variable
by trying the same thing with printenv:
  $ printenv | grep TEST_VAR
     # No output should be returned.

Accessing the value of shell or environmental variable:
$ echo $TEST_VAR
  Hello World!
As you can see, reference the value of a variable by
preceding it with a $ sign. The shell takes this to mean that
it should substitute the value of the variable when it comes
across this.

So now we have a shell variable TEST_VAR.
It shouldn't be passed on to any child processes.
We can spawn a new bash shell from within our current one to
demonstrate:
  $ bash # open child bash session inside the existing one
  $ echo $TEST_VAR
If we type bash to spawn a child shell, and then try to
access the contents of the variable, nothing will be
returned. This is what we expected.

Get back to our original shell by typing exit:
  $ exit

$ TEST_VAR="current BASH var"  # must be without spaces
                                 around !!!!
$ echo $TEST_VAR
  current BASH var
$ bash
$ echo $TEST_VAR
   # no output
$ exit  # returen to parent BASH shell
  exit
$ echo $TEST_VAR
 current BASH var
```

Creating Environmental Variables.

Now, let's turn our shell variable into an environmental variable.
We can do this by exporting the variable. The command to do so
is appropriately named:

```
$ export TEST_VAR
```

This will change our variable into an environmental variable.
We can check this by checking our environmental listing again:

```
$ printenv | grep TEST_VAR
  TEST_VAR=Hello World!
```

Or

```
$ export | grep TEST_VAR
  declare -x TEST_VAR="current BASH var"
```

This time, our variable shows up.
Let's try our experiment with our child shell again:

```
$ bash
$ echo $TEST_VAR
  Hello World!
```

Great! Our child shell has received the variable set by its
parent.
Before we exit this child shell, let's try to export another
variable.
We can set environmental variables in a single step like this:

```
$ export NEW_VAR="Testing export"
```

Test that it's exported as an environmental variable:

```
$ printenv | grep NEW_VAR
 NEW_VAR=Testing export
```

Now, let's exit back into our original shell:

```
$ exit
```

Let's see if our new variable is available:

```
$ echo $NEW_VAR
$
```

Nothing is returned.
This is because environmental variables are only passed
to child processes. There isn't a built-in way of setting
environmental variables of the parent shell. This is good
in most cases and prevents programs from affecting the
operating environment from which they were called.

Example of shell vars vs. env. vars:

```
$ a=5
$ b=7
$ export b
$ bash
$ echo $a
$
$ echo $b
```

```
  7
$ exit
 exit
$ echo $a
  5
$ echo $b
  7
```

3.17 "=" : assign value for Shell/Env. variables.

```
Every shell/env variable is string !

Example of define:
  $ TEST_VAR1 = "current BASH var"
    TEST_VAR1: command not found  # because of spaces
                                    around "=" !!!
  $ TEST_VAR1="current BASH var"  # must be entered without
                                    spaces around "=" !!!!
  $ echo $TEST_VAR1
    current BASH var
  $ TEST_VAR2="previous and $TEST_VAR1"
  $ echo $TEST_VAR2
    previous and current BASH var
  $ TEST_VAR3='previous and $TEST_VAR1'
  $ echo $TEST_VAR3
    previous and $TEST_VAR1
  $ a=5
  $ echo $a
    5
```

3.18 examples..of..system..vars

```
$ echo `whoami`
 amit
$ echo $(whoami)
 amit
$ echo aaa $(whoami) bbb
 aaa amit bbb
$ echo aaa $USER bbb
 aaa amit bbb
$ echo [$(pwd)]
 [/home/amit]
$ echo [$PWD]
 [/home/amit]
$ echo [$HOME]
 [/home/amit]
```

```
$ echo $HOSTNAME
 amits-vostro-3520
$ echo $SHELL # current shell name
 /bin/bash
```

3.19 $PATH : path system variable

If the command is not built in command in bash, bash performs the search for the command by using path in $PATH variable.
The search is done according to the order of directories in $PATH variable.

It might be a good idea to create a directory ~/scripts to hold your scripts. Add the directory to the contents of the PATH variable:

```
  $ export PATH="$PATH:~/scripts"
```

3.20 $SHELL : current shell type)

https://unix.stackexchange.com/questions/43499/difference-between-echo-shell-and-which-bash

```
$ echo $SHELL
 /bin/zsh
$ which bash
 /bin/bash
```

The first command tells me which shell will be executed by login when you log in. In my case, /bin/zsh.
The second command tells me the first occurrence in my $PATH the bash command can be found.

3.21 $SHLVL : SHell LeVeL check shell level

Allows to check the shell's level
bash command runs additional bash session within parent bash window.
To exit it use exit command.

$SHLVL presents shell level:

```
$ bash
$ echo $SHLVL
  2
$ bash
$ echo $SHLVL
  3
$ exit
  exit
$ echo $SHLVL
```

```
  2
$ exit
  exit
$ echo $SHLVL
  1
```

3.22 $PS1 : Prompt String 1

```
$ echo $PS1
$ echo $PS1 >> ~/.profile
```

3.23 $BASH_VERSION : result of last command execution

```
$ echo "Bash version $BASH_VERSION"
  Bash version 5.1.16(1)-release
```

3.24 $? : result of last command execution

```
$ ls -la
  total 772
  drwxrwxr-x 2 amits amits   4096 Feb 28 12:12 .
  drwxrwxr-x 8 amits amits   4096 Feb 28 11:42 ..
  -rw-rw-r-- 1 amits amits 256176 Feb 28 12:11 try.pdf
  -rw-rw-r-- 1 amits amits 148037 Feb 28 11:41 try.tex
$ echo $?
  0
https://www.baeldung.com/linux/check-if-command-executed-
successfully#:~:text=We%20can%20use%20a%20binary%20comparison%20operator%20eq,
the%20pwd%20command%20displays%20the%20present%20working%20directory.

  We can use a binary comparison operator eq to check the
  outcome of the previous command, stored in the $?
  variable:

    #!/bin/bash
    pwd
    if [ $? -eq 0 ]; then
       echo "Command Executed Successfully"
    else
       echo "Command Failed"
    fi{verbatim}
```

3.25 $! : PID of last executed program

```
Present PID of last program:
$ geany 1.txt & disown
```

```
  [1] 8816
$ echo $!
  8816
```

3.26 $$: current bash Process ID

```
$ echo $$     # present current BASH pid
  6425
```

3.27 $PROMPT_DIRTRIM : PROMPT DIRectory TRIMmer

```
For people looking for a much simpler solution and don't need
the name of the first directory in the path,
Bash has built-in support for this using the PROMPT_DIRTRIM
variable. For example:

$ PROMPT_DIRTRIM=2

If set to a number greater than zero, the value is used as the
number of trailing directory components to #retain when
expanding the \w and \W prompt string escapes (see Printing a
Prompt). Characters removed are replaced with an ellipsis.
```

3.28 which : Locate a program in the user's PATH.

```
$ type -a which
  which is /usr/bin/which
  which is /bin/which

which is not BASH built in
which locates a program in the user's $PATH.
Search the $PATH env. variable and display the location of
any matching executables.

- Search the $PATH environment variable and display the location
  of any matching executables:
    $ which {{executable}}
- If there are multiple executables which match, display all:
    $ which -a {{executable}}
 Example :
    $ which -a time
       /usr/bin/time
       /bin/time
```

3.29 export : set or present exported variables

```
Export command is used to add system variable.
It is session specific i.e., when Bash window is closed,
it became not valid. If you want you variable to be set
always in your shell, add it to .bashrc file:

    1. Make sure your .bashrc exists
  ls -A ~/.bashrc
    2. Add your variable at the bottom
  export VARIABLE=value
    3. Save, then open a new terminal and verify the variable is set
  $ echo $VARIABLE
  value

$ type export
export is a shell builtin

To present all exported variables :
$ export

Example of set env variable:
  $ NEW_VAR="Testing export"
  $ export NEW_VAR

We can set environmental variables in a single step like this:
  $ export NEW_VAR="Testing export"

If no names are supplied, or if the -p option is given, a
list of names of all exported variables is displayed:
$ export | head
  declare -x COLORTERM="truecolor"
  declare -x DESKTOP_SESSION="ubuntu"
  declare -x DISPLAY=":0"
  declare -x GDMSESSION="ubuntu"
$ export | grep TEST_VAR
  declare -x TEST_VAR="current BASH var"

also see declare that can set exported variable and
non exported variable.
```

3.30 .bashrc : (BASH Run Comm.) runs every time new bash opened

```
    .bashrc file location: File bash.bashrc
    is located in /etc/  folder.

    For the current user it must be located at home dir:
        ~/.bashrc
        /home/amits/.bashrc
```

```
The .bashrc file is a script file that's executed
when a user logs in.

The file itself contains a series of configurations for the
terminal session. This includes setting up or enabling: coloring,
completion, shell history, command aliases, exported variables
and more.

Export command when entered in bash terminal is used
to add system variable. It is session specific i.e.,
when Bash window is closed, it became not valid.
If you want you variable to be set
always in your shell, (as opposed to the superuser's),
add it to .bashrc file:

    1. Make sure your .bashrc exists
  ls -A ~/.bashrc
    2. Add your variable at the bottom
  export VARIABLE=value
    3. Save, then open a new terminal and verify the
       variable is set
   $ echo $VARIABLE
     value
.bashrc contains also alias commands.
see https://linuxhandbook.com/linux-alias-command/
for example add to ~/.bashrc file following text:
   alias lilearn='/home/amits/Documents/RTC/linux_administration/
             linux_learn_apps/linux_learn_apps_in_C/
             tex_file_data_extract'
```

3.31 .profile vs. .bash_profile vs. .bashrc

https://www.baeldung.com/linux/bashrc-vs-bash-profile-vs-profile

```
The order after login:
----------------------
1.  /etc/profile
2. .bash_profile
3  .bash_login and .profile (add: echo "Hello .profile")

On every run of bash terminal:
------------------------------
1 .bashrc  # runs every time we open new bash
           (add echo "Hello .bashrc")

Startup files contain commands that are to be executed on
shell startup. As a result, the shell executes commands
present in these files automatically to set up the shell.
```

```
This happens even before displaying the command prompt.

1. /etc/profile file
--------------------
In an interactive login shell, Bash first looks for the
/etc/profile file. If found, Bash reads and executes it
in the current shell. As a result, /etc/profile sets up
the environment configuration for ALL users.

2. .bash_profile file
---------------------
Bash then checks if .bash_profile exists in the home
directory. If it does, then Bash executes .bash_profile in
the current shell. Bash then stops looking for other files
such as .bash_login and .profile.

3. .bash_login and .profile
---------------------------
If Bash doesn't find .bash_profile, then it looks for
.bash_login and .profile, in that order, and executes the
first readable file only.

4. .bashrc
----------
runs at every terminal run.
```

3.32 alias : Creates aliases

```
alias command.
ll is alias to ls -alF , see .bashrc file

alias is defined at terminal level.
to allow run at sessin level, add alias to .bashrc file.

Note: alias does not work in bash script !!!


To see the list of aliases:
$ alias
  alias alert='notify-send --urgency=low -i "$([ $? = 0 ] && echo
  terminal || echo error)" "$(history|tail -n1|sed -e '\''s/^\s*[0-9]\+
  \s*//;s/[;&|]\s*alert$//'\'')"'
  alias egrep='egrep --color=auto'
  alias fgrep='fgrep --color=auto'
  alias grep='grep --color=auto'
  alias l='ls -CF'
  alias la='ls -A'
  alias liln='/home/amits/Documents/RTC/linux_administration/
  linux_learn_apps/linux_learn_apps_in_C/tex_file_data_extract'
```

```
  alias ll='ls -alF'
  alias ls='ls --color=auto'
```

```
To define alias:
$ alias x=‘echo ‘
$ x "hello world"
  hello world

To remove alias:
$ unalias x
$ x hello world
 x: command not found
```

3.33 unalias : Remove aliases

```
Define alias:
$ alias x=‘echo ‘
$ x "hello world"
  hello world

Remove alias:
$ unalias x
$ x hello world
 x: command not found
```

3.34 printenv : Print values of all/specific Env. variables.

```
Print values of all or specific environment variables
- Display key-value pairs of all environment variables:
    $ printenv | less

- Display the value of a specific variable:
    $ printenv USER
      amits

- Display the value of a variable and end with NUL
  instead of newline:
   $ printenv --null USER
   amits $
```

3.35 env : Show and set the env. / run a prog. in a modified environment

```
https://www.geeksforgeeks.org/env-command-in-linux-with-examples/
```

https://stackoverflow.com/questions/43793040/how-does-usr-bin-env-work-in-a-linux-

1. env is used to either print environment variables.
2. It is also used to run a utility or command in a custom environment.
3. Improtant : In practice, env has another common use.
   It is often used by shell
   scripts to launch the correct interpreter.
   In this usage, the environment is typically not changed (i.e. used current
   envirinment).

https://en.wikipedia.org/wiki/Env
env may also be used in the hashbang line of a script to allow
the interpreter to be looked up via the $PATH.
For example, here is the code of a Python script:

```
#!/usr/bin/env python3
print("Hello, World!")
```

In this example, /usr/bin/env is the full path of the env command.
The environment is not altered, i.e. the original $PATH will be used.
Note that it is possible to specify the interpreter without using env, by giving
the full path of the python interpreter. A problem with that approach is that on
different computer systems, the exact path may be different.
By instead using env as in the example, the interpreter is searched for
and located at the time the script is run

a. Without any argument : print out a list of all environment variables.
   $ env

b. With -i or -ignore-environment or only - : Ignores the inherited environment
   and starts with a clean slate :
   $ env -i your_command
   This completely clear out the environment, but it does not prevent
   your_command setting new variables.

c. With -u or -unset: Unsets a variable.Remove variable from the environment
   $ env -u variable_name

d. With -0 or -null: End each output line with NULL, not newline
   $ env -0

e. With -v: Prints verbose information for each step (For Debug).
   $ env -v echo hello world
     executing: echo
     arg[0]= ‘echo’
     arg[1]= ‘hello’

```
       arg[2]= ‘world’
       hello world


env [OPTION]... [-] [NAME=VALUE]... [COMMAND [ARG]...]



Script Debugging
-x  Print commands and their arguments as they are executed.
The set command is especially handy when you are trying to debug your
scripts. Use the -x option with the set command to see which command in
your script is being executed after the result, allowing you to
determine each command’s result.

Script Exporting
-a  Mark variables which are modified or created for export.
Automatically export any variable or function created using the -a
option. Exporting variables or functions allows other subshells and
scripts to use them.

Exit When a Command Fails
-e  Exit immediately if a command exits with a non-zero status.
Use the -e option to instruct a script to exit if it encounters an error
during execution. Stopping a partially functional script helps prevent
issues or incorrect results.

Prevent Data Loss
-C  If set, disallow existing regular files to be overwritten
    by redirection of output.
The default Bash setting is to overwrite existing files. However, using
the -C option configures Bash not to overwrite an existing file when
output redirection using >, >&, or <> is redirected to that file.

Report Non-Existent Variables
-u  Treat unset variables as an error when substituting.
The default setting in Bash is to ignore non-existent variables and work
with existing ones. Using the -u option prevents Bash from ignoring
variables that don’t exist and makes it report the issue.




Using + rather than - causes these flags to be turned off.


$ type env
  env is hashed (/usr/bin/env)
```

```
- Show the environment variables:
   $ env

 - Run a program.
 Often used in scripts after the shebang (#!) for looking up the
 path to the program:
   $ env {{program}}

 - Clear the environment and run a program:
   $ env -i {{program}}

 - Remove variable from the environment and run a program:
   $ env -u {{variable}} {{program}}

 - Set a variable and run a program:
   $ env {{variable}}={{value}} {{program}}

 - Set multiple variables and run a program:
   $ env {{variable1}}={{value}} {{variable2}}={{value}}
     {{variable3}}={{value}} {{program}}
```

3.36 declare : Declare variables and give them attributes

```
Declare different types of vars
see: https://tldp.org/LDP/abs/html/declareref.html

allows to cancell "export" variable
when using -x to set export , or +x to cancell export

 - Declare a string variable with the specified value:
     $ declare {{variable}}="{{value}}"

 - Declare an integer variable with the specified value:
     $ declare -i {{variable}}="{{value}}"

  declare -i number
  # The script will treat subsequent occurrences of "number"
    as an integer.

    $ number=3
    $ echo "Number = $number"     # Number = 3

    $ number=three
    $ echo "Number = $number"     # Number = 0
    # Tries to evaluate the string "three" as an integer.
    Certain arithmetic operations are permitted for
    declared integer variables without the need for expr or let.
    $ n=6/3
    $ echo "n = $n"       # n = 6/3
```

```
  $ declare -i n
  $ n=6/3
  $ echo "n = $n"       # n = 2

- Declare an array variable with the specified value:
    $ declare -a {{variable}}=({{item_a item_b item_c}})

-x export declares a variable as available for exporting outside
   the environment of the script itself.
     $ declare -x var3
     $ declare -x var3=373

 The declare command permits assigning a value to a variable in
 the same statement as setting its properties.

 $ type declare
   declare is a shell builtin

 To set exported variable :
 $ TEST_VAR="current BASH var"
 $ declare -x TEST_VAR   # set exported - env var
   or
 $ declare -x TEST_VAR="current BASH var"

 To unset exported variable :
 $ declare + x TEST_VAR   # set non exported - shell var
 $ export | grep TEST_VAR
 $

 $ echo Enter some text
 $ read myvar
 $   Hello world
 $ echo '$myvar' now equals $myvar
     $myva now equals Hello world
```

3.37 "."vc'.' : Quoting

```
  Normally BASH interprets spaces as argument separators:
      $ echo Hello                 World
        Hello World
        # Argument 0 is : echo
        # Argument 1 is : Hello
        # Argument 2 is : World

  " " used to prevent the shell from interpreting spaces
        as argument separators. The whole expression
        within quotes ".." is considered as one argument.
  All characters between " " are the same argument.
```

```
Also " " disables globing, but does not disable ! and $ :
     $ echo "You are logged as $USER"
        You are logged as amits
     $ echo *
        latex-document-prototype.tex try.docx try.html try.pdf try.tex
     $ echo "*"
       *
     $ echo *.pdf
       try.pdf
     $ echo "*.pdf"
       *.pdf
Single quotes preserve all special characters, while double quotes
allow some special characters to be interpreted:
     $ echo "You are logged as $USER"
        You are logged as amits
     $ echo 'You are logged as $USER'
        You are logged as $USER
see : https://saturncloud.io/blog/difference-between-single-
      and-double-quotes-in-bash/
```

3.38 "expansion" : made in echo command

```
    echo     $USER
      |        |
      v        v
    echo     shmuel
      |        |
      V        v
 argument0   argument1
 command !
1. check ALIAS
2. check PATH
3. check BUILT IN
```

```
$ a=Hello
$ b=World
$ echo "$a $b"
 Hello World
"echo" is not path - search for command named "echo".
Firsly expansion is made.
After it Hello is substituted as first argument
World is substituted as second argument.

$ echo $a$b
 ABCXYZ
$ echo "$a""$b"
```

```
 ABCXYZ
$ echo "$a"123"$b"
 ABC123XYZ


$ a=ec
$ b=ho
$ c=hel
$ d=lo
$ $a$b $c$d
  hello

$ some_path="aaa/bbb/ccc"
$ extended_path="$some_path"/ddd
$ echo $extended_path
  aaa/bbb/ccc/ddd
$ extended_path1=ddd/"$some_path"
$ echo $extended_path1
  ddd/aaa/bbb/ccc

$ # print undefined variable (default - NOT ERROR, empty value):
$ echo [$yyyy]
  []
$ a=
$ echo [$a]
  []
$ a=""
$ echo [$a]
  []

Problematic example:
$ a=123
$ a=$aABC      # problematic: $a here is not separate veriable !
$ echo [$a]    # BASH thinks that $aABC is not defined
$ []
Fixed example :
$ a=123
$ a=${a}ABC    # now it is fixed !
$ echo $a    # BASH thinks that $aABC is not defined
  123ABC

$ echo "hello $USER"
  hello amits
$ echo 'hello $USER'
  hello $USER

Problematic example:
$ a="Hello        World"
$ echo $a
  Hello World   # echo gets Hello as first argument
```

```
                # and Word as second argument
Fixed example :
$ a="Hello        World"
$ echo "$a"     # now the result of expansion is
                # considered as one alone string
$ Hello        World
```

3.39 ‘cmd‘(backtick) equals $(cmd): run cmd and substitute its text

Back tick (key beyond ~) ‘command‘ or $(command) means:
firstly, run command that is within ‘‘ or within $() and put its
output text here.

The expression $(command) is a modern synonym for ‘command‘ which
stands for command substitution;

```
$ echo whoami
  whoami

$ echo ‘whoami‘
  amits

$ x=‘whoami‘
$ echo $x
  amits

$ x=$(whoami)
$ echo $x
  amits

$ echo ‘whoami‘
  amits

$ x=$(tldr ls)
$ echo $x
      ls List directory contents.More information: https://
      www.gnu.org/software/coreutils/ls. - List files one
      per line:  ls -1  - List all files, including hidden
      files:  ls -a  - List all files, with trailing / added
      to directory names:  ls -F  - Long format list
      (permissions, ownership, size, and modification date)
      of all files:  ls -la  - Long format list with size
      displayed using human-readable units (KiB, MiB, GiB):  ls -
```

In the example below output of command 2 will not be presented on
the screen, it will act as input of command 1:

```
   $ # store in x the
   $ # PID of ping command that is running :
```

```
$ pgrep ping
  46034
$ x=‘pgrep ping‘
$ echo $x
  46034
$ x=$(pgrep ping)
  46034

Use of ‘‘ vs $() :
$ # COMMAND1 $(COMMAND2 $(COMMAND3))

Print hello to the user:
$ echo hello to $(whoami)
  hello to amit
$ echo hello to ‘whoami‘
  hello to amit
$ echo "You are at ‘pwd‘"
  You are at /home/amits/Desktop

Example with problem:
---------------------
$ my_var=Hi there
$ echo ‘$my_var‘ # after expansion it is ‘Hi there‘
                 # and "Hi" is the first argument
   Hi: command not found

Example without problem:
$ var=123
$ echo ‘echo $var‘
 123
    $ echo $(echo $(echo $var))
     123
```

3.40 math..$[...]...$((...))...((...)) : command to do simple arithmetic

```
$(( )) is the same as $[ ] : performs integer math operation


$ echo $[ 1 + 2 ]  # it is not a command
                   # but part of bash features !!!
  3
$ echo $[1+2]
  3
$ echo $((1+2))   # it is not a command,
                  # only part of bash features !!!        3
$ echo $((2*4))
  8
$ echo $((5/2))
  2
```

```
$ echo $((5>2))
  1  # NOTE : in this case true is 1,
     # it is different than test command.
$ echo $((5<2))
  0

     Operator     Operation
     +, -, *, /     addition, subtraction, multiply, divide
     %           Modulus (Return the remainder after division)
     >,<,>=,<=       Note : the result logic is this case is similar
                            to C language.

$ x=4
$ y=5
$ ((z=x+y))
$ echo $z
  9
$ ((x+=y))
$ echo x
  9
$ ((x++))
$ echo x
  10

Result of ((   )) returns status code:
$ ((5>17))        # this is COMMAND !!!
$ echo $?
  1       # regular "bash" logic
$ ((5<17))
$ echo $?
  0




 ((5>17)) logic of status code is       :  0 -  true,  1 - false
         -----------
$((5>17)) logic of output expression is :  1 -  true,  0 -  flase
         -----------------
note: if command in bash uses status code !!!
Two following expressions are equivalent :

if [ "$x" -gt "$y" ];then echo YES; else echo NO; fi
if ((x>y))          ;then echo YES; else echo NO; fi

Additional explanation:
((0)) --> fault   ---> status = 1
((1)) --> sucess  ---> status = 0

Recommendation:
```

```
---------------
   use  ((  )) only for boolean conditions :  if ((  )) ...
   use $((  )) only for math :  echo $((3+4)) ; x=$((5+8))


   use $(( )) only for math operations

-------------------
$ x=6;y=10
$ ((y=++x))
$ echo "x=$x, y=$y"

$ echo "x=$x, y=$y"
  x=7, y=7

$ x=6;y=10
$ ((y=x++))
$ echo "x=$x, y=$y"

$ echo "x=$x, y=$y"
  x=7, y=6    # y gets x value before increment of x

-------------------
((x--))
((--x))
```

3.41 let : math - command to do simple arithmetic

let is a builtin function of Bash that allows us to do simple arithmetic.
It follows the basic format:

```
  $ let <arithmetic expression>
```

Here is a table with some of the basic expressions you may perform.
There are others but these are the most commonly used.

Operator	Operation
+, -, *, /	addition, subtraction, multiply, divide
var++	Increase the variable var by 1
var--	Decrease the variable var by 1
%	Modulus (Return the remainder after division)

```
   $ let a=5+4
   $ echo $a # 9
   $ let "a = 5 + 4"
   $ echo $a # 9
   $ let a++
   $ echo $a # 10
   $ let "a = 4 * 5"
   $ echo $a # 20
   $ let "a = $1 + 30"
```

```
$ echo $a # 30 + first command line argument
```

3.42 true : Returns a successful exit status code of 0: $? = 0

```
- Return a successful exit code:
   $ true

 Example :
 $ ! true
 $ echo $?
  1
```

3.43 false : Returns a non-zero exit code: $? = 1

```
- Return a non-zero exit code:
   $ false

 Example :
 $ ! false
 $ echo $?
  0
```

3.44 tab : autocomplete

```
Tab   : automatic completion
Tab Tab  : all options for complete
```

3.45 type : Display the type of command the shell will execute.

```
The ‘type’ is a built-in shell command used to identify the kind of
command. For example it checks if it is shell built in or external
command.

 type without any key :
              if NAME is an alias,
          shell reserved word,
          shell function,
          shell builtin,
          disk file
          or not found, respectively

 -t output a single word which is one of
              ’alias’,
              ’keyword’,
              ’function’,
              ’builtin’,
              ’file’
```

```
-p Returns the disk file that would be executed
-a Returns all locations containing an executable.

 For example:
 $ type echo
     echo is a shell builtin
 $ type cat
     cat is /usr/bin/cat
 $ type code
     code is /usr/bin/code
 $ type -a time
     time is a shell keyword
     time is /usr/bin/time
     time is /bin/time
```

3.46 Comments in BASH

```
comments in BASH :
    #comment
```

3.47 source/"." : Execute commands from a file in the current shell

```
Execute commands from a file in the current shell

- Evaluate contents of a given file:
   $ source {{path/to/file}}

- Evaluate contents of a given file
   (alternatively replacing source with .):
   $ . {{path/to/file}}
```

3.48 read : retrieving data from stdin

```
Read a line from the standard input and split it into fields.
$ read a b c d
  aaa bbb ccc ddd
$ echo $a
  aaa
$ echo $b
  bbb
$ echo $c
  ccc
$ echo $d
  ddd
$ bash  # if parameters are not exported ,
```

```
        # the child process does not recognize them
 $ echo "x=$x, a=$a, b=$b, c=$c"
   x=456, a=, b=, c=
 $ exit
   exit
 $ echo "x=$x, a=$a, b=$b, c=$c"
   x=456, a=aaa, b=bbb, c=ccc
 $

 Shell builtin for retrieving data from stdin
 $ type read
   read is a shell builtin
 $ help read   # man read is something else
   -a array assign the words read to sequential indices of the array
   -p prompt output the string PROMPT without a trailing newline before
 attempting to read
  -s silent (without output to terminal, to enter passwords)
Example:
 $ read -p "What's your name? " name
 $ echo "Hello $name"

If we insert the input at the same line, the prompt does not appear:
 ex_04.sh:
       ---------------------------
      $ read -p "What's your name? " name
      $ echo "Hello $name"
      ---------------------------
$ echo shmuel | bash ex04.sh
    Hello shmuel   # without prompt
```

3.49 gnome-terminal : open new linux terminal emulator

gnome-terminal is a terminal emulator application for accessing a UNIX shell environment which can be used to run programs available on your system.

```
$ gnome-terminal . &  # may be . is redundant
```

3.50 arguments..of..process

```
$# - number of arguments
S0 - name of script file
$1 - the first argument
$2 - the second argument
$* is full list of arguments
$@ is full list of arguments
-----------------------------------------
ex05.sh
-----------------------------------------
echo "this program was run with $0"  # S0 is name of script file
```

```
echo "received $# arguments."        # $# is number of arguments
echo "first arguemnt is [$1]"        # $1 is the first argument
echo "second argument is [$2]"       # S2 is the second argument
echo "third argument is [$3]"

echo "full list: $*"                 # $* is full list of arguments
echo "full list: $@"                 # $@ is full list of arguments
----------------------------------------
$ bash ex05.sh aaaa bbbb ccccc
      this program was run with ex05.sh
      received 3 arguments.
      first arguemnt is [aaaa]
      second argument is [bbbb]
      third argument is [ccccc]
      full list: aaaa bbbb ccccc

Note : $# does not take into account the argument 0 (the command itself) !
       it counts arguments without zero argument !!!


 $ bash ex05.sh "aaaa bbbb ccccc" # now it treated as single argument !!!
    this program was run with ex05.sh
      received 1 arguments.
      first arguemnt is [aaaa bbbb ccccc]
      second argument is []
      third argument is []
      full list: aaaa bbbb ccccc
      full list: aaaa bbbb ccccc
```

3.51 clear :Equivalent to CTRL+L : clears the screen of the terminal.

```
Clears the screen of the terminal.
 - Clear the screen (equivalent to pressing Control-L in Bash shell):
 $  clear
```

3.52 process..output..to..stderr

```
    Trick to print ouput to stderr from process:
    ---------------
    ex07.sh:
      echo "normal output"
      echo "error output" >&2
    ---------------
      $ bash ex07*  >/dev/null
      $ bash ex07* 2>/dev/null
```

3.53 exit : exit the process (script) , exit the shell

```
  Example with exit the process:
  ----------------------
  echo "this program will fail"
  exit 1  # exit with status code other than 0 is: error
  # Note:
  #     exit
  # is the same as:
  #     exit 0
  ----------------------
  $ bash ex06.sh
    this program will fail
  $ echo $?
    1
Note : exit status is limited to 255

Example with exit the bash:

  $ whoami
    amits
  $ su david
    password:
  $ whoami
    david
  $ exit    # exit from david
  $ whoami
    amits
```

3.54 if : if command

```
if expression1; then  # if expression1 return status of sucess (exit status 0)
                      # then statement1 will be executed.
     statement1
elif expression2  # elif (!), not elsif
     then statement2
else
     statement3
fi

Note : elif can appear several times
       if and else can appear only once.


Examples (also see test command):
---------------------------------
if ! grep $USER /etc/passwd
       then echo "your user account is not managed locally";
```

```
fi

grep $USER /etc/passwd
if [ $? -ne 0 ] ; then
   echo "not a local account" ;
fi
--------------------------------
leap year example:

#! /usr/bin/bash

if [ -z "$1" ]; then
   echo "Error: no arguments" >&2
   exit 2
fi

if (( $# != 1 )); then
  echo "Error: must have exactly 1 argument for year" >&2
  exit 2
fi

if (( ($1%4)==0));then
   if (( ($1%100)==0 ));then
      if (( ($1%400)==0 ));then
        echo "$1 % 4 == 0, $1 % 100 == 0 and $1 % 400 == 0 "
        echo "leap year"
        exit 0
      else
        echo "$1%4==0 and $1%100==0, but not dividable by 400";
        echo "common year"
        exit 1
      fi
   else
      echo "$1%4==0 && $1%100 != 0";
      echo "it is leap year"
      exit 0
   fi
else
   echo "$1 % 4 != 0";
   echo "it is common year"
   exit 1
fi
------------------------------
$ bash leap_year_check_upd.sh 2000
$ echo $?
    0
$ bash leap_year_check_upd.sh 2002
$ echo $?
    1
$ bash leap_year_check_upd.sh 2002 2000
```

```
    Error: must have exactly 1 argument for year

$ if bash leap_year_check.sh 2020;then echo YES;else echo NO;fi
  2020%4==0 && 2020%100 != 0
  it is leap year
  YES
```

3.55 test..[...] : command that can be used as test in if command

```
Evaluate conditional expression.

Exits with a status of 0 (true) or 1 (false) depending on
the evaluation of EXPR.  Expressions may be unary or binary.
Unary expressions are often used to examine the status of a file.
There are string operators and numeric comparison operators as well.
The behavior of test depends on the number of arguments.

          File operators:
          ---------------
  -a FILE        True if file exists.
  -b FILE        True if file is block special.
  -c FILE        True if file is character special.

  -d FILE        True if file exists and is a directory.
  -e FILE        True if file exists.
  -f FILE        True if file exists and is a regular file.
         # Note: regular file: not dir and not soft link

  -g FILE        True if file is set-group-id.
  -h FILE        True if file is a symbolic link.
  -L FILE        True if file is a symbolic link.
  -k FILE        True if file has its ‘sticky’ bit set.
  -p FILE        True if file is a named pipe.
  -r FILE        True if file is readable by you.
  -s FILE        True if file exists and is not empty.
  -S FILE        True if file is a socket.
  -t FD          True if FD is opened on a terminal.
  -u FILE        True if the file is set-user-id.
  -w FILE        True if the file is writable by you.
  -x FILE        True if the file is executable by you.
  -O FILE        True if the file is effectively owned by you.
  -G FILE        True if the file is effectively owned by your group.
  -N FILE        True if the file has been modified since it was last read.

  FILE1 -nt FILE2  True if file1 is newer than file2 (according to
                   modification date).
  FILE1 -ot FILE2  True if file1 is older than file2.
  FILE1 -ef FILE2  True if file1 is a hard link to file2.
```

```
                (same device and inode numbers)

          String operators:
          -----------------
-z STRING           True if string is empty.
-n STRING           True if string is not empty.
STRING1 = STRING2   True if the strings are equal.
STRING1 != STRING2  True if the strings are not equal.
STRING1 < STRING2   True if STRING1 sorts before STRING2
                    lexicographically.
STRING1 > STRING2   True if STRING1 sorts after STRING2
                    lexicographically.

          Other operators:
          ----------------
-o OPTION      True if the shell option OPTION is enabled.
-v VAR         True if the shell variable VAR is set.
-R VAR         True if the shell variable VAR is set
               and is a name reference.
! EXPR         True if expr is false.
EXPR1 -a EXPR2 True if both expr1 AND expr2 are true.
EXPR1 -o EXPR2 True if either expr1 OR expr2 is true.

arg1 OP arg2   Arithmetic tests.  OP is one of -eq, -ne,
               -lt, -le, -gt, or -ge.
-------------------------------------------------------------------
$ which test
  /usr/bin/test

Options to perform test:
    test EXPRESSION
    test             # always true
    [ EXPRESSION ]   # ] is argument
    [ ]              # always true



Strings comparison :
   NOTE : put"" on compared operands when using test command !!!

      "="
      "<"
      ">"
      "<="
      ">="
      "!="

      -o : or
      -a : and
       ! : not
```

```
$ [ "$password_guess" = 1234 -a "$username_guess" = grisha ]
$ ["$password_guess" = 1234 ] && ["$username_guess" = grisha ]
         # check that both commands succeded !!!

Same options:
-------------
if [! "$password_guess" != 1234 ] && ["$username_guess" = grisha ]; then
if ! ["$password_guess" != 1234 ] || ! ["$username_guess" = grisha ]; then
if ! ["$password_guess" != 1234 ] && ["$username_guess" = grisha ]; then
if ["$password_guess" = 1234 ] && ["$username_guess" = grisha ]; then
echo "Access granted"
else
echo "Access denied"
exit 1
fi
---------------


$ [ 1 = 1 -o asd = 1234 ] # note backspaces
$ echo $?
  0




$ test "AAA" = "AAA" # note spaces around "="
         $1  $2  $3


$ test "AAA" = "AAA" # Note MUST enter spaces between arguments !!!
$ echo $?
  0
$ test "AAA" = "BBB"
$ echo $?
  1
$ test "AAA" != "AAA"
$ echo $?
  1
$ test "AAA" "<" "BBB"
$ echo $?
  0
$ test "AAA" ">" "BBB"
$ echo $?
  1

Numbers comparison (integer comparison):
      -lt (less than)
      -eq (equal)
      -ne (not equal)
```

```
    -le (less or equal)
    -gt (greater than)
    -ge (greater or equal)

$ test 100 -lt 50  # less than
$ echo $?
  1
$ test 100 -eq 50  # equal
$ echo $?
  1
$ [ 100 -lt 200 ] && echo TRUE || echo FALSE
  TRUE
$ [ 200 -lt 100 ] && echo TRUE || echo FALSE
  FALSE


  [            200  -lt  100 ]
  $0(command)  $1    $2  $3  $4


-------------------------------------------
Note : put " " around arguments within test/[ ]

Example1 (problematic, enter "1 = 1 -o asd" to ):

  # read passsword from the user
  # the password is 1234
  # if the password is correct print
  # "Acess granted"
  # else print
  # "Acess denied" and exit with failure
  read -p 'enter password:' password_guess
  if [ $password_guess = 1234 ]; then
echo "Access granted"
  else
echo "Access denied"
exit 1
  fi


Example2 (now it can not be hacked):

  # read passsword from the user
  # the password is 1234
  # if the password is correct print
  # "Acess granted"
  # else print
  # "Acess denied" and exit with failure
  read -p 'enter password:' password_guess
  if [ "$password_guess" = 1234 ]; then   # now password guess
```

```
                                                # is one single word
                                                # because of " "
 echo "Access granted"
    else
 echo "Access denied"
 exit 1
    fi

Example 3 :
  # read username and passsword from the user
  # the password is 1234
  # the username is grisha
  # if the credentials are correct print
  #  "Access granted"
  # else print
  #  "Access denied" and exit with failure
  read -p 'enter username:' username_guess
  read -p 'enter password:' password_guess

  if [ "$password_guess" = 1234 \
       -a "$username_guess" = grisha ]; then
echo "Access granted"
  else
echo "Access denied"
exit 1
  fi
  ---------------------------------------
  $ [ ]        # without any text always 1 !
  $ echo $?
    1
  $ [ anyText ] # with any text always 0
  $ echo $?
    0
```

3.56 for : loop command

```
https://www.cyberciti.biz/faq/bash-for-loop/

For does not work as classic prog. languages.
but using ((   )) we can do the same
as in C:

Three-expression bash for loops syntax
---------------------------------------
This type of for loop share a common heritage with the C
programming language. It is characterized by a three-parameter loop
control expression; consisting of an initializer (EXP1), a loop-
test or condition (EXP2), and a counting expression/step (EXP3).
```

```
for (( EXP1; EXP2; EXP3 ))
      ## The C-style Bash for loop ##
      for (( initializer; condition; step ))
      do
        shell_COMMANDS
      done
```

A representative three-expression example in bash as follows:

```
        #!/bin/bash
        # set counter 'c' to 1 and condition
        # c is less than or equal to 5
        for (( c=1; c<=5; c++ ))
        do
           echo "Welcome $c times"
        done
```

How do I use for as infinite loops? ↑

Infinite for loop can be created with empty expressions, such as:

```
        #!/bin/bash
        for (( ; ; ))
        do
           echo "infinite loops [ hit CTRL+C to stop]"
        done
```

Conditional exit with break

You can do early exit with break statement inside the for loop. You can exit from within a FOR, WHILE or UNTIL loop using break. General break statement inside the for loop:

```
        for I in 1 2 3 4 5
        do
          statements1      #Executed for all values of
                           #''I'', up to a disaster-condition
                           # if any.
          statements2
          if (disaster-condition)
                then
               break      #Abandon the loop.
          fi
          statements3    #While good and, no disaster-condition.
         done
```

Early continuation with continue statement

To resume the next iteration of the enclosing FOR, WHILE or UNTIL loop use continue statement.

```
  for I in 1 2 3 4 5
  do
    statements1      #Executed for all values of
                     # ''I'', up to a disaster-condition if any.
```

```
     statements2
     if (condition)
        then
      continue   #Go to next iteration of I
                 #in the loop and skip statements3
     fi
    statements3
   done

   ---------------------------------------
   *           for loop syntax           *
   ---------------------------------------
    Numeric ranges for syntax is as follows:

    for VARIABLE in 1 2 3 4 5 .. N
    do
command1
command2
commandN
    done

    OR

    for VARIABLE in file1 file2 file3
    do
command1 on $VARIABLE
command2
commandN
    done

    OR

    for OUTPUT in $(Linux-Or-Unix-Command-Here)
    do
command1 on $OUTPUT
command2 on $OUTPUT
commandN
    done
   ------------------------------------
      #!/bin/bash
      for i in 1 2 3 4 5
      do
        echo "Welcome $i times"
      done
   ------------------------------------
      #!/bin/bash
      for i in {1..5}
      do
        echo "Welcome $i times"
      done
```

```
 ------------------------------------
  #!/bin/bash
  echo "Bash version ${BASH_VERSION}..."
  for i in {0..10..2}
  do
    echo "Welcome $i times"
  done
 ------------------------------------
 Loop with strings (helps to divide string to separate words)

 #!/bin/bash
 #Say we have a variable, and we need to loop through
 #a list of strings to install those packages:
 TRY_VAR="Mor Rotem Amit"
 for p in $TRY_VAR
 do
    echo "$p is ready"
 done

 Result:
 $ bash try_for.sh
    Mor is ready
    Rotem is ready
    Amit is ready
 ------------------------------------
 #!/bin/bash
 ## $@ expands to the positional parameters, starting from one.
 for i in $@
 do
    echo "Script arg is $i"
 done

 Result:
 $ bash try_for_for_args.sh well bad good
    Script arg is well
    Script arg is bad
    Script arg is good
---------------------------------
for var in $(command)
do
  print "$var"
done

## example ##
## Show the name of every PDF file in the /nas directory:
for f in $(ls ./try_dir)
do
  printf "File is %s\n" "$f"
done
```

```
Result:
$ bash for_with_ls_command.sh
  File is 1.txt
  File is 2.txt
  File is my_hostname.txt
  File is my_name.txt
```

3.57 case : case command

1. Example of case :

```
#!/bin/bash
if [ $# -lt 2 ];then
    echo "Usage : $0 Signalnumber PID"
    exit
fi

case "$1" in
   1) echo "Sending SIGHUP signal"
      kill -SIGHUP $2
      ;;
   2) echo "Sending SIGINT signal"
      kill -SIGINT $2
      ;;
   9) echo "Sending SIGKILL signal"
      kill -SIGKILL $2
      ;;
   *) echo "Signal number $1 is not processed"
      ;;
esac
```

2. Example of case :

```
#! env bash
while true; do
echo 'Drinks Menu:
1) Water
2) Coffee
3) Cola
4) Tea
0) Exit
'
read -p 'enter drink number:' drink_number

 case "$drink_number" in
  1|water)
       echo "you got water"
       ;;
  2|coffee|java)
```

```
          echo "you got coffee"
          ;;
        3|cola)
            echo "you got cola"
            ;;
      4|tea)
           echo "you got tea"
           ;;
      # 0 or q or exit or quit or anything
      # that starts with "x"
      0|q|"exit"|quit|x*)
      echo "goodbye"
      break
      ;;
        *)
   echo "ERROR: invalid drink number"
   ;;
        esac
    done
    # if--fi case-esac  while-done for-done
```

3.58 while : while command

```
#! /usr/bin/bash
read -p "Enter base value: " base_value
read -p "Enter power value: " power_value
num_of_iterations=1
result=1;

while [ $power_value -ge $num_of_iterations ]; do
    echo "num_of_iterations = $num_of_iterations"
    num_of_iterations=$((num_of_iterations+1))
    result=$((result*base_value))
done

echo "result = $result"
exit
```

4 Processes/Jobs Control

4.1 process..input..output

```
How process gets information:
-----------------------------
1. arguments
2. read file (side effect)
```

```
  3. signals (KILL)
  4. env. variables
  5. stdin (0)

  How process outputs information:
  --------------------------------
  1. stdout(1)
  2. stderr(2)
  3. write file (side effect)
  4. exit code (0 - well, others - error)
```

4.2 arguments..of..process

```
$# - number of arguments
S0 - name of script file
$1 - the first argument
$2 - the second argument
$* is full list of arguments
$@ is full list of arguments
-----------------------------------------
ex05.sh
-----------------------------------------
echo "this program was run with $0"  # S0 is name of script file
echo "received $# arguments."        # $# is number of arguments
echo "first arguemnt is [$1]"        # $1 is the first argument
echo "second argument is [$2]"       # S2 is the second argument
echo "third argument is [$3]"

echo "full list: $*"                 # $* is full list of arguments
echo "full list: $@"                 # $@ is full list of arguments
-----------------------------------------
$ bash ex05.sh aaaa bbbb ccccc
      this program was run with ex05.sh
      received 3 arguments.
      first arguemnt is [aaaa]
      second argument is [bbbb]
      third argument is [ccccc]
      full list: aaaa bbbb ccccc

Note : $# does not take into account the argument 0 (the command itself) !
       it counts arguments without zero argument !!!


 $ bash ex05.sh "aaaa bbbb ccccc" # now it treated as single argument !!!
    this program was run with ex05.sh
      received 1 arguments.
      first arguemnt is [aaaa bbbb ccccc]
      second argument is []
```

```
third argument is []
full list: aaaa bbbb ccccc
full list: aaaa bbbb ccccc
```

4.3 Ctrl+Z : SIGTSTP - an inTeractive SToP signal

```
https://medium.com/@prateeksrivastav598/
understanding-ctrl-z-and-ctrl-c-in-the-
linux-terminal-c28bd1e00dbe

https://www.gnu.org/software/libc/manual/html_node/
Job-Control-Signals.html

Unlike SIGSTOP, this signal can be handled and ignored.
Your program should handle this signal if you have
a special need to leave files or system tables in a
secure state when a process is stopped.
For example, programs that turn off echoing should handle
SIGTSTP so they can turn echoing back on before stopping.

    Ctrl + Z: Suspending a Process
Ctrl + Z is used to suspend a running process in the terminal.
This means that you temporarily :
 1. pause the execution of a program and
 2. move it to the background, allowing you to continue using
    the terminal.
It's useful when you want to halt a process without terminating it.
Here's how to use it:

    Example 1: Pausing a Long-Running Process
    ----------------------------------------
Let's say you're compressing a large file using the gzip
command, and it's taking longer than expected.
You can press Ctrl + Z to pause the compression process and return
to the terminal prompt.
This doesn't terminate the gzip command; it simply stops it
temporarily.
  $ gzip -9 largefile.txt
  [Press Ctrl + Z]
  [1]+  Stopped       gzip -9 largefile.txt
  We should note that even though the shell says the job is Stopped,
  it is, in fact, paused - not terminated - and we'll be able
  to resume it. To resume the operation, we can use the fg command.
  So, the gzip process is paused and can be resumed later using
  the fg (foreground) command:
  $ fg

    Example 2: Running a Process in the Background.
```

```
    ------------------------------------------------
You can also use Ctrl + Z to start a command in the background.
For instance, you can run a command like this:
  $ my_long_running_command &
However, if you forget to use the ampersand (&) to run it in the
background, you can press Ctrl + Z to suspend it and then type bg
(background) to resume its execution in the background.
```

4.4 Ctrl+C : SIGINT <SIgnal INTerrupt> interrupting a process

```
https://medium.com/@prateeksrivastav598/
understanding-ctrl-z-and-ctrl-c-in-the-linux-terminal-c28bd1e00dbe

            SIGINT(^C) - The gentle nudge
            -----------------------------
SIGINT(^C), or <SIgnal INTerrupt>, is perhaps the most commonly
encountered signal for many users. This signal is typically associated with the
CTRL+C command you often use in your terminal to stop a running process.
The primary purpose of SIGINT is to notify a process that the user
has requested an interruption.
Example of SIGINT:
   $ kill -INT <process_id> # SIGINT (interrupt) signal
     or
   $ kill -2 <process_id>

Example :
   run on first terminal window:
    $ sleep 100
   run on second terminal window:
    $ ps aux | grep sleep
root   9416  0.0  0.0  3212  1792 ?     S  Feb26 0:00 sleep 3600
amits  0816  0.0  0.0  8376  1920 pts/0 S+ 00:16 0:00 sleep 100
amits 10822  0.0  0.0  9216  2432 pts/1 S+ 00:16 0:00 grep --color=auto sleep
     $ kill -2 10816  # this will kill the process
                      # 10816 on first terminal window.


Ctrl + C: Interrupting a Process
Ctrl + C is used to forcefully terminate a running process.
When you press Ctrl + C, the terminal sends an interrupt signal (SIGINT)
to the process, causing it to stop immediately.
It's a quick way to exit a process or command.

Example: Stopping an Unresponsive Program
   Imagine you're running a program that becomes unresponsive
   or is stuck in an infinite loop.
   You can use Ctrl + C to interrupt the program and
   return to the terminal prompt.
```

```
$ ./my_program
[Press Ctrl + C]
^C
The ^C represents the interrupt signal.
```

4.5 Ctrl+D : End of File / exit shell

```
Note1: ^D is not signal, it is simply End Of File.
       ^D closes Bash
Sometimes when we try to exit the shell by pressing Ctrl+D
or by using logout, we may receive an error message:
$ logout
  There are stopped jobs. # message when there are paused jobs

When bash shows this message, it also prevents logout.
This is because bash doesn't allow us to exit the shell while
there are paused jobs.

The current shell's process manages its jobs.
When we pause a job, it's left in an incomplete state. If we
exit the shell with jobs paused, we might lose some critical data.
So, we need to take care of these paused jobs before we can exit the shell.
```

4.6 ps : "Process Status" - snapshot of the current processes

```
see https://sites.ualberta.ca/dept/chemeng/AIX-43/share/man/info/C
/a_doc_lib/cmds/aixcmds4/ps.htm

The ps program (short for "process status") displays info on the
currently-running processes. A related Unix utility named "top"
provides a real-time view of the running processes.

This version of ps accepts 3 kinds of options:
  1   UNIX options, which may be grouped and must be preceded by a dash.
  2   BSD options, which may be grouped and must not be used with a dash.
  3   GNU long options, which are preceded by two dashes.

Options of different types may be freely mixed, but conflicts can
appear.  There are some synonymous options, which are functionally
identical, due to the many standards and ps implementations that this
ps is compatible with.
--------------------------------------------------------------------------------
To see only the processes owned by a specific user on Linux run:
   $ ps -u {USERNAME}

   $ ps -u  | grep bash # present all processes of CURRENT USER !
     amits       6425  0.0  0.0  12056  5760 pts/0    Ss   11:31   0:00 bash
     amits       7263  0.0  0.0  11408  5120 pts/1    Ss+  12:27   0:00 /bin/bash
```

```
     amits      13283  0.0  0.0  10108  3840 pts/0    T    14:49   0:00 bash ex04
     amits      15526  0.0  0.0  10108  3840 pts/0    T    15:06   0:00 bash ex04
-------------------------------------------------------------------------------
 -a Select all processes except:
      1. session leaders (see getsid(2))
      2. processes not associated with a terminal.
 -u userlist Select by effective user ID (EUID) or name.
      This selects the processes whose effective user
      name or ID is in userlist.

 -x is for tty :
     Today in Linux, the tty is a legacy name used to refer
     to the user interface for text-based input and output,
     otherwise known as a "terminal".
     In Linux systems, there can be multiple tty device
     "consoles", to support potentially dozens of serial
     ports or more.
     tty0 is the current one in use, but Linux allows you
     to switch to another session by changing to a different tty,
     e.g., tty1. Linux (e.g., Ubuntu) supports up to 6 ttys by default,
     but this number is configurable

 EXAMPLES
To see every process on the system using standard syntax:
   $ ps -e    #  Select all processes.
   $ ps -ef   #  Do full-format listing.
   $ ps -eF   #  Extra full format.
   $ ps -ely  #  -l : BSD Long format.
                 -y : Do not show flags; show rss in place of addr.
                    This option can only be used with -l.
By running $ ps l, you can see which processes are running,
and which are sleeping. If a process is sleeping, the WCHAN column
("wait channel", the name of the wait queue) will tell you what
kernel event the process is waiting for.
   $ ps  l # display BSD long format.
   $ ps -l # Long format.  The -y option is often useful with this

To see every process on the system using BSD syntax:
   $ ps ax
   $ ps aux

Example :
   $ ps aux | grep bash
      amits       5464  0.0  0.0  11408  5120 pts/0    Ss+  13:03   0:00 bash
      amits       6966  0.0  0.0  11408  5248 pts/1    Ss   14:32   0:00 bash
      amits       6974  0.0  0.0   9216  2432 pts/1    S+   14:32   0:00 grep
                                                                --color=auto bash
To print a process tree:
   ps -ejH
   ps axjf
```

To get info about threads:
ps -eLf
ps axms

Column	Contents
%CPU	How much of the CPU the process is using
%MEM	How much memory the process is using
ADDR	Memory address of the process
C or CP	CPU usage and scheduling information
COMMAND*	Name of the process, including arguments, if any
NI	Nice value
F	Flags
PID	Process ID number
PPID	ID number of the process's Parent Process
PRI	Priority of the process
RSS	Resident set size (process memory occupied in RAM)
S or STAT	Process status code
START or STIME	Time when the process started
VSZ	Virtual memory usage
TIME	The amount of CPU time used by the process
TT or TTY	Terminal associated with the process
UID or USER	Username of the process's owner
WCHAN	Memory address of the event the process is waiting for

Note 1 : In computing, resident set size (RSS) is the portion of memory (measured in kilobytes) occupied by a process that is held in main memory (RAM). The rest of the occupied memory exists in the swap space or file system, either because some parts of the occupied memory were paged out, or because some parts of the executable were never loaded.
Note 2 : nice is a program found on Unix and Unix-like operating systems such as Linux. It directly maps to a kernel call of the same name.
nice is used to invoke a utility or shell script with a particular CPU priority, thus giving the process more or less CPU time than other processes.
A niceness of -20 is the lowest niceness, or highest priority.
Note 3 : The SZ field displays the size of the process in memory.
The value of the field is the number of pages the process is occupying.
On Linux machines, a page is 4,096 bytes.

PROCESS STATE CODES
https://askubuntu.com/questions/1144036/process-status-of-s-s-s-sl
https://www.linusakesson.net/programming/tty/index.php
Here are the different values that the s, stat and state output specifiers (header "STAT" or "S") will display to describe the state of a process:

D uninterruptible sleep (waiting for some event)
R running or runnable (on run queue)
S interruptible sleep (waiting for some event or signal)
T stopped by job control signal

```
        t    stopped by debugger during the tracing
        Z    defunct "zombie" process, terminated but not yet
             reaped by
             its parent
   A Linux process can be in one of the following states:

        R------------->Z
   D<-->R     S------->Z
        R<--->S--->T-->Z
        R<-------->T

   For BSD formats and when the stat keyword is used, ADDITIONAL
   CHARACTERS may be displayed:

        <    high-priority (not nice to other users)
        N    low-priority (nice to other users)
        L    has pages locked into memory (for real-time and custom IO)
        s    This process is a session leader
        l    This process is multi-threaded (using CLONE_THREAD,
             like NPTL pthreads do)
        +    This process is in the foreground process group

    Job control is what happens when you press ^Z to suspend a program,
    or when you start a program in the background using &.
    A job is the same as a process group.
    Internal shell commands like jobs, fg and bg can be used to manipulate
    the existing jobs within a session.
    Each session is managed by a session leader, the shell, which is
    cooperating tightly with the kernel using a complex protocol of signals
    and system calls.

-----------------------------------------------------------------
The default niceness for processes is inherited from its parent process
and is usually 0.

    Users can pipeline ps with other commands,
    such as less to view the process status output one
    page at a time:

    $ ps -A | less

    Users can also utilize the ps command in conjunction with
    the grep command (see the pgrep and pkill commands) to find
    information about a single process, such as its id:

    $ # Trying to find the PID of ‘firefox-bin‘ which is 2701
    $ ps -A | grep firefox-bin
      2701 ?        22:16:04 firefox-bin
Custom columns choose :
    $ ps a -o user -o comm -o pid -o nice
```

```
  USER     COMMAND             PID  NI
  amits    gdm-wayland-ses    2342   0
  amits    gnome-session-b    2345   0
  root     agetty             3891   0
  root     agetty             3931   0
  amits    tex_file_data_e   14438   0
  amits    htop              30232   0
  root     sudo              30469   0
  root     sudo              30471   0
  root     geany             30472 -20
  root     bash              30520 -20
  root     dbus-launch       30529 -20

AIX FORMAT DESCRIPTORS
   This ps supports AIX format descriptors, which work
   somewhat like the
   formatting codes of printf(1) and printf(3).
   For example, the normal default output can be
   produced with this: ps -eo "%p %y %x %c".  The
   NORMAL codes are described in the next section.

   CODE   NORMAL   HEADER
   %C     pcpu     %CPU
   %G     group    GROUP
   %P     ppid     PPID
   %U     user     USER
   %a     args     COMMAND
   %c     comm     COMMAND
   %g     rgroup   RGROUP
   %n     nice     NI
   %p     pid      PID
   %r     pgid     PGID
   %t     etime    ELAPSED
   %u     ruser    RUSER
   %x     time     TIME
   %y     tty      TTY
   %z     vsz      VSZ

Example:
  $ ps a -o user -o comm -o pid -o nice
it is the same as :
  $ ps a -o "%U %c %p %n"


 The use of pgrep simplifies the syntax and avoids potential race
 conditions:
```

4.7 pstree : display a tree of processes

```
https://www.geeksforgeeks.org/pstree-command-in-
linux-with-examples/

pstree [options] [pid or username]

A convenient tool to show running processes as a tree
The pstree command is a handy little utility that shows the
processes currently running on your system and it show them in
a tree diagram.
To display process tree:
   $ pstree

To include command line arguments in output
   $ pstree -a

To display PIDs
   $ pstree -p

To force pstree to expand identical subtrees in output.
   $ pstree -c

To sort processes
   $ pstree -n

To see who is the owner/user of a process
   $ pstree -u

To highlight the current process or any other process
   $ pstree -h

To show process group IDs in output
   $ pstree -g

To make pstree display process tree specific to a user.
   $ pstree amits
```

4.8 pgrep : ps+grep

```
Find or signal processes by name

Search for a Linux process by name run:
  $ pgrep -u {USERNAME} {processName}
  Example :
  $ pgrep -u amits sleep # only for "amits" user
    19267
  $ pgrep sleep # all users
```

```
18588
19267
```

```
Return only PID number , pgrep = ps + grep :
   $ pgrep ping = ps aux | grep -w [p]ing:
```

4.9 top : "Table Of Processes"

```
To show all processes for a specific user using top/htop
    $ top -U amits

is a task manager or system monitor program,
found in many Unix-like operating systems,
that displays information about CPU and memory utilization


The program produces an ordered list of running processes
selected by user-specified criteria,
and updates it periodically.
Default ordering is by CPU usage,
and only the top CPU consumers are shown.
top shows how much processing power and memory are being used,
as well as other information about the running processes.
Some versions of top allow extensive customization of the display,
such as choice of columns or sorting method.
top is useful for system administrators, as it shows which
users and processes are consuming the most system
resources at any given time.

top view can be listed with help of arrows.

   To get help :
   $ top
   ^Z         # suspend and move the process to bg
   $ man top
   $ fg       # move the process to fg
```

4.10 htop : similar as top, but more user friendly

```
More user friendly than top,
The title is always on the top of the table.

Press F4 to insert filter word,
 for example, "amits" word in filter will cause to present user "amits"
 "bash" word in filter will present processes of bash.
```

```
Press F5 to see process tree !!!
Press F5 again to present process list.

NI:  niceness in htop window.
   The higher the NI value, the lower the priority-
   the "nicer" the process is to other processes.
```

4.11 nice : run a program with a custom priority (niceness)

```
Niceness values range from -20 (highest priority - most favorable
to the process) to 19 (lowest priority - least favorable
to the process).

∘ The nice value determines the priority of a process,
  with lower values indicating higher priority.

 NOTE:  your shell may have its own version of nice,
 which usually supersedes the version described here.

 Please refer to your shell's documentation  for
 details about the options it supports.

 $ type nice
    nice is /usr/bin/nice # not built in


 nice [OPTION] [COMMAND [ARG]...]

 - Launch a program with altered priority:
     $ nice -n {{niceness_value}} {{command}}
   Example:
     $ sudo nice -n -20 geany
       [sudo] password for amits: ****

https://www.scaler.com/topics/linux-nice/
∘ The Nice command in linux allows users to prioritize
  process execution.
∘ The nice value determines the priority of a process,
  with lower values indicating higher priority.
∘ To check the nice value of a running process,
  use the ps -l <process_id> command.
∘ execute a process with given nice value :
    $ nice -n <nice_value> <command> syntax.

    $ sudo nice -n -20 geany  # this caused to
        [sudo] password for amits:
    $ ps a -o user -o comm -o pid -o nice
       USER     COMMAND             PID  NI
       amits    gdm-wayland-ses    2342   0
```

```
amits    gnome-session-b    2345   0
root     agetty             3891   0
root     agetty             3931   0
root     agetty             3937   0
root     login              4195   0
amits    bash               4294   0
amits    bash               5964   0
amits    bash               6101   0
amits    tex_file_data_e   14438   0
amits    htop              30232   0
root     sudo              30469   0
root     sudo              30471   0
root     geany             30472 -20
root     bash              30520 -20
root     dbus-launch       30529 -20
```

4.12 renice : alters priority of already running process

```
The Renice command can be used to alter a process's priority while it is
already running. Open a terminal and execute the following command:
   $ renice <nice_value> -p <process_id>
Replace <nice_value> with the new priority level you want to assign
and <process_id> with the ID of the running process you wish to modify.
For example, to increase the priority of a process with ID 1234 to 5, you
would run:
   $ renice 5 -p 1234

Niceness values range from -20 (most favorable to the
process) to 19 (least favorable to the process)

- Change priority of a running process:
    $ renice -n {{niceness_value}} -p {{pid}}

- Change priority of all processes owned by a user:
    $ renice -n {{niceness_value}} -u {{user}}

- Change priority of all processes that belong to
  a process  group:
    $ renice -n {{niceness_value}} --pgrp {{process_group}}
```

4.13 eval : combines arguments into a single string and runs

```
https://www.geeksforgeeks.org/eval-command-in-
linux-with-examples/

https://www.educative.io/answers/what-is-the-
eval-command-in-linux

eval runs string as a bash command
```

```
eval is a built-in Linux command which is used to execute
arguments as a shell command.
It combines arguments into a single string and uses it as an
input to the shell and execute the commands.

When you have Linux command saved in a variable and wish to run
that command, the eval command is useful.

Example:
$ COMM="ls"
$ eval $COMM

Example:
$ eval echo hello
```

4.14 jobs : Execute arguments as a single command in shell

```
Display status of jobs in background in the current session

The jobs command marks the last paused job with the + sign and the immediate previo
with the - sign. If we use fg and other job commands without a job number, the last
implied.
 - Show status of all jobs:
   jobs

 - Show status of a particular job:
   jobs %{{job_id}}

 - Show status and process IDs of all jobs:
   jobs -l

 - Show process IDs only of all jobs:
   jobs -p

 Example :
 $ man ed
   ^Z  # ctrl+Z by user
   [1]+  Stopped                 man ed
 $ man cp
   ^Z  # ctrl+Z by user
   [2]+  Stopped                 man cp
 $ nano 2.txt
   ^T^Z # ctrl T + ctrl+Z
   [3]+  Stopped                 nano 2.txt
 $ man pwd
   ^Z
   [4]+  Stopped                 man pwd
 $ jobs
   [1]   Stopped                 man ed      # "stopped" means paused !!!
```

```
  [2]   Stopped                 man cp
  [3]-  Stopped                 nano 2.txt  # job before the last
  [4]+  Stopped                 man pwd     # last job
$ jobs -l    # including job number
  [1]   6065 Stopped                 man ed
  [2]   6084 Stopped                 man cp
  [3]-  6120 Stopped (signal)        nano 2.txt
  [4]+  6123 Stopped                 man pwd
$ fg %1
  man ed
```

4.15 kill : send a signal to a process

```
NAME
    kill - send a signal to a process

SYNOPSIS
     kill [options] <pid> [...]

DESCRIPTION
    The  default  signal  for kill is TERM.
    Use -l or -L to list available signals.
    Particularly useful signals include HUP, INT, KILL, STOP,
    CONT, and 0.

    Alternate signals may be specified in three ways: -9, -SIGKILL or -KILL.

    Negative PID values may be  used to choose whole process groups;
    see the PGID column in ps command output.

    A PID of -1 is special;
    it indicates all processes except the kill process itself and init.

kill sends a signal to a process, usually related
to stopping the process.
All signals except for SIGKILL/KILL and SIGSTOP/STOP can
be intercepted by the process to perform a clean exit.

To present signals list (to be used without the SIG prefix,
first 15 signals are mandatory):
    $ kill -l
```

```
1) SIGHUP       2) SIGINT (1,^C)  3) SIGQUIT(^\)  4) SIGILL       5) SIGTRAP
                ----------------
6) SIGABRT      7) SIGBUS         8) SIGFPE       9) SIGKILL(3)  10) SIGUSR1
                                                  -------------
11) SIGSEGV     12) SIGUSR2       13) SIGPIP      14) SIGALRM     15) SIGTERM (2)
                                                                  ---------------
16) SIGSTKFLT   17) SIGCHLD       18) SIGCONT     19) SIGSTOP     20) SIGTSTP(^Z)
```

```
Note1: ^D is not signal, it is simply End Of File.
       ^D closes Bash
Note2: To respond on kill signal the process need to be run,
       if the process receive the ^C signal while it is suspended with ^Z ,
       it will finish its work only after resume from suspend with fg command.

               SIGINT(^C) - The gentle nudge
               ----------------------------
SIGINT(^C), or <SIgnal INTerrupt>, is perhaps the most commonly
encountered signal for many users. This signal is typically associated with the
CTRL+C command you often use in your terminal to stop a running process.
The primary purpose of SIGINT is to notify a process that the user
has requested an interruption.
Example of SIGINT:
   $ kill -INT <process_id> # SIGINT (interrupt) signal
     or
   $ kill -2 <process_id>

   Example :
      run on first terminal window:
       $ sleep 100
      run on second terminal window:
       $ ps aux | grep sleep
 root   9416  0.0  0.0  3212  1792 ?     S  Feb26 0:00 sleep 3600
 amits  0816  0.0  0.0  8376  1920 pts/0 S+ 00:16 0:00 sleep 100
 amits 10822  0.0  0.0  9216  2432 pts/1 S+ 00:16 0:00 grep --color=auto sleep
       $ kill -2 10816  # this will kill the process
                        # 10816 on first terminal window.

               SIGTERM - The polite request
               ----------------------------
  see: https://linuxhandbook.com/sigterm-vs-sigkill/
  The SIGTERM (SIGnal and TERMinate) signal is used for terminating a program.
  SIGTERM is more forceful than SIGINT but still gives a process the opportunity
  to perform clean-up tasks before it ends. It allows the process to catch the signa
  and manage its termination elegantly - saving data or finishing essential tasks.
  The SIGTERM can also be referred as a soft kill because the process
  that receives the SIGTERM signal may choose to ignore it.
  SIGTERM doesn't kill the child processes, With SIGTERM, a process gets
  the time to send the information to its parent and child processes.
  Its child processes are handled by init.
  To send SIGTERM:
     $ kill <process_id>   # default SIGTERM (terminate)
       or
     $ kill -s TERM <process id>

              SIGKILL - The last resort
               ------------------------
  see: https://linuxhandbook.com/sigterm-vs-sigkill/
```

```
Now, what if a process does not respond to the SIGTERM signal,
or is stuck in an endless loop and not releasing the resources?
This is where SIGKILL comes in. SIGKILL, as the name suggests,
kills the process immediately. The system does not give the process
any chance to clean up or free up resources

SIGKILL (the last option) is used for immediate termination of a process.
This signal cannot be ignored or blocked !

The process will be terminated along with its threads (if any).
It is a brutal way of killing a process, and it should only be
used as a last resort.
Suppose you have an unresponsive process that you want to close.
The SIGKILL can be used in such a case.
SIGKILL kills the child processes as well.
Use of SIGKILL may lead to the creation of a zombie process because
the killed process doesn't get the chance to tell its parent process
that it has received a kill signal.
  $ kill -9 <process id>
     or
  $ kill -KILL <process id>

Terminate a background job:
   $ kill %{{job_id}}

see https://www.gnu.org/software/libc/manual/html_node/Standard-Signals.html
    https://www.gnu.org/software/libc/manual/html_node/Termination-Signals.html
    https://www.gnu.org/software/libc/manual/html_node/Job-Control-Signals.html

INT  is for ^C
QUIT is for ^/ , it is similar to ^C but performes the core damp.
TERM is a generic signal used to program termination. Unlike
     KILL, this signal can be blocked, handled, and ignored.
KILL is for immediate process termination. It cannot be handled or ignored,
     and is therefore always fatal. It is also not possible to block this signal.
TSTP is ^Z signal The SIGTSTP signal is an interactive stop signal.
     Unlike STOP, this signal can be handled and ignored
STOP is The STOP signal stops the process. It cannot be handled,
     ignored, or blocked.
```

4.16 pkill : Signal process by name

```
Signal processes by process's name

see https://www.gnu.org/software/libc/manual/html_node/Standard-Signals.html
https://www.gnu.org/software/libc/manual/html_node/Termination-Signals.html
https://www.gnu.org/software/libc/manual/html_node/Job-Control-Signals.html
```

```
INT  is for ^C
QUIT is for ^/ , it is similar to ^C but performes the core damp.
TERM is a generic signal used to program termination. Unlike
     KILL, this signal can be blocked, handled, and ignored.
KILL is for immediate process termination. It cannot be
     handled or ignored, and is therefore always fatal.
     It is also not possible to block this signal.
TSTP is ^Z signal The SIGTSTP signal is an interactive stop signal.
     Unlike STOP, this signal can be handled and ignored
STOP is The STOP signal stops the process. It cannot be handled,
     ignored, or blocked.

First option:
  $ pkill -QUIT ping
  $ pkill -INT ping   # ^C
Second option:
  $ pkill -TERM ping
The Last option:
  $ pkill -KILL ping
To stop process:
  ^Z: $ pkill -TSTP xed

- Force kill matching processes (can't be blocked):
     pkill -9 "{{process_name}}"
- Send KILL signal to processes which match:
     pkill -KILL "{{process_name}}"
Example :
# on first terminal :
 $ sleep 500
# on second terminal
 $ ps aux | grep sleep
root       14276  0.0  0.0   3212  1664 ?        S    01:41   0:00 sleep 3600
amits      18341  0.0  0.0   8376  1920 pts/0    S+   02:36   0:00 sleep 500
 $ pkill -KILL sleep
 $ pkill -KILL sleep
pkill: killing pid 14276 failed: Operation not permitted # because of root !
```

4.17 xkill : Kill a window interactively in a graphical session

```
- Display a cursor to kill a window when pressing
  the left mouse button (press any other mouse button to cancel):
   $ xkill
```

4.18 bg : resume suspended process to run it in the background.

```
https://www.scaler.com/topics/how-to-run-process-in-background-linux/

- Resume the most recently suspended job and run it in the background:
```

```
$ bg

- Resume a specific job (use jobs -l to get its ID) and run it in the background:
$ bg %{{job_id}}

How To Run a Process in Background in Linux?
Running processes in the background is a common requirement in Linux systems.
Running a process in the background allows you :
      1. to continue using the terminal
      2. execute other commands while the process runs independently.
This can be particularly useful for :
      1.  long-running tasks or
      2.  when you want to execute multiple commands simultaneously.

What is a Background Processes
Let's first understand what a background process is.
In Linux, a background process refers to a process that runs independently
of the terminal.
When you execute a command, it typically runs in the foreground,
meaning it occupies the terminal until it completes.
On the other hand, running a process in the background allows you
to execute other commands while the process continues to run silently.

                How to Use the bg Command in Linux
                ----------------------------------
One way to run a process in the background is by using the bg command.
The bg command is built-in to most Linux distributions and allows you to send
a running process to the background. Following steps are involved to run the
process in background:
   1. Start a process in the foreground by executing a command.
      For example, let's say we want to compress a large file using the gzip com
        $ gzip largefile.txt
   2. While the process is running, press Ctrl+Z to pause the process
      and bring it to the background.
   3. Use the bg command to resume the process in the background:
        $ bg
   4. The process will now continue running in the background,
      and you will see its job ID displayed on the terminal.
      You can execute other commands while it progresses silently.
      However, keep in mind that the process may still produce output,
      which can appear on the terminal.
```

4.19 & symbol : start process in the background immediately

```
& Symbol
Another method to run a process in the background is by appending the
ampersand symbol & to the command you execute.
This symbol tells the shell to start the process
in the background immediately.
```

To just start the command in the background at the beginning:
all you need to do is put an ampersand at the end of any Linux command.
This will allow to use terminal along with running the command:

Example:

```
$ gzip largefile.txt &
```

In this example, the gzip command starts running in the background
as soon as you execute it.
The process ID (PID) will be displayed on the terminal,
allowing you to continue using the shell while the process completes silently.

It's important to note that when using the & symbol,
the process may still produce output, which can interfere with
the prompt on the terminal.

Note:

To avoid this, it's recommended to redirect the output
to a file or to /dev/null, which discards the output.
You can do this by appending > filename or > /dev/null
to the command, respectively.

4.20 disown : Signal process by process's name

Allow sub-processes to live beyond the shell that they are attached to.
- Disown the current job:
```
$ disown
```

- Disown a specific job:
```
$ disown %{{job_number}}
```

- Disown all jobs:
```
$ disown -a
```

If you close current terminal, the jobs that it run will be also closed,
because the jobs are terminal's children.
Terminal itself is the child of graphical system (that presents windows).
If you need to close your terminal, and don't want these commands to stop,
then you need to ‘‘disown'' the job(s).

Disown command will cause the job to be child of graphical system,
in place of child of terminal.

If you only have one job in the background, the following command will work:
```
$ disown
```
If you have multiple, then you'll need to specify the job ID.
```
$ disown %1
```
You'll no longer see the job in your jobs table when you execute the jobs command.

```
Now it's safe to close the terminal and your command will continue running.
   $ jobs -l
You can still keep an eye on your running command by using the ps command.
   $ ps aux | grep sleep
linuxco+    1650  0.0  0.0   8084   524 pts/0    S    12:27   0:00 sleep 10000
And if you want to stop the command from running, you can use the kill
command and specify the process ID.
   $ kill 1650

Examples :
 $ geany | disown

 $ geany & disown  # send to bg and disown
```

4.21 time : Run the command and print the time measurements

```
Users of the bash shell need to use an explicit path in
order to run the  external time command "/bin/time" and not the
shell builtin variant "time" !!!

   $ which -a time
      /usr/bin/time
      /bin/time
   $ time sleep 2
     real 0m2.003s
     user 0m0.002s
     sys 0m0.000s
   $ /bin/time --format="%E real, %U user, %S sys" sleep 2
     0:02.00 real, 0.00 user, 0.00 sys

On system where
time is installed in /usr/bin, the first example would become
/usr/bin/time wc /etc/hosts

"time" runs the program "COMMAND" with any given arguments "ARG"....
When COMMAND finishes, time displays information about resources
used by COMMAND (on the standard error output, by default).
If COMMAND exits with non-zero status, time
displays a warning message and the exit status.

Run the command and print the time measurements to stdout:
     $ time <command> <args>
```

4.22 fg : resume a specific suspended job by its job number.

```
The fg command allows us to resume a specific suspended job by
its job number. The job's name is output when it starts:
```

```
Resume last job:
  $ fg
  Resume job number 1:
  $ fg %1
```

4.23 $? : result of last command execution

```
 $ ls -la
   total 772
   drwxrwxr-x 2 amits amits   4096 Feb 28 12:12 .
   drwxrwxr-x 8 amits amits   4096 Feb 28 11:42 ..
   -rw-rw-r-- 1 amits amits 256176 Feb 28 12:11 try.pdf
   -rw-rw-r-- 1 amits amits 148037 Feb 28 11:41 try.tex
 $ echo $?
   0
https://www.baeldung.com/linux/check-if-command-executed-
successfully#:~:text=We%20can%20use%20a%20binary%20comparison%20operator%20eq,
the%20pwd%20command%20displays%20the%20present%20working%20directory.

  We can use a binary comparison operator eq to check the
  outcome of the previous command, stored in the $?
  variable:

    #!/bin/bash
    pwd
    if [ $? -eq 0 ]; then
       echo "Command Executed Successfully"
    else
       echo "Command Failed"
    fi{verbatim}
```

5 Commands envistigation

5.1 type : Display the type of command the shell will execute.

```
The ‘type’ is a built-in shell command used to identify the kind of
command. For example it checks if it is shell built in or external
command.

 type without any key :
              if NAME is an alias,
          shell reserved word,
          shell function,
          shell builtin,
          disk file
          or not found, respectively
```

```
-t output a single word which is one of
             'alias',
             'keyword',
             'function',
             'builtin',
             'file'

-p Returns the disk file that would be executed
-a Returns all locations containing an executable.

 For example:
 $ type echo
     echo is a shell builtin
 $ type cat
     cat is /usr/bin/cat
 $ type code
     code is /usr/bin/code
 $ type -a time
     time is a shell keyword
     time is /usr/bin/time
     time is /bin/time
```

5.2 which : Locate a program in the user's path.

```
 Locate a program in the user's path.
 - Search the PATH environment variable and display the location
   of any matching executables:
     $ which {{executable}}
 - If there are multiple executables which match, display all:
     $ which -a {{executable}}
  Example :
     $ which -a time
        /usr/bin/time
        /bin/time
```

5.3 whereis : locate the source, and manual page for a command.

```
$ type whereis
   whereis is /usr/bin/whereis
```

5.4 command : locate the source, and manual page for a command.

5.5 history : command-line history

```
history   : Display the history list with line numbers.
history -c: delete all of the entries
history N : Only list the last N commands.
history 10: list last 10 entries
-------------------------------------------------------
!!        : to execute the last command you typed.
!n        : to execute command number n.
!-n       : execute the command n times before
!!==!-1   :equivalent commands
!"string" : execute the last command starting with
            that ‘‘string’’ (last command with given name)
-------------------------------------------------------
CTRL-R    : to perform a ‘‘reverse-i-search’’. For
            example, if you wanted to use the command
            you used the last time you used snort,
            you would type: CTRL-R then type ‘‘snort’’.

An argument of N lists only the last N entries.
```

5.6 man : display manual pages

```
The table below shows the section numbers of the manual
followed by the types of pages they contain.

  1 General commands
  2 System calls
  3 Library functions, covering in particular
       the C standard library
  4 Special files (usually devices, those found in /dev) and drivers
  5 File formats and conventions
  6 Games and screensavers
  7 Miscellaneous
  8 System administration commands and daemons

When you invoke the man command, you can optionally
specify the section of the manual to read :
   $ man 3 printf

To see all the man pages available in Ubuntu use
   $ man -k .

-f : to find page that contains the command :
   $ man -f signal
      signal (7)           - overview of signals
      signal (2)           - ANSI C signal handling
   $ man -f print
      print (1)            - execute programs via entries
                             in the mailcap file
```

```
-k : to find command and its page that contains the word:
   $ man -k pgrep
      pgrep (1)            - look up, signal, or wait for processes
                             based on name a...
   $ zipgrep (1)           - search files in a ZIP archive for lines
                             matching a pat...

man is an interface to the system reference manuals
  $ man -k : the same as apropos
  $ man -f : the same at whatis
  $ -h  - help on intercative commands

find words in man manual:
 to find "name" option in man manual choose :
     / "name"
 to find next search result press :
     press "/" and press ‘‘enter’’
```

5.7 tldr : "Too Long; Didn't Read" - simplified man pages

```
 simplified man pages

 To update :
   $ tldr -update

 To install:
   $ sudo apt install tldr


 Update in Ubuntu (there is a problem to update tldr,
 since tldr -u did not work ):

    $ tldr --version
      0.6.4 # this is the old version
    $ apt remove tldr
    $ pip install tldr
    $ tldr --version
      tldr 3.2.0 (Client Specification 1.5)


    see: https://forums.linuxmint.com/viewtopic.php?t=407665
```

5.8 whatis : breaf command description

```
whatis is a command for obtaining the brief description
of a specific command whose exact name is already known
```

5.9 apropos : command to search the man page files

```
Often a wrapper for the man -k command.
Used to search the "name" sections of all manual pages
for the specified string or strings (called keywords).
The output is a list of all manual pages containing
the search term (case insensitive) in their name or description.
This is often useful if one knows the action that is desired,
but does not remember the exact command or page name.
```

6 Text and text files manupulations

6.1 pandoc : Convert documents between various formats

```
Convert documents between various formats:
- List all supported input formats:
   $ pandoc --list-input-formats
- List all supported output formats:
   $ pandoc --list-output-formats
- Convert file to PDF (the output format is determined by file extension):
   $ pandoc {{input.md}} -o {{output.pdf}}

Examples :
  $ pandoc latex-document-prototype.tex -o try.html
  $ pandoc latex-document-prototype.tex -o try.pdf
  $ pandoc latex-document-prototype.tex -o try.docx
```

6.2 tab : autocomplete

```
Tab   : automatic completion
Tab Tab  : all options for complete
```

6.3 | : pipe - redirect stdout of one cmd to stdin of another cmd.

- ‘‘Pipe’’ redirects standard output of one command to standard input of another command
- t1, t2, tL are ‘‘temporary files’’ in this explanation.
- ‘‘|’’ acts as temporary file.
- Pipe redirects only REGULAR input/output, not error

```
keyboard --> command1 ---> | ---> command2 ---> |--> command3---> display
command1 > t1
command2 <t1 >t2
command3 > tL
```

- Example 1:
  Redirect output of (print of all files *.log)

to (grep -i (insensitive) that search lines with ‘‘error’’)
and sort all this lines:
$ cat *log|grep -i error|sort

- Example 2:
  Redirect (result of recursive insensitive search ,
  starting with current directory) to (search of lines that not
  include word ‘‘ignored’’), the (sort the results with ‘‘unic’’)
  and send the result to file serious_errors.log:
  $ grep -ri error .|grep -v ‘‘ignored’’|sort u> serious_errors.log

- Example 3 (YES command usage):
  $ cat dir1_rm_inputs.txt
  y
  y
  n
  n
  y
  y
  y
  $ cat dir1_rm_inputs.txt | rm ./dir1/*.* -i

  $ rm -i dir2/*
  rm: remove regular empty file 'dir2/6.txt? n
  rm: remove regular empty file 'dir2/7.txt? n
  rm: remove regular file 'dir2/errors2.log? n
  rm: remove regular file 'dir2/errors3.log? n
  rm: remove regular file 'dir2/errors.log? n
  $ yes | rm -i dir2/*
  rm: remove regular empty file ‘dir2/6.txt’? rm: remove regular
  empty file ‘dir2/7.txt’? rm: remove regular file ‘dir2/
  errors2.log’? rm: remove regular file ‘dir2/errors3.log’? rm:
  remove regular file ‘dir2/errors.log’?

- Example: automation to enter ‘‘yes’’ when install:

  $ yes | sudo apt install some_programs

- apt has -flag to allow install without interrupts.

6.4 Redirections of I/O

Process:

```
We can control input and output direction of process:
  0 ---> stdin
  1 ---> stdout
  2 ---> err
• Standard output can be redirected to file using >
  (remove previous file and creates from the start of file).
• Standard output can be appended to file using >>
  (appends to the end of existing file).
-------------------------------
Redirect input to file:
    $ cat operations.txt
      2+3
      4*5
      x=7
      x*100

    $ bc < operations.txt
      5
      20
      700
---------------------------------
Redirect output to file :

Redirect result of ls command to file ls_comm_result.txt:
  $ ls -l 1.txt 7.txt > ls_comm_result.txt
     ls: cannot access '7.txt': No such file or directory

Redirect regular output of ls command and also ERR output
to the same file ls_comm_result.txt as regular output:
  $ ls -l 1.txt 7.txt > ls_comm_result.txt 2>&1
     ls: cannot access '7.txt': No such file or directory

    # or:
  $ ls -l 1.txt 7.txt 1>ls_comm_result.txt 2>&1
     ls: cannot access '7.txt': No such file or directory

Redirect ERR output to ls_comm_result.txt and
regular output to the same file as error output:
  $ ls -l 1.txt 7.txt 2>ls_comm_result.txt 1>&2
    # or:
  $ ls -l 1.txt 7.txt 2>ls_comm_result.txt  >&2

In This case regular output is redirected to errors output:
  $ echo "error" >&2

Send grep command output to file :
  $ grep error errors.log > grep_result.txt

Append grep command output to the end of the file:
  $ grep error errors.log >> grep_result.txt
```

```
Append cat command output to file:
  $ cat 1.txt
    this is very interesting.
    what ?
    this.
  $ cat 2.txt
    also this is very nice and interesting
    exectly
  $ cat 2.txt >> 1.txt
  $ cat 1.txt
    this is very interesting.
    what ?
    this.
    also this is very nice and interesting
    exectly

Print cat command output to file:
  $ cat 1.txt > 2.txt
  $ cat 1.txt
    this is very interesting.
    what ?
    this.
    also this is very nice and interesting
    exectly
  $ cat 2.txt
    this is very interesting.
    what ?
    this.
    also this is very nice and interesting
    exectly

Print ls command output to file:
  $ ls
    1.txt  bc    dir2        fruits.txt       numbers.txt  sizes.txt
    2.txt  dir1  errors.log  grep_result.txt  out1
  $ ls > out
  $ cat out
    1.txt
    2.txt
    bc
    dir1
    dir2
    errors.log
    fruits.txt
    grep_result.txt
    numbers.txt
    out
    out1
    sizes.txt
```

```
Use echo command output to prepare file in place of nano:
  $ echo 'this
  > is a file
  > that try to use echo command
  > in place of nano command'> my_file.txt
  $ cat my_file.txt
    this
    is a file
    that try to use echo command
    in place of nano command


Append line to file with echo command output:
  $ echo "ups fogot this line" >> my_file.txt
  $ cat my_file.txt
    this
    is a file
    that try to use echo command
    in place of cat command
    ups fogot this line

Redirect errors output to additional file errors.txt:
  $ ls 5.txt z.txt
 ls: cannot access ‘z.txt’ No such file or directory # err output
 5.txt                                               # regular output
  $ ls 5.txt z.txt >out.txt 2>error.txt # same as 1>out.txt
  $ cat out.txt
 5.txt
  $ cat error.txt
 ls: cannot access ‘z.txt’: No such file or directory
```

6.5 echo : display line of text

```
Display line of text/string that are passed as an argument.
Mostly used in shell scripts and batch files
to output status text to the screen or a file
Helps to see a coomand after expansion

  $ echo hello
  hello

  $ echo hello   world   # 2 different arguments : Hello and World
  hello world

  $ echo "hello   world" # 1 argument "Hello world"
  hello   world
```

```
$ echo ~
/home/amit

$ echo "~"
~


echo ls *.txt # to see the command after expansion

? replaces exactly one character:
$ echo ?.txt # ? replace exactly one character
   1.txt 3.txt 5.txt 7.txt

The list of files with extention of 4 characters :
$ echo *.????
   2.html 5.html

$ echo ??.txt
   11.txt 12.txt

$ echo ???.txt
   111.txt

$ echo 1?.txt
   11.txt 12.txt

$ echo 1??.txt
   111.txt

$ echo [0-9][0-9][0-9].txt
   111.txt

^ is for "not"  :
$ echo [^0-9].txt  # not(numbers 0 to 9)
   a.txt b.txt g.txt i.txt

$ echo [a-z].txt
   a.txt b.txt g.txt i.txt

$ echo [1a2x3456bc].txt
   1.txt 3.txt 5.txt a.txt b.txt

$ echo {1..9}
   1 2 3 4 5 6 7 8 9

$ echo {1..100..5}  # from 1 to 100 with step 5
   1 6 11 16 21 26 31 36 41 46 51 56 61 66 71 76 81 86 91 96

$ echo a{,x,y,z}4  # {,x,y,x} means : nothing,x,y,z
   a4 ax4 ay4 az4
```

```
$ echo a{‘“’,x,y,z}4
   a4 ax4 ay4 az4

$ echo {,.}*.txt # with or without "."
   *.txt .hidden.txt .secret.txt

$ echo *.txt .*.txt
   *.txt .hidden.txt .secret.txt

$ echo */*.txt
     docs/111.txt docs/11.txt docs/12.txt docs/1.txt
     docs/3.txt docs/5.txt docs/7.txt docs/a.txt
     docs/b.txt docs/g.txt docs/i.txt
$ echo "a{b,c,d}e"
     a{b,c,d}e
$ echo a{b,c,d}e
     abe ace ade

  to enable ** recursive glob :
     $ shopt -s globstar  // set recursive glob on
     $ shopt globstar
        globstar        on
     $ echo **/*.txt   # recursive glob:
                       # find in the current dir (!)
                       # and all sub dirs under it
       1.txt 2.txt 3.txt 4.txt 5.txt   # files from current dir also
       temp1/a.txt temp1/b.txt temp1/c.txt temp1/d.txt
       temp2/10.txt temp2/1.txt temp2/2.txt temp2/3.txt
       temp2/4.txt temp2/5.txt temp2/6.txt temp2/7.txt
       temp2/8.txt temp2/9.txt
     $ shopt -u globstar
     $ shopt globstar
        globstar        off
     $ echo **/*.*    # now it will search only in sub dirs
       temp1/a.txt temp1/b.txt temp1/c.txt temp1/d.txt
       temp2/10.txt temp2/1.txt temp2/2.txt temp2/3.txt
       temp2/4.txt temp2/5.txt temp2/6.txt temp2/7.txt
       temp2/8.txt temp2/9.txt

    search in subdirs only :
     $ echo */*.txt
       temp1/a.txt temp1/b.txt temp1/c.txt temp1/d.txt
       temp2/10.txt temp2/1.txt temp2/2.txt temp2/3.txt
       temp2/4.txt temp2/5.txt temp2/6.txt temp2/7.txt
       temp2/8.txt temp2/9.txt

    split text in several strings:
      $ echo ’this
      > is afile
```

```
  > that is vary long'
  this
  is afile
  that is vary long

Trick to print ouput to stderr from process:
---------------
ex07.sh:
  echo "normal output"
  echo "error output" >&2
---------------
  $ bash ex07*  >/dev/null
  $ bash ex07* 2>/dev/null
```

6.6 printf : display line of text

```
Format and print text.

Similar to echo command, but does not output
new line, if not defined particularly.
  $ printf ‘‘\rA’’   - return to start of the same line and
                       print A.
  $ printf ‘‘A\n’’ - print A and add enter.


  $ printf "Hello World\n"
    Hello World
  $ printf "Hello World"
    Hello World $

  This will output A, wait 3 sec., clear line and output B:
  $ printf "A"; sleep 3; printf "\rB\n"
```

6.7 grep : "Global Regular Expression search and Print"

```
https://en.wikipedia.org/wiki/Grep

print lines that match patterns
grep allows to enter regular expression for the search.

returns status  $? = 0 if string is found,
returns status  $? = 1 if string is not found
returns status  $? = 2 if error occured


grep is a command-line utility for searching
```

```
 plain-text data sets for lines that match a regular expression
 i.e. to search files for certain patterns
 Filter only lines with desired contents

 - Search for a pattern within a file:
      grep "{{search_pattern}}" {{path/to/file}}

      $ grep "is" *.txt
1.txt:this is an adittional try
try1.txt:This is very convinient editor,
try1.txt:Its name is tilde.

      $ grep is *.txt
1.txt:this is an adittional try
try1.txt:This is very convinient editor,
try1.txt:Its name is tilde.
 ------------------------------------------------------
 Options:
     -L : list of FILES that don't include the word
     -l : list of FILES that include the result
     -i : Insesitive to upper case
     -w : Select  only  lines with whole word.
     -o : present ONLY WORD we search
     -r : recursive search
     -v : (inVerterd) list of rows that does not include the word
     -c : Count number of lines with word
     -q : quiet, exit immediately with 0 status if any match is found.
 ------------------------------------------------------

 - Search for an exact string (disables regular expressions):
      grep --fixed-strings "{{exact_string}}" {{path/to/file}}

 - Search for a pattern in all files recursively in a directory, showing
   line numbers of matches, ignoring binary files:
      grep --recursive --line-number --binary-files={{without-match}}
      "{{search_pattern}}" {{path/to/directory}}

     Global Regular Expression "error" search:
         $ grep error errors.log
            error: aeasasadf
            dfdfdfdf error sfsadgasdg
            ldksldjks no errors

     -i : Insesitive to upper case
        $ grep error errors.log -i
            error: aeasasadf
            dfdfdfdf error sfsadgasdg
            ERROR dssdsdsdsd
            ldksldjks no errors
```

```
-w : exact word
   $ grep errors errors.log -i -w
      ldksldjks no errors

-o : present only word we search, without text
   $ grep error errors.log -o -i
      error
      error
      ERROR
      error

-r : recursive search in all subdirectories :
   $ grep error -r
       dir2/errors2.log:error: aeasasadf
       dir2/errors2.log:dfdfdfdf error sfsadgasdg
       errors.log:dfdfdfdf error sfsadgasdg
       errors.log:ldksldjks no errors

-l : list of files that include the result :
   $ grep error -r -l
        dir2/errors2.log
        dir2/errors.log
        dir2/errors3.log
        errors.log

-L : list of files that don't include the word :
   $ grep error -r -L
       .secret.txt
       .hidden.txt
       dir1/5.txt

-v : (inVerterd) list of rows that does not include the word "error":
   $ grep error errors.log -v
       aaaaa
       bbbb

       dfdf

       ERROR dssdsdsdsd
       sddsd
       dsdsds

-v -i : (inVerterd) list of rows that does not include the word "error"
         insensitive :
   $ grep error errors.log -v -i
     aaaaa
     bbbb

     dfdf
```

```
      sddsd
      dsdsds

List all files, that contain at least one row without word error:
   $ grep error -v -r -l
     dir1/b.txt
     dir1/dudu.txt
     dir2/errors2.log


List all files that does not contain any word ‘‘error’’:
   $ grep error -r -L
     .secret.txt
     .hidden.txt
     dir1/5.txt
     dir1/b.txt

Count number of lines with word "error":
    $ grep error errors.log -c
        3
    $ grep error errors.log -i -c
        4
```

6.8 cat : "ConcATenite" : concatenite and print.

```
The cat utility serves a dual purpose: concatenating and printing.

With a single argument, it is often used to print a file to the
user's terminal emulator.

With more than one argument, it concatenates several files

- Print the contents of a file to stdout:
      cat {{path/to/file}}

- Concatenate several files into an output file:
      cat {{path/to/file1 path/to/file2 ...}} > {{path/to/output_file}}

- Append several files to an output file:
      cat {{path/to/file1 path/to/file2 ...}} >> {{path/to/output_file}}
```

6.9 more : present text file in parts.

```
Same as cat , but presents the contents
in several parts with more option

$ cat /usr/include/stdio.h
$ more /usr/include/stdio.h
```

6.10 less : present text file in parts and allows search

```
Same as cat , presents the contents
in option to move with arrows
allows search

$ less /usr/include/stdio.h

allows to find with
/ prin....
/+''enter''
```

6.11 head : Display n rows at the head of file

```
Present n rows at the head of fil


Present 3 rows at the head:
      $ head /usr/include/stdio.h -n 3
```

6.12 tail : Display n rows at the end of file

```
  Present n rows at the end of file

 - Show last 'count' lines in file:
         tail --lines {{count}} {{path/to/file}}

      $ tail --lines 3 /usr/include/stdio.h
```

6.13 sort : Display lines in sorted order

```
    prints the lines of its input or concatenation
    of all files listed in its argument list in sorted order
 regular sort:
    $ sort fruits.txt
      apple
      apple
      banana
      banana
```

```
pineapple
strawberry
tomato
watermelon
```

-u: unic sort remove the same words:

```
$ sort fruits.txt -u
  apple
  banana
  pineapple
  strawberry
  tomato
  watermelon
```

-r : reverse sort :

```
$ sort fruits.txt -u -r
  watermelon
  tomato
  strawberry
  pineapple
  banana
  apple
```

-h: sort file sizes in human order :

```
$ sort files sizes.txt -h
  10
  30K
  200K
  500K
  5M
  1G
  2G
```

- sort numbers:

```
$ sort numbers.txt
  1
  1
  1212
  19898990
  2
  33
  45
```

6.14 wc : "Word Count" - standard input statistics

```
Count lines, words, and bytes.
The program reads either standard input or a list of computer
files and generates one or more of the following statistics:
     1. newline count,
     2. word count,
```

```
    3. byte count.
    4. charachters count
If a list of files is provided, both individual file and total
statistics follow.
  - Count all lines in a file:
      wc --lines {{path/to/file}}
 - Count all words in a file:
      wc --words {{path/to/file}}
 - Count all bytes in a file:
      wc --bytes {{path/to/file}}
 - Count all characters in a file (taking multi-byte characters into account):
      wc --chars {{path/to/file}}
 - Count all lines, words and bytes from stdin:
    {{find .}} | wc
 - Count the length of the longest line in number of characters:
    wc --max-line-length {{path/to/file}}

  Example:
  $ wc ex1.c
    8  20 138 ex1.c

    #8   lines
    #20  words,
    #138 characters
```

7 Compressing and archiving

7.1 gzip/gunzip : shrink huge files and save space

```
  GNU zip compression utility.
  Creates .gz files (original file is deleted)
  Ordinary performance (similar to Zip).
  Very popular in Linux, almost every Linux has gzip.
  Relative fast and well decompression
  Has no password

  NOTE: Does not work with directories !

  $ gzip <file>    # to compress
  $ gunzip <file>  # to decompress


  Example :
  Suppose in current directory there are 3 files:
  1.txt , 2.txt , 3.txt

  $ gzip 1.txt 2.txt 3.txt
```

```
Normally this will replace the original file by compressed file :
  1.txt.gz in place of 1.txt
  2.txt.gz in place of 2.txt
  3.txt.gz in place of 3.txt

Note: if the file has a hard link, the compression will not be
done by gzip:
 $ gzip 1.txt 2.txt 3.txt
   zip: 1.txt has 1 other link  -- unchanged
 $ file 2.txt
 $ file 2.txt.gz
    2.txt.gz: gzip compressed data, was "2.txt",
    last modified: Thu Feb 15 11:40:21 2024, from Unix, truncated
 $ gunzip 2.txt # produce 2.txt
 $ gunzip 3.txt # produce 3.txt

 -k : keep input files during compression (don't replace)
 $ gzip -k 2.txt # now there are 2 files : 2.txt and 2.txt.gz
```

7.2 bzip2/bunzip2 : better compression than gzip

```
Similar usage as gzip.
Compression level is higher than in in gzip
Decompression sometimes problematic (may be).
Does not work with directories


$ bzip2 <file>    # to compress
$ bunzip2 <file>  # to decompress

Example:
$ bzip2 3.txt        # replace 3.txt with 3.txt.bz2

$ bunzip2 3.txt.bz2  # replace 3.txt.bz2 with 3.txt

$ bzip2 -k  3.txt    # generates 3.txt.bz2
                     # and dont change 3.txt
```

7.3 tar : Tape ARchive - back up set of files

```
Archiving utility (without compressing !!!).
Often combined with a compression method, such as gzip or bzip2

So for extraction of information within *.tar.gz we perform
following steps:
*.tar.gz ---> gzip decompress ---> *.tar ----> tar extract
```

```
Note : jar in Java has the same flags as tar in linux

Useful to backup or release a set of files into a single file
In the past used to move files from Linux to tape.
tar stores also structure of directories, inods , etc. and send
it to tape archive.

Note : tar stores the file system along with inode data
so it stores the original permissions of the files !!!

Creating an archive:
--------------------
  $ tar -c  -f <archive_file_name>  <files or directories>
Extracting archive:
-------------------
        -x
Listing archive:
----------------
        -t
    -c: create. The opposite action is extract (-x)
    -x: extract
    -t: test (listing) check that archive is OK
    In the past tasr used >> to sent to tape
    now we use -f to tell tar to use file:
    -f: file. Archive created in/extracted from file
             (tape used otherwise if no -f).
     Note, that if we don't add -f flag
     tar will read from stdin and write the result to stdout
    --------------------------------
    Also:
    -v: verbose. Useful to follow archiving progress.
    --------------------------------
    Creating an archive:
       $ tar -cvf <archive> <files or directories>
    Extracting an archive:
       $ tar -xvf <archive> <files or directories>
    Viewing the contents of an archive or integrity check:
       $ tar -tvf <archive>

    Combination of compressing and archiving on the fly.
    Useful to avoid creating huge intermediate files
    Much simpler than with tar and bzip2\gzip!

    -j --bz2  option: [un]compresses on the fly with bzip2
    -z --gzip option: [un]compresses on the fly with gzip
    -J --xz   option: [un]compresses on the fly with xz

    create tar.gz file (- czf):
     $ tar -czfv files.tar.gz 1.txt 2.txt dir1 dir2 # create z file
```

```
unzip  tar.gz file:
 $ tar -xzf files.tar.gz # eXtract z file
list tar.gz file :
 $ tar -tzf files.tar.gz # list z file

--------------------------------
tar -czfv files.tar.gz 1.txt
tar -cfv 1.txt | gzip > files.tar.gz  # the same operation in
                                      # another form
--------------------------------
Files or directories are given with paths relative to
the archive's directory.

Use -C DIR or --directory=DIR, to specify the
extraction directory location.

Suppose we have following directories tree:
Desktop
|
|--- linux_2312
       |
       |---- playground
                |
                |--- 1.txt
                |--- 2.txt

And we want like to compress playground directory,
including name playground:
  amits:Desktop$ tar -czf play.tar.gz -C linux_2312 -v playground/
     playground/
     playground/2.txt
     playground/1.txt
  amits:Desktop$ ls
     Documents     LinuxAdminLearn.sh     play.tar.gz
  amits:Desktop$ tar -tzf play.tar.gz
    playground/
    playground/2.txt
    playground/1.txt

  amits:Desktop$ mkdir try_dir
  amits:Desktop$ mv play.tar.gz ./try_dir
  amits:Desktop$ cd try_dir
  amits:Desktop$ tar -xzf play.tar.gz
  amits:Desktop$ ls
    playground  play.tar.gz
  amits:try_dir$ cd playground/
  amits:playground$ ls
    1.txt  2.txt
--------------------------------
Example :
```

```
create archive "text files.tar":
  $ tar -c -f text_files.tar 1.txt 2.txt ./dir1 ./dir2

to check what is inside the archive text_files.tar :
  $ tar -t -f text_files.tar
  or
  $ tar -t <text_files.tar

 Now if we want to extract only 1.txt from tar frile
 text_files.tar :
  $ tar -x -f text_files.tar 1.txt
  ------------------------------------------------
    tar   -x    -f text_files.tar        1.txt
  command flag  flag with its argument   argument
  ------------------------------------------------
  Now we run gzip on file text_files.tar:
  $ gzip text_files.tar # in place of text_files.tar
                        # it place text_files.tar.gz
  $ gunzip text_files.tar
  ------------------------------------------------
  $ mkdir try_dir
  $ mv text_files.tar try_dir
  $ gzip text_files.tar  # run gzip on tar
  $ ls
    text_files.tar.gz    # this file stores inode and compressed
  $ gunzip text_files.tar.gz
  $ ls
    text_files.tar
  $ tar -xf text_files.tar
  $ ls
    1.txt  2.txt  dir1  dir2  text_files.tar
  ----------------------------------------
  Example with xz :
  $ tar --xz -cf data.xz 1.txt
  $ tar --xz -xf data.xz
  ----------------------------------------
```

7.4 xz/ : relative new, better compression

```
New application with huigh compression level.
```

7.5 7zip/ : better compression than bzip2

```
7zip supports strong AES256 encryption.
No need to encrypt in a separate pass.
At last a solution available for Unix and Windows!
The tool supports most other compression formats:
zip, cab, arj, gzip, bzip2, tar, cpio, rpm , deb.
```

Option 1 to install :

https://www.linuxcapable.com/how-to-install-7-zip-on-ubuntu-linux/
Download 7-Zip using the wget command below:

```
$ wget https://www.7-zip.org/a/7z2301-linux-x64.tar.xz
$ mkdir 7zip
$ tar -xJf 7z2301-linux-x64.tar.xz -C 7zip
$ cd 7zip
$ ./7zz
```

Option 2 to install (preferred):
https://www.geeksforgeeks.org/how-to-install-7zip-in-ubuntu/

```
$ sudo apt-get update
$ sudo apt-get install p7zip-full
```

Much better compression ratio than bzip2 (up to 10 to 20%).
Actually its compression is the best of all options.
Free of charge, but was developed by company

```
$ man 7z      # 7zip manual
```

Add files to 7z :

```
$ 7z a data.7z 1.txt # add files to archive

7-Zip [64] 16.02 : Copyright (c) 1999-2016 Igor Pavlov : 2016-05-21
p7zip Version 16.02 (locale=en_IL,Utf16=on,HugeFiles=on,64 bits,12
CPUs 12th Gen Intel(R) Core(TM) i7-1255U (906A4),ASM,AES-NI)
Scanning the drive:
1 file, 474 bytes (1 KiB)
Creating archive: data.7z
Items to compress: 1
Files read from disk: 1
Archive size: 409 bytes (1 KiB)
Everything is Ok
```

List data in archive:

```
$ 7z x py2403_day03.zip  # to see contents of zip file
$ 7z l data.7z

            Date      Time    Attr         Size   Compressed  Name
------------------- ----- ------------ ------------  ------------------
2024-02-26 15:18:03 ....A          474          295  1.txt
------------------- ----- ------------ ------------  ------------------
2024-02-26 15:18:03                474          295  1 files
```

Decompress files:

```
$ 7z x data.7z

Unzip zip file:
$ 7z x py2403_day03.zip
```

8 File comparison and file integrity

8.1 file..integrity..check

```
We can look at file data as single lage number.
We can define function that translates this large number
to another number called "checksum"

Can another file generate the same checksum ?
crc32 number present 4.294.967.296 different numbers
there are more than 4e9 different files

According to shovah ha yonim there are two different files with the
same crc number.
```

8.2 crc32 : calculate crc32

```
install:
$ sudo apt install libarchive-zip-perl

crc32 - compute CRC-32 checksums for the given files
crc32 filename [ filename ... ]

For example:
$ cat 1.txt
  this is David's file
$ crc32 1.txt
  7b6accd4
```

9 Find files and check directory size

9.1 du : "Disk Usage" - file and dir usage space

```
https://phoenixnap.com/kb/show-linux-directory-size

Display the current directory size by running du without any arguments:
The system lists the current directory content and subdirectories with a
```

```
number representing the object size in kilobytes.
The current directory size is the last line.
 $ du
Print the Current Directory Size in a Human-Friendly Format
Add the -h option to the du command to make the output easily readable:
 $ du -h

Find the Size of a Specific Directory
 $ du -h ./example1

Show Only the Total Disk Usage of a Directory
To print an extra total line at the end of the output and summarize the disk
space usage for all files and directories, use the -c argument:
 $ du -c ./example1

Scan up to a Chosen Subdirectory Level
Limit the scan to a certain subdirectory level with the --max-depth option.
For example, to scan only the top-level directory (example1),
use --max-depth=0:
 $ du -h --max-depth=0 ./example1
To list the top directory (example1) and the first layer of subdirectories
(example2 and example3), use --max-depth=1:
 du -h --max-depth=1 ./example1
--------------------------------------------------------------------------------

Disk usage: estimate and summarize file and directory space usage
 - List the sizes of a directory and any subdirectories, in human-
   readable form (i.e. auto-selecting the appropriate unit for each
   size):
      $ du -h {{path/to/directory}}

 - List the human-readable sizes of a directory and of all the files and
   directories within it:
      $ du -ah {{path/to/directory}}


  Summary of disk usage for two directories:
     $ du -h
       8.0K ./dir2/2024
       12K ./dir2
       8.0K ./dir1/2024
       20K ./dir1
       76K .
     $ du -h | sort -h
       8.0K ./dir1/2024
       8.0K ./dir2/2024
       12K ./dir2
       20K ./dir1
       76K .
     $ du -h -s ./dir1 ./dir2  # -s is for summary of size
```

```
    20K ./dir1
    12K ./dir2
```

Apparent size is the actual amount of data that can be read from the file. Block-oriented devices can only store in terms of blocks, not bytes. As a result, the disk usage is always rounded up to the next highest block

```
  $ du -h -s --apparent-size ./dir1 ./dir2
```

9.2 locate : command

```
Uses database fo find files.
Note: the database is recomputed periodically (usually weekly or daily).
Manual : the database is located in /var/lib/mlocate/mlocate.db
If we add file to disk or delete file from disk we need to update the data
 base. To update database:
    $ sudo updatedb
- locate a file by its exact filename
   $ locate stdio.h | less
- locate all text files in current directory:
   $ locate "$PWD/*.txt"
- locate file *.txt starting with root directory:
   $ locate "/*1.txt" | head
```

9.3 find : command

Find filenames quickly.

Find files or directories under the given directory tree, recursively

```
Suppose we need to find file stdio.h
$ cd /
$ find . -name "stdio.h" 2>/dev/null

Suppose we need to find all *.txt files starting from ~
$ cd ~
$ find . -name "*.txt"
$ cd ~/Desktop
```

There are many flags in find command ,

```
  the flags are divided in categories :
Actions (-print by default)
Operators (-o logical OR, -a logical AND)
Tests (+n(>n) -n(<n)  n (=n)  ) check size, time , etc.
          For example:
             -mtime +5 : if file modified in 5 last day

   List all files that modified in 14 last days :
   $ find -mtime -14


   -mtime n
        File's data was last modified less than, more  than  or  exactly
        n*24  hours  ago.

   List all files that size of them is larger that 2G :
   $ find -size +2G 2>/dev/null

   List all files from ~ that are at least 100Mb :
   $ find ~ -size +100M -name "*cl*" 2>/dev/null
```

9.4 tree : show curr. dir as tree.

```
Show current directory as tree ,
with sizes in human style and permissions :


Combine tree with du :
----------------------
$ tree --du -h -p

-L     Level
-s     Print the size of each file in bytes along with the name.
-h     Print the size of each file but in a more  human  readable  way,
       e.g.  appending  a size letter for kilobytes (K), megabytes (M),
       gigabytes (G), terabytes (T), petabytes (P) and exabytes (E).

--du   For each directory report its size as the accumulation of  sizes
       of  all  its  files and sub-directories


--inodes
       Prints the inode number of the file or directory

-p     Print  the  file  type  and permissions for each file (as per ls
       -l).


-u     Print the username, or UID # if no username is available, of the
```

```
        file.
 -g     Print the group name, or GID # if no group name is available, of
        the file.

- Print files and directories up to 'num' levels of
  depth (where 1 means the current directory):
     $ tree -L {{num}}

- Print directories only:
     $ tree -d

- Print hidden files too with colorization on:
     $ tree -a -C
```

10 Dirs and files manupulation

10.1 directory.permissions.in.Linux

see https://unix.stackexchange.com/questions/21251/execute-vs-read-bit-how-do-directory-permissions-in-linux-work

When applying permissions to directories on Linux, the permission bits have different meanings than on regular files.

The read bit (r) allows the affected user to list the files within the directory

The write bit (w) allows the affected user to create, rename, or delete files within the directory, and modify the directory's attributes

The execute bit (x) allows the affected user to enter the directory, and access files and directories inside

The sticky bit (T, or t if the execute bit is set for others) states that files and directories within that directory may only be deleted or renamed by their owner (or root)

10.2 basename : store the name of dir based on full path

```
Remove leading directory portions from a path.
 - Show only the file name from a path:
    $ basename {{path/to/file}}

 - Show only the rightmost directory name from a path:
```

```
$ basename {{path/to/directory/}}
```

10.3 nautilus : file explorer in Ubuntu

```
nautilus : native file explorer in Ubuntu
```

10.4 chmod : CHange file MODe (permissions and spec. modes)

- At the begin of -rwxrwxrwx is the type of file
(soft link (l), dir(d), regular file(-))

  r - read rights (ex: present the contents with cat)
  w - write rights (ex: write with nano)
  x - execute rights (ex: run executable)

  u - user (file's owner)
  g - group
  o - others

  Process knows who is its owner and what are groups that run it.
  When process wants TO USE FILE it can check the rights to use it.
  inode of file contains info of file's owner and what is the file's
  group. Sometimes operational system prevents from use the files, and
  sometimes process checks the match between process's owner and group
  again file's inode information.
  If there is no match - the process stops its execution.
  In case of files: all the above info relates only to the CONTENTS
  of the file, not INODE and not FILE's NAME.

  In case of file of type "directory", the content of file contains the
  list of files in the folder that can be read with "ls" command.
  Read the file of type "directory" is to run ls command.
  File of type "directory" contains the list of files and directories
  within the directory.

FILE PROTECTION must be done at two (!) levels :
1. Permission of the file itself.
2. Permissions of the folder that contains the file (to protect the
   info about the file).

Otherwise the PROTECTED file can be DELETED from the directory's list
of files (by delete the directory) !!!
Because "rm" command only deletes the registration of the file within
the directory.

Delete of the write protected file related to directory permissions.
If directory of the file has write permissions and at the same time the
file has no write permissions - it will be deleted !!!:

```
$ ls -l 1.txt
-rw-rw-r-- 1 amit amit 146 Sep 22 12:54 1.txt
$ chmod u-w 1.txt
$ ls -l 1.txt
-r--rw-r-- 1 amit amit 146 Sep 22 12:54 1.txt
$ rm 1.txt
  rm: remove write-protected regular file '1.txt'? Y
$ ls -l 1.txt
ls: cannot access '1.txt': No such file or directory
```

Only write protection to file's directory really protects the files in this directory from being deleted. But it does not protect the files from being edited !!! So both protection is needed : the file protect and the directory protect.
So, protection of the file from being deleted must be done in two levels simulataneously:

1. Protection of file's folder (write option - off).
2. Protection of the file by setting the file's own protection option (write option off).

CMOD command:

```
$ chmod [symbolic permissions/octal premissions] filename
```

Symbolic permissions :

Who:	Action:	Permission type:
u - user	+ (turn on)	r(read)
g - group	- (turn off)	w(write)
o - others	= (set)	x(execute)
a - all		

Take combination of u,g,o,a ; one of +/-/= ; combination of r,w,x

- u+r : turn on "r" of user
- additional permissions separated by ",":
  u+r,g-wx : set on r to user and set off wx to group
- ug+x : set on x for u and g
- unset all r : ugo-r ; a-r
- u= #u equals nothing
- unset rxw for u : u-rxw
- use of = : u=r

Octal permissions:

```
$ chmod 777 1.txt
   U    G    O
  rwx  rwx  rwx
   7    7    7

$ chmod 746 1.txt
   U    G    O
  rwx  rw-  r--
   7    6    4
```

```
-------------------------------------------------------------------------
                -R : Apply chmod recursively.
Note : uppercase X will be set only for folders or executable files.
       always if you want to permit execution to folders and
       files - use X not x !!!
-R : change files and directories recursively
     $ chmod -R o-rwx ./  # Unset rwx permissions for o(thers) recursevely
     $ chmod -R o+rwX ./  # Set rwx permissons recuresevely
                          # Uppercase X : x will be set only for folders or
                          # executable files !
     $ tree -p            # present permissions also


To make directory dir1 and everything in it available to everyone:
$ chmod -R a+rX dir1
$ ls -liR dir1

-------------------------------------------------------------------------

See https://linuxhandbook.com/suid-sgid-sticky-bit/
In addition of 9 options {u(r,w,x), g(r,w,x), o(r,w,x)} there are 3
additional options
     • Set uid (SUID)
     • Set gid (SGID)
     • Sticky bit
-------------------------------------------------------------------------
SUID : -rwsr-xr-x # s here is SUID permission
When the SUID bit is set on an executable file, this means
that the file will be executed with the same permissions as the
<owner of this file> (for example - root).
Practical example: If you look at the binary executable file
of the passwd command, it has the SUID bit set:
   $ ls -lai /usr/bin/passwd
     2885961 -rwsr-xr-x 1 root root 59976 Feb  6 14:54 /usr/bin/passwd
This means that any user running the passwd command will
be running it with the same permission as root.
The passwd command needs to edit files like /etc/passwd,
/etc/shadow to change the password. These files are owned
by root and can only be modified by root.
But thanks to the setuid flag (SUID bit), a regular user will also be
able to modify these files (that are owned by root) and change his/her passwo
This is the reason why you can use the passwd command to
change your own password despite of the fact that the files
this command modifies are owned by root.
$ stat /usr/bin/passwd
     File: /usr/bin/passwd
     Size: 59976      Blocks: 120        IO Block: 4096   regular file
     Device: 10303h/66307d Inode: 2885961     Links: 1
     Access: (4755/-rwsr-xr-x)  Uid: (    0/    root)   Gid: (    0/    root)
How to set SUID bit?
```

I find the symbolic way easier while setting SUID bit.
You can use the chmod command in this way:

```
$ chmod u+s file_name
```

Difference between small s and capital S as SUID bit
The S as SUID flag means there is an error that you should look into.
You want the file to be executed with the same permission as the owner
but there is no executable permission on the file.
Which means that not even the owner is allowed to execute
the file and if file cannot be executed, you won't get
the permission as the owner.

SGID # example is "s" within : drwx rws r-x
is similar to SUID. With the SGID bit is set,
any user executing the file will have same permissions
as the group owner of the file.
It's benefit is in handling the directory.
When SGID permission is applied to a directory, all sub directories
and files created inside this directory will get the same group
ownership as main directory (not the group ownership of the user that
created the files and directories).
How to set SGID?
You can set the SGID bit in symbolic mode like this:

```
$ chmod g+s directory_name
```

The Sticky Bit # Its example is "t" within drwx rwx rwt
Steaky bit:

- for the files: in the past used for file mark
- for the directory: the sticky bit sets shared directory:
  every user can create in this directory file, but the other user
  cannot delete this files that are not of the user. The user
  can update the files of other users but cannot delete them (because
  the deletion is related to directory permissions).

Sticky Bit works mostly on the directory.

With sticky bit set on a directory, all the files
in the directory can only be DELTED or RENAMED by the
file owners ONLY or the root.

This is typically used in the /tmp directory that works
as the trash can of temporary files. This means that a user (except root)
cannot delete the temporary files created by other users in the /tmp director

```
$ ls -ldi /tmp
   5242881 drwxrwxrwt 25 root root 20480 Feb 20 12:45 /tmp
$ stat /tmp
   File: /tmp
```

```
     Size: 20480      Blocks: 48         IO Block: 4096   directory
     Device: 10303h/66307d Inode: 5242881     Links: 24
     Access: (1777/drwxrwxrwt)  Uid: (    0/    root)   Gid: (    0/    root)
How to set the sticky bit?
As always, you can use both symbolic and numeric mode to set the sticky
bit in Linux.
   $ chmod +t my_dir
```

10.5 ln : Creates links to files and directories

```
Creates links to files and directories.
Two types of links in Linux:

• Symbolic links
• Hard links

- Create a symbolic link to a file or directory:
     ln -s {{/path/to/file_or_directory}} {{path/to/symlink}}

- Create a hard link to a file:
     ln {{/path/to/file}} {{path/to/hardlink}}

- Overwrite an existing symbolic link to point to a different file:
     ln -sf {{/path/to/new_file}} {{path/to/symlink}}


To create symbolic link (-s option):
   $ ln -s 1.txt 1_soft.txt

To create hard link:
   $ ln 1.txt 1_hard.txt

Linux file system EXT and hard link
How files are stored in disk:
Linux file system name is EXT (Extended).
All files are stored with 3 linked parts:
     file name-->inode(information node)-->data of the file
-------------------------------------------------------------
• Hard link functions as original file ,
  Hard link is simply new <file name> that linked
  to the inode of the original file.
  Hard link is only additional name to the same file (inode + data) !!!
  Hard link is difficult to recognize. It can be recognized only
  by inode number. If inode number of the file is similar to inode
  number of another file - it is hard link.
  Sector on disk is analogues to address.
  So hard link must be located on the same partition,
  Hard link can't link to another partition or the network address of
```

folders!!! For example, we cannot create hard link on USB disk with
link to file on the hard disk.
User cannot create hard link for folders, because recursive command on
folders tree with hard link would cause to infinite loop.
But he can create soft link to
folders.
Operational system Linux creates hard link to directories:
"." is a hard link to current directory.
".." is the hard link with same inode in subdirs as "." in top dir.
Every folder contain at list one hard link named ‘‘.’’.
Every subfolder contains hard link ‘‘..’’.

Hard link counter of folder less 1 shows how many subfolders has the folder.

```
$ mkdir dir31
$ mkdir dir31/dir32
$ mkdir dir31/dir32/dir33

$ ls -li

1502894 -rw-rw-r-- 1 amit amit   86 Sep 22 05:42 2_txt
1502893 drwxrwxr-x 3 amit amit 4096 Sep 22 07:53 dir31  # 2 subfolders
```

rm of hard link does not delete the original file !
rm of orogonal file does not delete the hard link and file contents !!!
In the inode there is inode counter. If we delete the hard link
(additional name of the same file), inode counter is diminished by 1.
IF inode counter reaches zero - the operational system knows that the
space in the disk in empty and it can be used for other purposes.

```
$ ls -li  # here inode counter is 2
 7753157 -rw-rw-r-- 2 amits amits       45 Feb 19 19:26  1.txt
 7753157 -rw-rw-r-- 2 amits amits       45 Feb 19 19:26  hard.1.txt


  $ ln 1.txt 1.hard

   File name     inode        data
   1.txt    ---> 123     ---> How do you do ?
                 Shemuel
                   ^
   1.hard   _______|

  Example :
  $ ln 1.txt hard.1.txt
  $ stat hard.1.txt
      File: hard.1.txt
      Size: 45         Blocks: 8          IO Block: 4096   regular file
      Device: 10303h/66307d Inode: 7764243     Links: 2
      Access: (0664/-rw-rw-r--)  Uid: ( 1001/   amits)   Gid: ( 1001/   amits)
      Access: 2024-02-19 15:46:06.573765125 +0200
```

```
        Modify: 2024-02-15 16:31:55.646124081 +0200
        Change: 2024-02-19 15:45:28.276807762 +0200
        Birth: 2024-02-15 16:31:55.646124081 +0200
    $ stat 1.txt
        File: 1.txt
        Size: 45         Blocks: 8          IO Block: 4096   regular file
        Device: 10303h/66307d Inode: 7764243     Links: 2
        Access: (0664/-rw-rw-r--)  Uid: ( 1001/   amits)   Gid: ( 1001/   amits)
        Access: 2024-02-19 15:46:06.573765125 +0200
        Modify: 2024-02-15 16:31:55.646124081 +0200
        Change: 2024-02-19 15:45:28.276807762 +0200
        Birth: 2024-02-15 16:31:55.646124081 +0200

    $ ls -i # returns inode numbers of files !!!
       $ ls -i
           7764243  1.txt
           7764243  hard.1.txt

  ------------------------------------------------------------
  • Soft link has <its own inode> and the data that includes
    path to linked file.
    Soft link is separate file with data that contain address of
    linked file.
    Soft link is signed with "l" at the head of list of permissions:
    Link counter of soft link is always 1.
     $ ls -la
      lrwxrwxrwx 1 amits amits    5 Feb 19 21:15 soft1.txt -> a.txt

      File name     inode        data
      1.txt    ---> 123     ---> How do you do ?
                    Shemuel

      1.soft   ---> 456     ---> ./1.txt
                    L
  ------------------------------------------------------------
  Sector on disk is analogues to address.
So hard link must be located on the same partition,
Hard link can't link to another partition or the network address of
folders!!!  For example, we cannot create hard link on USB disk with
link to file on the hard disk.
  ------------------------------------------------------------
```

10.6 find : Search for a file

```
    The find command in UNIX is a command line utility for
    walking a file hierarchy.
    It can be used to find files and directories and perform
    subsequent operations on them.
    It supports searching by file, folder, name, creation date,
```

```
modification date, owner and permissions.
By using the ‘-exec’
other UNIX commands can be executed on files or folders
that found.

Syntax:
    $ find [where to start searching from]
           [expression determines what to find]
           [-options] [what to find]

1. Search for a file with a specific name.
      $ find ./GFG -name sample.txt

2. Search for a file with pattern.
      $ find ./GFG -name *.txt

3. Find and delete a file with confirmation.
      $ find ./GFG -name sample.txt -exec rm -i {} \;

4. Find empty files and directories.
      $ find ./GFG -empty

5. Search for file with given permissions.
      $ find ./GFG -perm 66

6. Displays repositories and sub-repositories
      $ find . -type d

7. Search text 'Geek' within multiple text files.
      $ find ./ -type f -name "*.txt" -exec grep 'Geek'  {} \;

- Find files by extension:
    find {{root_path}} -name '{{*.ext}}'

- Find files matching multiple path/name patterns:
       find {{root_path}} -path '{{**/path/**/*.ext}}'
       -or -name '{{*pattern*}}'

- Find directories matching a given name,
  in case-insensitive mode:
       find {{root_path}} -type d -iname '{{*lib*}}'

- Find files matching a given pattern, excluding
  specific paths:
       find {{root_path}} -name '{{*.py}}' -not
       -path '{{*/site-packages/*}}'

- Find files matching a given size range, limiting
  the recursive depth to "1":
       find {{root_path}} -maxdepth 1 -size
```

```
        {{+500k}} -size {{-10M}}

- Run a command for each file (use {} within the command to
  access the filename):
       find {{root_path}} -name '{{*.ext}}'
       -exec {{wc -l {} }}\;

- Find files modified in the last 7 days:
      find {{root_path}} -daystart -mtime -{{7}}

- Find empty (0 byte) files and delete them:
       find {{root_path}} -type {{f}} -empty -delete

Examples :

Find all files *.c starting from current directory:
     $ find . -name "*.c"

Find all files *.jpg starting with root directory:
     $ find / -name "*.jpg" -print

Find all files *.jpg starting from home directory:
     $ find ~ -name "*.jpg" -print

Insert '"' to make globing at find command level,
no at bash level,  before expansion for find command:

     $ find -name "*.txt"
        ./amit.txt
        ./docs/11.txt
        ./docs/.secret.txt
        ./docs/5.txt
        ./docs/7.txt
```

10.7 touch : Create files or update access times

```
Update the access and modification times if the file exists
A FILE argument that does not exist is created empty, unless -c
or -h is supplied.

1. In its default usage, it is the equivalent of creating
   or opening a file and saving it without any change to the file
   contents. touch avoids opening, saving, and closing the file.
   Instead it simply updates the dates associated with the file
   or directory. An updated access or modification date can be
   important for a variety of other programs such as backup
   utilities.
2. The touch command can also be useful for quickly creating
   files for programs or scripts that require a file with a
```

```
      specific name to exist for successful operation of the
      program, but do not require the file to have any specific
      content.

Example:
     create files 1.txt , 2.txt, ... , 5.txt
     touch {1..5}.txt
```

10.8 stat : display file (inod) and file system info

```
stat() is a Unix system call that returns file attributes
about an inode.

The semantics of stat() vary between operating systems.
As an example, Unix command ls uses this system call to retrieve
information on files that includes:


atime: time of last access (ls -lu)
mtime: time of last modification (ls -l)
ctime: time of last status change (ls -lc)


The stat() and lstat() functions take a filename argument.
If the file is a symbolic link, stat() returns attributes of the
eventual target of the link,
while lstat() returns attributes of the link itself.

File system information:
$ stat -f try.py
    File: "try.py"
    ID: 5a2e6f11fdd5900a Namelen: 255     Type: ext2/ext3
    Block size: 4096       Fundamental block size: 4096
    Blocks: Total: 120454536  Free: 91353786   Available: 85216660
    Inodes: Total: 30670848   Free: 29126190
```

10.9 ls : "LiSt" files and dirs

```
Lists the contents of files and directories.
Without options and arguments displays a list
of the names of all files in the current working directory
in alphabetical order
   -i : return inode numbers of files !!

   -l : displays detailed information about files and directories
        in a long format
```

```
   -a : includes hidden files and directories in the listing

   -t : sorts files and directories by their last modification time,
        most recently modified ones first

   -S : sorts files and directories by their sizes,
        listing the largest ones first

   -r : reverse order (most recent at the end)

   -R : lists files and directories recursively, including subdirectories

   -h : prints file sizes in a human-readable format (e.g., 1K, 234M, 2G)

   -d : info for directories only.


ls command examples:
  ls           # files list (alef bet order)
  ls /         # root dir
  ls .         # current dir
  ls           # current dir
  ls ~         # current dir
  ls -l        # display detailed info: long format
  ls -a        # include hidden files
  ls -lh       # size of files in human format
  ls -lah      # detailed info long format, include hidden files
                 size of files in human format
  ls [0-9].txt #
  ls -R        # include subdirs
  ls -ld docs  # dir info only
  ls -t        # most recent first
  ls -r        # oldest first
  ls -S        # sort by size (largest first)
  ll           # alias of ls -l

file types:
     - : regular file
     d : directory (file listing a set of files)
     l : soft link file (Files referring to the name of another file
     s : socket file (inter process communication)
     p : pipe file (Used to cascade programs)
     c : char file
     b : block file driver (disk) communication files

     search hidden files:
     ls *.txt .*.txt

     $ ls [0-9].txt
```

```
    1.txt  3.txt  5.txt  7.txt

amits:temp1$ ls -l [a-c].txt
    -rw-rw-r-- 1 amits amits 0 Feb 15 13:55 a.txt
    -rw-rw-r-- 1 amits amits 0 Feb 15 13:55 b.txt
    -rw-rw-r-- 1 amits amits 0 Feb 15 13:55 c.txt
amits:temp1$ ls -l {a..c}.txt                    # same result
    -rw-rw-r-- 1 amits amits 0 Feb 15 13:55 a.txt
    -rw-rw-r-- 1 amits amits 0 Feb 15 13:55 b.txt
    -rw-rw-r-- 1 amits amits 0 Feb 15 13:55 c.txt
$ ls -i # returns inode numbers of files !!!
  7764243  1.txt
  7764243  hard.1.txt
```

10.10 du : "Disk Usage" - file and dir usage space

```
https://phoenixnap.com/kb/show-linux-directory-size

Display the current directory size by running du without any arguments:
The system lists the current directory content and subdirectories with a
number representing the object size in kilobytes.
The current directory size is the last line.
 $ du
Print the Current Directory Size in a Human-Friendly Format
Add the -h option to the du command to make the output easily readable:
 $ du -h

Find the Size of a Specific Directory
 $ du -h ./example1

To print an extra total line at the end of the output and summarize the disk
space usage for all files and directories, use the -c argument:
 $ du -c ./example1



Disk usage: estimate and summarize file and directory space usage
 - List the sizes of a directory and any subdirectories, in human-
   readable form (i.e. auto-selecting the appropriate unit for each
   size):
      $ du -h {{path/to/directory}}

 - List the human-readable sizes of a directory and of all the files and
   directories within it:
      $ du -ah {{path/to/directory}}


  Summary of disk usage for two directories:
     $ du -h
```

```
  8.0K ./dir2/2024
  12K ./dir2
  8.0K ./dir1/2024
  20K ./dir1
  76K .
$ du -h | sort -h
  8.0K ./dir1/2024
  8.0K ./dir2/2024
  12K ./dir2
  20K ./dir1
  76K .
$ du -h -s ./dir1 ./dir2  # -s is for summary of size
  20K ./dir1
  12K ./dir2
```

```
Apparent size is the actual amount of data that can be read from the
file. Block-oriented devices can only store in terms of blocks, not
bytes. As a result, the disk usage is always rounded up to the next
highest block
  $ du -h -s --apparent-size ./dir1 ./dir2
```

10.11 ncdu : "Norton Commander and Disk Usage" - file and dir usage space

```
- Analyze the current working directory:
      $ ncdu

- Colorize output:
      $ ncdu --color {{dark|off}}  # dark is color , off is regular

- Analyze a given directory:
      $ ncdu {{path/to/directory}}

- Save results to a file:
      $ ncdu -o {{path/to/file}}
```

10.12 fdisk : partitions in hard disk

```
- List partitions:
   $ sudo fdisk -l
```

10.13 lsblk : Lists information about block(storage) devices

```
- List all storage devices in a tree-like format:
    $ lsblk
```

```
- Also list empty devices:
   $ lsblk -a

- Print the SIZE column in bytes rather than in a
  human-readable format:
   $ lsblk -b

- Output info about filesystems:
   $ lsblk -f
```

10.14 df : overview of the filesystem disk space usage

```
- Display all filesystems and their disk usage in human-readable form:
  $ df -h

- Display all filesystems and their disk usage:
   $ df
```

10.15 cd : "Change Dir"

```
go to parent directory : cd ..
go to home directory   : cd ~
go to root directory   : cd /
go to prev. dir.       : cd -

$ cd -
  /home/amit
$ cd -
  /home/amit/Documents
```

10.16 pwd : "Print Working Directory"

10.17 mkdir : "MaKe DIRectory"

```
The mkdir command in Linux allows the user to create directories.
This command can create multiple directories at once
as well as set the permissions for the directories.

This command can create multiple directories at once
as well as set the permissions for the directories.

- Create specific directories:
    mkdir {{path/to/directory1 path/to/directory2 ...}}

- Create specific directories and their [p]arents if needed:
```

```
    mkdir -p {{path/to/directory1 path/to/directory2 ...}}
-p to create parent folders :
    $ mkdir -p aa/bb/cc

- Create directories with specific permissions:
    mkdir -m {{rwxrw-r--}} {{path/to/directory1 path/to/directory2 ...}}



  $ mkdir aa/bb/cc
    mkdir: cannot create directory ‘aa/bb/cc’: No such file
           or directory
  $ mkdir -p aa/bb/cc  # now it is performed
```

10.18 cp : "CoPy" file to file, file to dir, dir to dir

```
    Copy files and directories

  By default, the cp command copies files without preserving
  their attributes such as permissions, timestamps, and ownership.

  Recursively copy a directory’s contents :
           cp -R {source_directory} {target_directory}
           cp -r {source_directory} {target_directory}
    When the program has two arguments of path names to <files>, the program
  copies the contents of the first <file> to the second <file>, creating
  the second <file> if necessary.

    When the program has one or more arguments of path names of <files> and
  following those an argument of a path to a <directory>, then the program
  copies <each source file> to <the destination directory>, creating any
  files not already existing (!).

    When the program’s arguments are the path names to two <directories>, cp
  copies all files in the source <directory> to the destination
  <director>y, creating any files or directories needed. This mode of
  operation requires an additional option flag, typically -r, to indicate
  the recursive copying of directories. If the destination directory
  already exists, the source is copied into the destination, while a new
  directory is created if the destination does not exist.

  Example:
    mkdir temp{1..5}


Renaming a file by copying and deleting it
Linux users copy a file by using the ‘‘cp’’ command.
When you copy a file, you give the source files and rename the files.
  $ cp old_file new_file
```

As an example, if we were to rename ‘‘august.png’’ to ‘‘september.png’’:
  $ cp august.png september.png
This will create a copy of the file with a new name.
But now you’ll have two copies.
You’ll resolve this by deleting the old file.
You delete with ‘‘rm.’’

  $ rm august.png

10.19 mv : "MoVe" files or dirs from one place to another.

    If both filenames are on the same filesystem, this results
in a simple file rename; otherwise the file content is copied
to the new location and the old file is removed.

    Using mv requires the user to have write permission for
the directories the file will move between. This is because mv
changes the content of both directories (i.e., the source
and the target) involved in the move.
When using the mv command on files located on the same filesystem,
the file’s timestamp is not updated.

 rm of hard link does not delete the original file !
 rm of original file does not delete the hard link and file contents that is linked

Example : move interactive (-i) :
$ mv 1.txt b.txt -i
    cp: overwrite ‘b.txt’? y

$ mv dir1 dir2   # move dir1 under dir2
$ mv dir2/dir1 ./ # move dir1 under current dir

You can rename a file by ‘‘moving it’’ to a new file name.
    $ mv august.png september.png
This actually does what you would think of as a ‘‘rename’’ command and in a single st
But it should be noted that it’s also the general ‘‘move’’ command.
So, you are moving the file at the same time.
If you move it to a different directory, it will be moved as well as renamed.

10.20 rm : "ReMove" removes references(!) to objects from the filesystem

The rm command removes references to objects
from the filesystem using the unlink system call,
where those objects might have had multiple references
(for example, a file with two different names),

```
and the objects themselves are discarded only when
all references have been removed and no programs
still have open handles to the objects.

The command generally does not destroy file data,
since its purpose is really merely to unlink references,
and the filesystem space freed may still contain
leftover data from the removed file.
This can be a security concern in some cases,
and hardened versions sometimes provide for
wiping out the data as the last link is being cut,
and programs such as <shred> and <srm> are available
which specifically provide data wiping capability.

rm    : Remove specific files

rm -i : Remove specific files [i]nteractively
        prompting before each removal.
rm -r : (r)ecursively delete a directory and all its contents (!)

Note : normally rm will not delete directories,
       while rmdir will only delete empty
       directories). But -r allows to remove directories:

       $ rm dir3
            rm: cannot remove ‘‘dir3’’ Is a directory
       $ rm dir3 -r # remove dir3 and all child directories
       $            # Done directory remove

       $ rm dir1/{2..4}.txt
       $ rm dir2/{1,4,5}.txt

 Remove all files with answer "yes" to all prompts :
       $ yes | rm -r Documents/
```

10.21 rmdir : "ReMove DIR" remove empty(!) dir from filesystem.

```
Note : removes only empty directories !!!

$ rmdir dir1
   rmdir: failed to remove ‘‘dir1’’: Directory not empty

rmdir     : remove specific empty directory
rmdir -p  : remove specific directory and all its ancestors - "AVOT"
```

10.22 tree : show curr. dir as tree.

```
  Show current directory as tree ,
  with sizes in human style and permissions :
```

```
$ tree --du -h -p

-L     Level
-s     Print the size of each file in bytes along with the name.
-h     Print the size of each file but in a more  human  readable  way,
       e.g.  appending  a size letter for kilobytes (K), megabytes (M),
       gigabytes (G), terabytes (T), petabytes (P) and exabytes (E).

--du   For each directory report its size as the accumulation of  sizes
       of  all  its  files and sub-directories


--inodes
       Prints the inode number of the file or directory

-p     Print  the  file  type  and permissions for each file (as per ls
       -l).


-u     Print the username, or UID # if no username is available, of the
       file.
-g     Print the group name, or GID # if no group name is available, of
       the file.

- Print files and directories up to 'num' levels of
  depth (where 1 means the current directory):
     $ tree -L {{num}}

- Print directories only:
     $ tree -d

- Print hidden files too with colorization on:
     $ tree -a -C
```

10.23 file : determine file type.

```
- Give a description of the type of the
  specified file. Works fine for files with no
  file extension:
       file {{path/to/file}}
  -b, -brief : This is used to display just
   file type in brief mode



  Examples:
     $ type -a cat
```

```
$ file [a-z]*
$ file  /bin/cat
$ file tex_file_data_extract

$ file $(which test) # $( ) means substitute command
  /usr/bin/test: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV),
  dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2,
  BuildID[sha1]=68ccce12c6a2a96d3a40fb5868f8b8c45e6015a5, for GNU/Linux 3.2.0,
  stripped
$ file `which test`  # ` ` means substitute command
  /usr/bin/test: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV),
  dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2,
  BuildID[sha1]=68ccce12c6a2a96d3a40fb5868f8b8c45e6015a5, for GNU/Linux 3.2.0,
  stripped
```

11 User Groups

11.1 CreateNewUser - steps to create new user

```
1. Create Home    : $ sudo useradd tal -m # new user with home directory
2. Skeleton files : copy files from /etc/skel to /home/tal
                    Note : copy without "." and ".."
3. set shell type : $ sudo usermod -s /bin/bash tal
4. set pw         : sudo passwd tal
5. add user to sudo group
                  : $ sudo usermod -aG sudo david
   Debian : "sudo"
   RedHat : "wheel"
      or...
   visudo (edit sudoer file)
```

11.2 visudo - edit sudoer file

```
 Safely edit the sudoers file.

 - Edit the sudoers file:
   sudo visudo

 - Check the sudoers file for errors:
   sudo visudo -c

- Edit the sudoers file using a specific editor:
  sudo EDITOR={{editor}} visudo

- Display version information:
  visudo --version
```

11.3 passwd : Change a user's PASSWorD.

```
Change user password

- Change the password of the current user interactively:
    passwd

- Change the password of a specific user:
    passwd {{username}}

- Get the current status of the user:
    passwd -S

- Make the password of the account blank
  (it will set the named account passwordless):
    passwd -d
```

11.4 useradd : add user to the system

```
It is not simple, more complicated
relative to adduser (simple perl script that uses useradd)

When we use useradd, it creates the account without password :
  $ sudo useradd tal
    password for amit:
  $

But we cannot enter the account with su - tal
because it has not user:
  $ su - tal
    Password:
    scdsds
    ssu: Authentication failure

To give to user tal password, run following command (with sudo):
  $ passwd tal
    passwd: You may not view or modify password information for tal.
  $ sudo passwd tal
    New password:
    Retype new password:
    passwd: password updated successfully

 Create user with home directory as his name use $ useradd -m
  $ sudo useradd tal -m
  $ su - tal
    Password:
  $ sudo passwd tal
    New password:
    Retype new password:
```

```
  passwd: password updated successfully
$ su - tal
  Password:
$
$ pwd
  /home/tal
$ echo $SHELL  # when we create user with useradd , shell will be
  /bin/sh
```

By default the user tal uses shell "dash", to define shell to user we need define option -s /bin/bash when we define new user with useradd command
But now, when the user already exists, we can change the shell type with usermod command.

```
$ sudo usermod -s /bin/bash tal
$ su - tal
  Password:
$ ll
  total 36
  drwxr-x--- 4 tal  tal  4096 Sep 27 16:02 ./
  drwxr-xr-x 5 root root 4096 Sep 27 15:58 ../
  -rw-r--r-- 1 tal  tal   220 Jan  6  2022 .bash_logout
  -rw-r--r-- 1 tal  tal  3771 Jan  6  2022 .bashrc
  drwx------ 2 tal  tal  4096 Sep 27 16:02 .cache/
  drwxr-xr-x 5 tal  tal  4096 Jul 11 19:21 .config/
  -rw-r--r-- 1 tal  tal    22 Sep  8  2011 .gtkrc-2.0
  -rw-r--r-- 1 tal  tal   516 Dec 17  2013 .gtkrc-xfce
  -rw-r--r-- 1 tal  tal   807 Jan  6  2022 .profile
```

Above skeleton files are within /etc/skel directory,
and can be copied from it manually. But when we copy the file from one user to another user, it will have the user name of previous user !!!
There are two ways to solve this:

- Allow the user to copy files.
- Change the user of the files.

Note: don't run command cp -r /etc/skel/.* because it will copy also ‘.’ And ‘..’ that is not desirable.

So it is preferred to copy files by exact list of its names.

Thinks to concern when we create new user:

- User's home
- User's skeleton files
- User's shell
- User's password

Note: for useradd don't use -P option !!!

```
-P, --prefix PREFIX_DIR  # Gibrish
   Apply changes in the PREFIX_DIR directory and use the
   configuration files from the PREFIX_DIR directory.
   This option does not chroot and is intended for preparing a
   cross-compilation target. Some limitations: NIS and LDAP
   users/groups are not verified. PAM authentication is using
   the host files. No SELINUX support.
```

11.5 $SHELL : env. variable (check current shell type)

https://unix.stackexchange.com/questions/43499/difference-between-echo-shell-and-which-bash

```
$ echo $SHELL
 /bin/zsh
$ which bash
 /bin/bash
 The first command tells me which shell will be executed by login
 when you log in. In my case, /bin/zsh.
 The second command tells me the first occurrence in my
 $PATH the bash command can be found.
```

11.6 adduser : perl script, that uses useradd

```
simple perl script (that uses useradd command internally):

  $ sudo adduser david
    [sudo] password for amit:
    Adding user ‘david’ ...
    Adding new group ‘david’ (1001) ...
    Adding new user ‘david’ (1001) with group ‘david’; ...
    Creating home directory ‘/home/david’ ...
    Copying files from ‘/etc/skel’ ...
    New password:
    Retype new password:
    passwd: password updated successfully
    Changing the user information for david
    Enter the new value, or press ENTER for the default
 Full Name []: David
 Room Number []: 1
 Work Phone []: 046778723
 Home Phone []: 0584266865
 Other []: no
    Is the information correct? [Y/n] Y

  $ cat /etc/passwd | grep bash # check users of bash
     root:x:0:0:root:/root:/bin/bash
```

```
amits:x:1001:1001:AmitS,,,:/home/amits:/bin/bash
david:x:1000:1000:David,1,046778723,0584266865,no:/home/david:/bin/bash
```

11.7 userdel : delete user from the system

```
it is not simple, more complicated
relative to deluser (simple perl script that uses userdel)

Remove the user:
   $ userdel -r moshe

  Files in the user's home directory will be removed
  along with the home directory itself and the user's
  mail spool. Files located in other file systems
  will have to be searched for and deleted manually.
```

11.8 deluser : perl script, that uses userdel

```
 It is simple Perl script (that uses userdel command)


 Delete a user account and related files.
 The userdel command modifies the system account files,
 deleting all entries that refer to the user name LOGIN.
 The named user must exist.

 Try to delete user tal:
   $ sudo deluser --remove-all-files tal
    ....
    /usr/sbin/deluser: Cannot handle special file /proc/3614/fd/1
    /usr/sbin/deluser: Cannot handle special file /proc/3614/fd/2
    /usr/sbin/deluser: Cannot handle special file /etc/systemd/
    system/sudo.service
    Removing user 'tal' ...
    Warning: group 'tal' has no more members.
    Done.
```

11.9 /etc/group - file with list of groups

```
 $ head /etc/group
   root:x:0:
   daemon:x:1:
   bin:x:2:
   sys:x:3:
   adm:x:4:syslog,amits
   tty:x:5:syslog
   disk:x:6:
   lp:x:7:
```

```
mail:x:8:
news:x:9:

$ cat /etc/group | grep amits
adm:x:4:syslog,amits
cdrom:x:24:amits
sudo:x:27:amits
dip:x:30:amits
plugdev:x:46:amits
lpadmin:x:120:amits
sambashare:x:131:amits
amits:x:1001:
```

```
check who has permission to use sambashare (program to share files):
    $ cat /etc/group | grep sambashare
      sambashare:x:136:amit
```

11.10 /etc/passwd : conteins info what users are in the system

```
$ cat /etc/passwd | grep amit
    amits:x:1001:1001:AmitS,,,:/home/amits:/bin/bash
                # /bin/bash : what shell the user uses
```

11.11 groups : present all groups of the user

```
groups - print the groups a user is in

$ groups david
  david : david sudo
$ groups amits
  amits : amits adm cdrom sudo dip plugdev lpadmin sambashare
$ groups
  amits adm cdrom sudo dip plugdev lpadmin sambashare
```

11.12 chown : CHange OWNer (user/group) of files and dirs

```
Change user and group ownership of files and directories.

Unprivileged (regular) users who wish to change the group membership
of a file that they own may use chgrp.

- Change the owner user of a file/directory:
   chown {{user}} {{path/to/file_or_directory}}

- Change the owner user and group of a file/directory:
   chown {{user}}:{{group}} {{path/to/file_or_directory}}
```

```
- Recursively change the owner of a directory and its contents:
   chown -R {{user}} {{path/to/directory}}

- Change the owner of a symbolic link:
   chown -h {{user}} {{path/to/symlink}}

- Change the owner of a file/directory to match a reference file:
   chown --reference={{path/to/reference_file}} {{path/to/file_or_directory}}
```

```
$ mkdir project_dir
$ sudo chown david:family project_dir  # david : owner , family : group
```

11.13 members : present all users of the group

```
$ groups
   amits adm cdrom sudo dip plugdev lpadmin sambashare
$ members lpadmin
   cups-pk-helper amits
$ members sudo
   amits david
```

11.14 groupadd : Add user groups to the system

```
Add user groups to the system. See also: groups, groupdel, groupmod
 - Create a new group:
    $ sudo groupadd {{group_name}}

 - Create a new system group:
    $ sudo groupadd --system {{group_name}}

 - Create a new group with the specific groupid:
    $ sudo groupadd --gid {{id}} {{group_name}}

 $ sudo groupadd family
 $ sudo usermod -aG family amits
 $ groups amits
   amits : amits adm cdrom sudo dip plugdev lpadmin sambashare family
 $ groups
   amits adm cdrom sudo dip plugdev lpadmin sambashare
```

11.15 passwd : change password

```
Change password for current user :
   passwd
```

11.16 whoami : currently logged-in user

```
allows users to see the currently logged-in user.
```

11.17 su : "Substitute User"

```
The 'su' command, short for 'substitute user', is a simple
yet powerful command in Linux. It allows you to switch to
another user account on your system, right from the command
line. This can be particularly useful for system
administrators who need to perform tasks on behalf of other
users.

To enter David's user account including David's environment
variables:
$ su - david

enter and exit from user david :
$ whoami
  amits
$ su david
  password:
$ whoami
  david
$ exit
$ whoami
  amits

To connect as root user:
   $ sudo su - root # or sudo su -.
It allows to work without sudo for every command.
Don't forget to run exit.

   $ sudo su - root
     [sudo] password for amit:
   #
   #
   # whoami
     root
   #

 Note: "#" signs that the user is root.
 Note : sudo is not user, sudo is group
```

11.18 exit : exit the shell

```
$ whoami
  amits
$ su david
  password:
$ whoami
  david
$ exit    # exit from david
$ whoami
  amits
```

11.19 usermod : MODifies a USER account/add user to group.

```
- Change a username:
    sudo usermod --login {{new_username}} {{username}}

- Change a user id:
    sudo usermod --uid {{id}} {{username}}

- Change a user shell:
    sudo usermod --shell {{path/to/shell}} {{username}}

- Add a user to supplementary groups (mind the lack of whitespace):
    sudo usermod --append --groups {{group1,group2,...}} {{username}}

-a, --append  # the same as --append --groups
    Add the user to the supplementary group(s).
    Use only with the -G option (!) : -aG

-G, --groups GROUP1[,GROUP2,...[,GROUPN]]]
    A list of supplementary groups which the user is also a member of.
    Each group is separated from the next by a comma, with no intervening
    whitespace. The groups are subject to the same restrictions as
    the group given with the -g option.

- Change a user home directory:
    sudo usermod --move-home --home {{path/to/new_home}} {{username}}

$ echo $SHELL # check the shell type
   /bin/sh

Change the shell type for user "david" with usermod command:
     $ sudo usermod -s /bin/bash david

Add the user "david" to group "sudo":
  amits $ su - david
       Password:
  david $ sudo echo hello
```

```
            [sudo] password for david:
            david is not in the sudoers file.  This incident will be reported.
        david $ exit
            logout
        amits $ sudo usermod -aG sudo david
            [sudo] password for amits:
        amits $ su - david
            Password:
    To run a command as administrator (user "root"), use "sudo <command>".
    See "man sudo_root" for details.
        david $ sudo echo hello # to check that david is sudoer
            [sudo] password for david:
            hello

        $ sudo usermod -aG family david # add user david to group family
        $ groups david # check in what groups is david ?
          david : david sudo family
```

11.20 usermod : add Tal to sudo group and remove tal from sudo group.

```
to add user to sudo group :
 amits$ sudo usermod -a -G sudo tal

to remove tal from sudo group:
 amits$ sudo usermod -r -G sudo tal




\subsection{sudo   : run command as administrator.}
\begin{verbatim}

   To run a command as administrator (user
   "root"), use "sudo <command>".
   See "man sudo_root" for details.
   To be "sudoer" the user must be added to
   the group "sudo".

   $ su - david
     Password:
   $ sudo echo hello
     [sudo] password for david:
     david is not in the sudoers file.  This incident will be reported.
   $ exit
     logout
   amits:Temp_Amit$ sudo usermod -aG sudo david
     [sudo] password for amits:
```

```
   $ su - david
     Password:
     To run a command as administrator (user "root"), use "sudo <command>".
     See "man sudo_root" for details.

david@amits-vostro-3520:~$ sudo echo hello # to check that david is sudoer
        [sudo] password for david:
        hello
```

12 Special Device files

12.1 Device..files with special behaviour

These files are related to drivers, placed in /dev
directory and are of two types:

- char file
- block file

Read or write to char or block file causes to some system action.
These both files are related to writing data and reading data from
one place to other place.
But the difference between them is how they read/write data.

Block file:

A block file is a hardware file which read/write data in blocks
instead of character by character. This type of files are very
much useful when we want to write/read data in bulk fashion.
All our disks such are HDD, USB and CDROMs are block devices.
This is the reason when we are formatting we consider block size.
The write of data is done in asynchronous fashion and it is
CPU intensive activity. These devices files are used to store data
on real hardware and can be mounted so that we can access the data
we written. Have a look at our other post on getting block size of a
device.

Block file examples:/dev/loop0, /dev/ram0, /dev/sda1, /dev/sr0.

Character file:

A char file is a hardware file which reads/write data in character by
character fashion. Some classic examples are keyboard, mouse, serial
printer. If a user use a char file for writing data no other
user can use same char file to write data which blocks access
to other user.
Character files uses synchronise technic to write data.
Of you observe char files are used for communication purpose
and they can not be mounted.
Example character files:
                        /dev/autofs,

```
                              /dev/console,
                              /dev/crash,
                              /dev/lp0,
                              /dev/null,
                              /dev/ppp,
                              /dev/random,
                              /dev/tty ,etc
        ls -l output for a char file:
             $ ls -l /dev/null
                 crw-rw-rw- 1 root root 1, 3 Sep 21 06:54 /dev/null
             $ ls -l /dev/zero
                 crw-rw-rw- 1 root root 1, 5 Sep 21 06:54 /dev/zero
             $ ls -l /dev/random
                 crw-rw-rw- 1 root root 1, 8 Sep 21 06:54 /dev/random
             $ ls -l /dev/urandom
                 crw-rw-rw- 1 root root 1, 9 Sep 21 06:54 /dev/urandom

          Note: stdout, stdin and stderr are also placed in /dev directory:
          (fd/0,1,2 - file descriptor 0,1,2; stdout,stdin,stderr are soft links)

$ ls -li /dev
482 lrwxrwxrwx   1 root  root   15 Feb 20 12:41 stderr -> /proc/self/fd/2
480 lrwxrwxrwx   1 root  root   15 Feb 20 12:41 stdin  -> /proc/self/fd/0
481 lrwxrwxrwx   1 root  root   15 Feb 20 12:41 stdout -> /proc/self/fd/1
```

12.2 /dev/null and /dev/zero : data sink files

```
Data written to the /dev/null and /dev/zero special files is discarded

Reads  from  /dev/null file always return end of file ^D,
whereas reads from /dev/zero file always return bytes
containing zero ('\0' characters with 0x0 ASCII code).

Read 20 zeros in decimal format from /dev/zero file :
$ od -tu1 -N20 /dev/zero
  0000000   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
  0000020   0   0   0   0
  0000024

$ od -tu1 -An -N20 /dev/zero # without present byte count
  0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
  0   0   0   0


See: $ man null

If you want to sent the output to nowhere, send it to /dev/null:
```

```
        $ echo "this will not be output to any place" > /dev/null

If you don't want to see error messages, you can send errors to /dev/null:
        $ ls -l dir1 dir2
             ls: cannot access 'dir2': No such file or directory
             dir1:
             total 4
             -rw-r--r-- 1 amit amit 13 Sep 23 22:51 a.txt
        $ ls -l dir1 dir2 2>/dev/null # rederect errors to dev/null
             dir1:
             total 4
             -rw-r--r-- 1 amit amit 13 Sep 23 22:51 a.txt

If you want to see only error messages, you can send regular output to /dev/null:
        $ ls -l dir1 dir2 1>/dev/null
             ls: cannot access 'dir2': No such file or directory

If we don't want to see messages of any type (error and regular)
we can send both outputs to /dev/null:
        $ ls -l dir1 dir2 1>/dev/null 2>&1
```

12.3 /dev/random file - random numbers generator file

```
Read 5 random numbers from /dev/random file:
Random is quality generator !
  $ od -tu1 -An -N5 /dev/random
    253  85  48 224 101
  $ od -tu1 -An -N5 /dev/random
    166   2 125 190 224
  $ od -tu1 -An -N5 /dev/random
    215 185 237  29 144

4 byte numbers :
 $ od -tu4 -An -N4 /dev/random # -tu4 is 4 byte decimal number
   1255275280
 $ od -tu4 -An -N4 /dev/random # another random number
   1409517909
 $ od -tu4 -An -N4 /dev/random
   1099118913

Randomization of /dev/random is of less quality
Randomization of /dev/urandom in of high quality
```

12.4 /dev/full - file for device full simulation

```
If we try to write to /dev/full the system returns that device is full.
$ echo 'hello' > /dev/full
   bash: echo: write error: No space left on device
```

12.5 /etc/group - file with list of groups

```
$ head /etc/group
  root:x:0:
  daemon:x:1:
  bin:x:2:
  sys:x:3:
  adm:x:4:syslog,amits
  tty:x:5:syslog
  disk:x:6:
  lp:x:7:
  mail:x:8:
  news:x:9:

 $ cat /etc/group | grep amits
  adm:x:4:syslog,amits
  cdrom:x:24:amits
  sudo:x:27:amits
  dip:x:30:amits
  plugdev:x:46:amits
  lpadmin:x:120:amits
  sambashare:x:131:amits
  amits:x:1001:


check who has permission to use sambashare (program to share files):
    $ cat /etc/group | grep sambashare
      sambashare:x:136:amit
```

12.6 /etc/passwd - info what users are in the system

```
$ head /etc/group
  root:x:0:
  daemon:x:1:
  bin:x:2:
  sys:x:3:
  adm:x:4:syslog,amits
  tty:x:5:syslog
  disk:x:6:
  lp:x:7:
  mail:x:8:
  news:x:9:

 $ cat /etc/group | grep amits
  adm:x:4:syslog,amits
  cdrom:x:24:amits
  sudo:x:27:amits
  dip:x:30:amits
  plugdev:x:46:amits
  lpadmin:x:120:amits
  sambashare:x:131:amits
```

```
amits:x:1001:
```

```
check who has permission to use sambashare (program to share files):
    $ cat /etc/group | grep sambashare
      sambashare:x:136:amit

Users that are using BASH:
   $ cat /etc/passwd | grep bash
       root:x:0:0:root:/root:/bin/bash
       amit:x:1000:1000:amit,,,:/home/amit:/bin/bash
```

12.7 /etc/skel - skeleton files for new user

Skeleton files (.bashrc , .profile) for new user are
within /etc/skel directory, and can be copied from it manually.
But when we copy the file from one user to another user,
it will have the user name of previous user !!!

There are two ways to solve this:

- Allow the user to copy files.
- Change the user of the files.

Note: don't run command cp -r /etc/skel/.* because it will copy
also ‘‘.’’ And ‘‘..’’ that is not desirable.

So it is preferred to copy files by exact list of its names.

```
$ ls -lai /etc/skel
  total 28
  9699438 drwxr-xr-x   2 root root  4096 Nov 15 10:14 .
  9699329 drwxr-xr-x 154 root root 12288 Feb 20 23:18 ..
  9701502 -rw-r--r--   1 root root   220 Feb 25  2020 .bash_logout
  9701503 -rw-r--r--   1 root root  3771 Feb 25  2020 .bashrc
  9701504 -rw-r--r--   1 root root   807 Feb 25  2020 .profile
```

13 Binary files treatment

13.1 binary files compare

1. Convert files in hex format and compare as unicode files:

```
$ hexdump file1.bin > file1.hex
$ hexdump file2.bin > file2.hex
$ diff file1.hex file2.hex
```

```
2. A visual binary diff tool that allows you to compare binary files
   interactively.

$ vbindiff file1.bin file2.bin
```

13.2 nm :"Name Mangling" damp symbol table from object files.

```
https://en.wikipedia.org/wiki/Nm_(Unix)

- List global (extern) functions in a file (prefixed with T):
     nm -g {{path/to/file.o}}

- List only undefined symbols in a file:
     nm -u {{path/to/file.o}}

- List all symbols, even debugging symbols:
     nm -a {{path/to/file.o}}

- Demangle C++ symbols (make them readable):
     nm --demangle {{path/to/file.o}}

  Example:
  amit@amit-vm-mint21:~/C_programs$ nm ex1
     000000000000038c r __abi_tag
     0000000000004010 B __bss_start
     0000000000004010 b completed.0
            w __cxa_finalize@GLIBC_2.2.5
     0000000000004000 D __data_start
     ...

 Symbol Type Description
   A Global absolute symbol.
   a Local absolute symbol.
   B Global bss symbol.
   b Local bss symbol.
   D Global data symbol.
   d Local data symbol.
   f Source file name symbol.
   L Global thread-local symbol (TLS).
   l Static thread-local symbol (TLS).
   R Global read only symbol.
   r Local read only symbol.
   T Global text symbol.
   t Local text symbol.
   U Undefined symbol.
```

13.3 readelf : ELF file info.

```
display information about ELF files ONLY
including their metadata
readelf [-a|--all]
        [-h|--file-header]
        [-l|--program-headers|--segments]
        [-S|--section-headers|--sections]
        [-g|--section-groups]
        [-t|--section-details]
        [-e|--headers]
        [-s|--syms|--symbols]
```

13.4 od : "Octal Damp" - display binary file.

```
od is a command on various operating systems
for displaying ("dumping") data in various human-readable
output formats. The name is an acronym for "octal dump"
since it defaults to printing in the octal data format.

Display file contents in octal, decimal or hexadecimal
format. Optionally display the byte offsets and/or
printable representation for each line.

Default settings: Octal format, 8 bytes per line,
                  ------------
byte offsets in octal, and duplicate lines replaced with *:
      $ od tex_file_data_extract

  -tu1 - decimal format
  -tx1 - hexadecimal format
  -N   - limit dump to N bytes:
  -v   - do not use * to mark line duplicates

Decimal:
   $ od -tu1 a.txt
Hexadecimal:
 - Display file in hexadecimal format (1-byte units),
   and 4 bytes per line:
   $ od --format={{x1}} --width={{4}} -v {{path/to/file}}
   $ od -tx1 tex_file_data_extract -N 1024

Read text file with od to see ASCII codes :
   $ echo "ABCD" > a.txt
   $ cat a.txt
```

```
  ABCD
$ od -tu1 a.txt # decimal ASCII values
  0000000  65  66  67  68  10
  0000005
$ od -tx1 a.txt # 1 Byte (8 bit) hexadecimal ASCII values
  0000000 41 42 43 44 0a
  0000005
  Note : 0x0a is for line feed character
$ od -tx4 a.txt # 4 BYTE (32 BIT) hexadecimal ASCII values
  0000000 44434241 0000000a
  0000005
```

14 Editors, text operations

14.1 nano : run text editor.

```
Run text editor
nano ex1.c
```

14.2 vim :v improved editor.

```
vim (v improved) , to run :
    vim 1.txt

2 modes of work :
    • Edit (:i to enter insert mode),
      i.e. type ":" and after it "i"
    • Commands (esc to enter command mode)
       > Shift +:e [file] - Opens a [file] that you want
         to open. Here [file] is the filename that you want to open
       > Esc, :w    - Save the file but do not exit.
       > Esc, :q!   - To quit the file without first
         saving what you were working on.
       > Esc, :wq   - To save the file and exit from Vim.
       > Esc, :help - To get help
```

14.3 tilde : intuitive text editor.

```
tilde - an intuitive text editor
for the console/terminal
```

15 Install applications

15.1 dpkg : Debian PaKaGe install

Command to install debian package (adds it to apt)

- -i OR --install: Install a package using the dpkg command.
The command will extract all control files for the specified
package, remove any previously installed older
instance of the package, and install the new package on our system.
- -r OR --remove: Remove an installed package
from our system. It removes every file belonging to the
specific package except the configuration files.
This can be seen as the uninstallation option.
- -P OR --purge: An alternative way to remove an in
stalled package from our system. It completely removes
every fie belonging to the specific package, including
the configuration files. This can be seen as the
'complete uninstallation' option.
- -l OR --list: Lists all installed packages on your system.
- -L OR --listfiles: List installed package's file locations.
- -s OR --status: Shows the status of an installed package.
- -S OR --search: Search for a pattern in installed packages.

Example of VS Code install:
$ sudo dpkg -i code_1.81.1-1691620686_amd64.deb

DPKG - s command : shows the status of an installed package:
$ dpkg -s code

sudo dpkg -i *.deb

sudo dpkg -install *.deb

see APT to find differences between APT and DPKG

15.2 apt : "Advanced Package Tool"

https://en.wikipedia.org/wiki/APT_(software)

APT provides a high-level commandline interface
for the package management system.

- Update the list of available packages and versions
(it's recommended to run this before other apt commands):
sudo apt update

- Search for a given package:

```
     apt search {{package}}
     Example : apt search code # for Visual Studio
- Show information for a package:
     apt show {{package}}
     Example : apt show code   # for Visual Studio
- Install a package, or update it to the latest available version:
     sudo apt install {{package}}

- Remove a package (using purge instead also removes its configuration files):
     sudo apt remove {{package}}

- Upgrade all installed packages to their newest available versions:
     sudo apt upgrade

- List all packages:
     apt list

- List installed packages:
     apt list --installed



*********Apt vs. Dpkg*********************************
https://www.geekbits.io/what-is-the-difference-between-apt-and-dpkg/

Apt (Advanced Package Tool) is a newer package manager
that offers more features than dpkg. It can
automatically handle dependencies, so you don't have to
worry about manually installing them.
Apt also has a wider range of repositories from which
you can install packages.

Dpkg (Debian Package Management System) is the older
package manager. It doesn't have as many features as apt,
but it's still a powerful tool.
Dpkg can't automatically handle dependencies,
so you'll need to install them manually.
Dpkg also has a limited range of repositories,
but you can still find the most common packages.


Apt is a higher-level tool that provides a more straightforward
interface to work with than dpkg.
While dpkg can only install, remove, or query individual packages,
apt offers additional functionality, such as automatically
handling dependencies,retrieving and installing packages from
remote locations, and upgrading all installed packages to
their latest versions.

Additionally, apt offers several useful commands
```

for managing repositories and upgrading the system.
For example, the apt-get command can install or remove packages,
while the apt-cache command can search for available packages.

Another difference is that dpkg is primarily a command-line tool,
while apt also provides a graphical user interface (GUI)
in the form of the aptitude tool.

Which Package manager should you use?
So, which package manager should you use? It depends on your needs.
If you want a more powerful tool with automatic dependency management,
then apt is the way to go.
On the other hand, if you want a more straightforward package manager
or if you need to install packages from a specific repository,
then dpkg is the better choice.

Overall, apt (Recommended tool) is a more user-friendly and robust
option than dpkg, making it the preferred package manager
for most users. It provides a simpler and more automated way to
manage packages on your system.

However, dpkg can still be helpful in certain situations,
such as when working with old or custom packages
that are unavailable through apt.

16 GCC and binary files operations

17 Ethernet

17.1 Wireshark install

https://askubuntu.com/questions/700712/how-to-install-wireshark

1.Add the stable official PPA. To do this, go to terminal by pressing Ctrl+Alt+T
and run:

```
sudo add-apt-repository ppa:wireshark-dev/stable
```

2.Update the repository:

```
sudo apt-get update
```

3.Install wireshark 2.0:

```
sudo apt-get install wireshark
```

4.Run wireshark:

```
    sudo wireshark
```

5.If you get an error
"couldn't run /usr/bin/dumpcap in child process: Permission Denied", go to the terminal again and run:

```
    sudo dpkg-reconfigure wireshark-common
```

Say YES to the message box. This adds a wireshark group. Then add user to the group by typing

```
    sudo adduser $USER wireshark
```

Then restart your machine and open Wireshark. It works. Good Luck.

17.2 packet.names.at.different.levels.of.TCP/IP

```
packet names at different levels of TCP/IP

application  |  message
transport    |  segment
network      |  datagram
link         |  frame
```

17.3 TCP.vs.UDP

```
TCP is Transmission Control Protocol
UDP is User Datagram Protocol

TCP use processes to check that the packets are correctly sent
to the receiver.

UDP send packets without any checks but with the advantage
of being quicker than TCP.
```

17.4 sysctl : change kernel Networking params

```
All TCP/IP tuning parameters are located under /proc/sys/net/

important tuning parameters:
/proc/sys/net/core/rmem_max - Maximum TCP Receive Window
/proc/sys/net/core/wmem_max - Maximum TCP Send Window
/proc/sys/net/ipv4/tcp_rmem - memory reserved for TCP receive buffers
/proc/sys/net/ipv4/tcp_wmem - memory reserved for TCP send buffers

we dont change the above parameters directly,
sysctl command is used to modify kernel parameters at runtime.
```

```
The parameters available are those listed under /proc/sys/.

Example for modifying kernel receiver buffer to 16Mbyte:
   $ sysctl -q -w net.core.rmem_max = 16777216
   $ sysctl -q -w net.core.rmem_default=16777216

   -q : quiet
   -w : write
```

17.5 IP..netmask..explanation

```
Four basic elements are combined to make an Ethernet system
   1. The FRAME, which is a standardized set of bits used to carry
      data over the system.

   2. The MEDIA ACCESS CONTROL protocol, which defines how multiple
      computers access the shared Ethernet channel in a fair manner.

   3. The SIGNALING COMPONENTS, which consists of standardized
      electronic devices that send and receive signals over an
      Ethernet channel.

    4. The PHYSICAL MEDIUM, which consists of the cables and other
       hardware used to carry the digital Ethernet signals between
       computers attached to the network.

The Ethernet frame (MAC Frame)
------------------------------
7 bytes  Preamble (for synchronizing)
1 byte   SFD      (Start Frame Delimeter)
6 bytes  DESTINATION ADDRESS (MAC address)
6 bytes  SOURCE ADDRESS
2 bytes  LENGTH/TYPE (frame type or frame length)

Data field:
46 TO 1500 or 1504  DSAP and SSAP (Destination and Source
or 1982 bytes       Service Access Point)+CTRL (1 byte each)


 4 bytes FRAME CHECK SEQUENCE (CRC 32)



The Media Access Control protocol
---------------------------------
MAC protocol defines how the physical medium is shared among
stations connected.
```

Defines how multiple computers access the shared Ethernet
channel in a fair manner.

Ethernet-equipped computer operates independently of all other
stations on the network; there is no central controller

Ethernet uses a broadcast delivery mechanism, in which each frame
that is transmitted is heard by every station.

Every frame sent has an Destination Address and Source Address

MAC address (true HW address, used for switching):

MAC refers to a unique hardware address assigned to network devices
for identification at the data link layer.
Every Ethernet Station has a unique 48 bit MAC Address (6 bytes).

IP address (can be changed, used for routing):

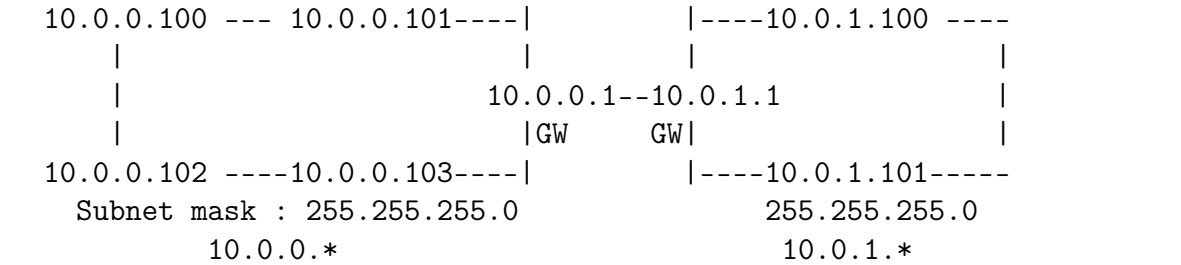
Two computers with different MAC addresses can
have the same IP address . Also for example, in case when we change
HW adapter in the same computer

This use of m.a.c. address rise some problems :
1. there are network hw adapters that allow to put different m.a.c.
   address.
2. In the case when in the network one of network adapters is damaged
   and replaced by new one (for ex. fix the adaptor problem).

That is why defined IP (Internet Protocol), and IP address that was
defined by Internet Protocol. There are IP adresses
ver. 4 and ver. 6 (4 is more common).

Internet protocol ---> IP ----> IP address

IP address consists of two parts : (subnet number)&(Host number)



GW : Gate Way (Router)

IP address ver. 4 consists of 4 bytes:

0-255 | 0-255 | 0.255 | 0-255
192 . 168 . 0 . 1

Subnet mask number
---

netmask/subnet number helps to computer to understand if the package will be send inside the subnet or directly to GW router to pass the message outside of subnet.

We would like to identify between local networks. For this purpose network mask (netmask) is defind.
We tell that some bits of the IP number present the network number - netmask number is used for this purpose.
The bits in netmask number that are equal to 1 define area field of network number.
The bits in netmask number that are equal to 0 define area field of computer number within local network.
If the computer 10.0.0.100 wants to send message to computer 10.0.0.103 - it knows that it must send the message in local network 10.0.0 and there is no need to pass the message out of the local network 10.0.0. Enough that 10.0.0.100 will send broadcast to all the computers in its local network 10.0.0...until it will get the answer from 10.0.0.103.
But if the computer 10.0.0.100 will try to send the message to 10.0.1.101...Because it knows that network 10.0.1 is not its own network, it will send the request to Gate Way computer 10.0.0.1.
Gate Way computer 10.0.0.1 sends the message to Gate Way computer 10.0.1.1 and the Gate Way of 10.0.1 will send broadcast on its local network 10.0.1.
After number of hopes the message from 1.0.0.100 will arrive to computer 1.0.1.101 At home network we connected by WiFi to router that si the GW of local home newtwork. The router is connected to Internet provider... and so on , until the message arrives its target.
In the following example the network number can be 24 bit and the computer number can be 8 bits.

```
10  .  0   .  0   . 100                IP address
255 . 255  .  255 .  0                  Subnet mask
  Network number  | Computer Number     255.255.255.0 <--> /24
  (subnet mask)   | (Host number)
-----------------------------------
                  | Num. of computers that can be connected:
                  |          2^8 =256
                  | 255 is for brodcast
                  | 1   is for GW

   10  .  0   .  0   . 100                IP address
   255 . 255  .  0   .  0                  Subnet mask
Network number | Computer Number           255.255.0.0 <--> /16
```

```
(subnet mask)  | (Host number)
-------------------------------------
               | Num. of computers that can be connected:
               |             2^16 = 65000
               | 255 is for brodcast
               | 1   is for GW

        10  .  0   .  0   . 100               IP address
        255 . 240  .  0   .  0                Subnet mask
Network number | Computer Number              255.240.0.0 <--> /12
     ---------------------------------
               | Num. of computers: 2^20 = 1048576
               | 255 is for brodcast
               | 1   is for GW
 240d=11110000b


 Loopback channel
 ----------------
 Intended for inter process communication - IPC:
 In some computers 127.0.0.1 is the same computer (for loopback
 perform), in other computers it is address 0.0.0.0 (for loopback
 perform).We can take 2 programs in the same computer, without common
 memory and send messages from one program to another (inter process
 communication - IPC).


IF one of computers has the same IP than another computer in the local
network, we can change of IP and netmask of one of computers with
ifconfig command with following options :
  $ ifconfig eth0 192.168.0.102               # only IP number chagne
  $ ifconfig eth0 192.168.0.102 255.255.255.0 # IP + Mask
  $ ifconfig eth0 192.168.0.102 /24 # define how many bits takes
                                    # Network address
An additional example:
  $ ifconfig wifi0 10.0.0.100 255.255.0.0
  $ ifconfig wifi0 10.0.0.100 /16   # define how many bits takes
                                    # Network address
```

17.6 DHCP : Dynamic Host Configuration Protocol

https://en.wikipedia.org/wiki/Dynamic_Host_Configuration_Protocol

DHCP server runs within router (Gate Way).
DHCP client runs within computer in the same subnet as router.

DHCP server manage internal table with all computers in subnet:
Leas time helps to manage time given to computer to have some IP.

MAC addr.	IP addr	Time
a1-a1-a1-b1-c2-d4	10.0.0.100	~~~~~~
b2-c4-d2-12-28-aa	10.0.0.101	~~~~~~

We can use DHCP (Dynamic Host Configuration Protocol) to define IP numbers automatically.

DHCP stands for Dynamic Host Configuration Protocol.

It is a protocol that allows devices on a network (DHCP clients) to get an IP address and other network settings automatically from a DHCP server. This makes it easier to connect multiple devices to a network without manually configuring each one.

The DHCP protocol also provides a mechanism whereby a client can learn important details about the network to which it is attached, such as the location of a default router (GW), the location of a name server, and so on.

Some of the benefits of using DHCP are:

- It reduces the chance of IP address conflicts, where two devices have the same IP address.
- It simplifies network management, as the DHCP server can assign and reclaim IP addresses centrally.
- It supports mobility, as devices can move from one network to another and get a new IP address automatically.

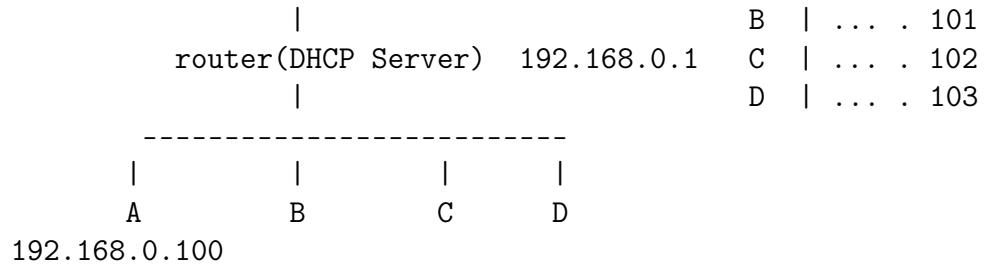
Some of the drawbacks of using DHCP are:

- It depends on the availability and reliability of the DHCP server. If the server is down or unreachable, devices cannot get an IP address or renew their lease.
- It may introduce security risks, as malicious users can spoof DHCP messages or set up rogue DHCP servers to intercept or redirect network traffic.
- It may not be suitable for some devices that need a fixed or static IP address, such as servers or printers.

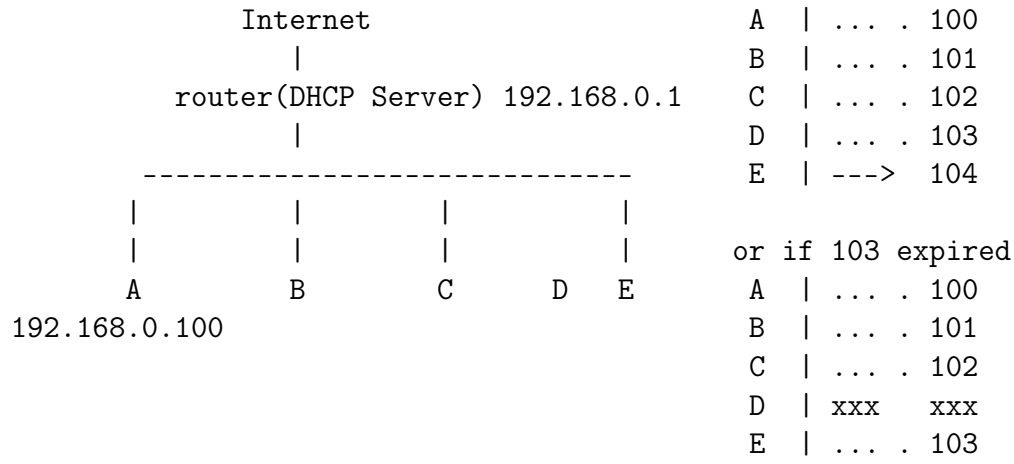
Router usually is used as DHCP server.
He is also Gate Way.
It's address is usually constant : 192.168.0.1.
Operation systems are familiar with such addresses.

Computer that is turned on for the first time in the network, send messages to check which computer is GateWay.
After it finds it , the computer ask the G.W. router to get IP address (this operation is calles IP lease).
The router manages internally the table of mac address, IP , expiration date of IP , etc.

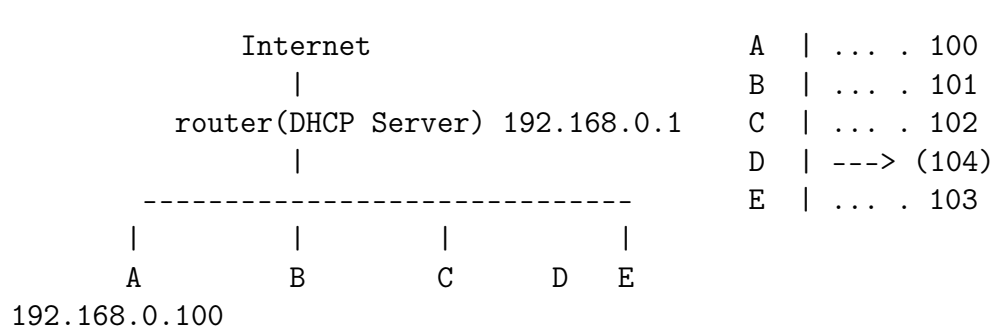
Internet A | 100



Now computer D is disconnected and in place of D connected
new computer E. If address of D is not expired, router will set
addrerss of E ...104, but if address of D is expired,
then the router can set the address of E to be ...103.

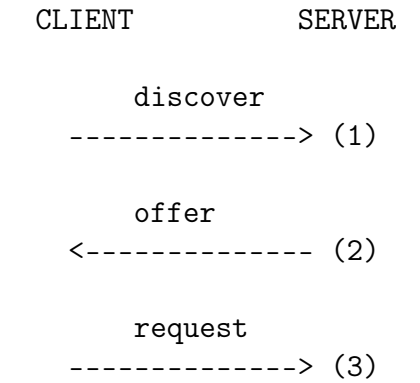


But now the computer D connected again and D asks from
DHCP server to reftesh its IP address. Now DHCP server will set its
address to ...104:



If the address ...103 in any case is important for D,
it can ask E computer to release its address, to allow ifconfig, in
order not to make collision with existing IP.
E is calling the router and ask him to release the IP address.
In this case D can also make release to get the address.
Additional option is to define IP address per m.a.c. address
within router (i.e. edit routing table within the router).

DHCP protocol



DHCP operations fall into four phases:
server discovery,
IP lease offer,
IP lease request,
IP lease acknowledgement.
These stages are often abbreviated as DORA for
Discovery, Offer, Request, and Acknowledgement.

Discovery
The DHCP client broadcasts a DHCPDISCOVER message on the network subnet using the destination address 255.255.255.255 (limited broadcast) or the specific subnet broadcast address (directed broadcast). A DHCP client may also request an IP address in the DHCPDISCOVER, which the server may take into account when selecting an address to offer.

Offer
When a DHCP server receives a DHCPDISCOVER message from a client, which is an IP address lease request, the DHCP server reserves an IP address for the client and makes a lease offer by sending a DHCPOFFER message to the client. This message contains the client's client id (traditionally a MAC address), the IP address that the server is offering, the subnet mask, the lease duration, and the IP address of the DHCP server making the offer.

Request
In response to the DHCP offer, the client replies with a DHCPREQUEST message, broadcast to the server, requesting the offered address. A client can receive DHCP offers from multiple servers, but it will accept only one DHCP offer.

Acknowledgement
When the DHCP server receives the DHCPREQUEST message from the client, the configuration process enters its final phase. The acknowledgement phase involves sending a DHCPACK packet to the client. This packet includes the lease duration and any other configuration information that the client might have requested. At this point, the IP

```
configuration process is completed.
```

17.7 dhclient : obtain/release the IP address drom DHCP server

```
DHCP server runs within router (Gate Way).
DHCP client runs within computer in the same subnet as router.

- Get an IP address for the eth0 interface:
     $ sudo dhclient {{eth0}}

- Release an IP address for the eth0 interface:
     $ sudo dhclient -r {{eth0}}

To obtain fresh IP address from DHCP server (router (Gate Way)):
   $ sudo dhclient enp0s3

To release IP address within DHCP server:
   $ sudo dhclient -r enp0s3
Ones current lease has been released, the client exits.


Example:
$ ip address
  2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc
  fq_codel state UP group default qlen 1000
  link/ether 08:00:27:78:7a:16 brd ff:ff:ff:ff:ff:ff
  inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic
$ sudo dhclient -r enp0s3
$ ip address
  2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc
  fq_codel state UP group default qlen 1000
  link/ether 08:00:27:78:7a:16 brd ff:ff:ff:ff:ff:ff
  inet6 fe80::e013:b6a6:6af3:9b07/64 scope link noprefixroute
$ sudo dhclient enp0s3
$ ip address
  2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc
  fq_codel state UP group default qlen 1000
  link/ether 08:00:27:78:7a:16 brd ff:ff:ff:ff:ff:ff
  inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic
```

17.8 SSH : OpenSSH remote login client

```
ssh (SSH client) is a program for logging into a remote
machine and for executing commands
on a remote machine.  It is intended to provide secure
encrypted communications between two
```

untrusted hosts over an insecure network.

Used to access a remote computer on most Linux/Unix systems
(write shell commands just as if you were sitting at the
workstations) just like telnet.

SSH is Secure SHell (connection to remote computer),
SCP is Secure CoPy (copy to remote computer).
SSH allows connect another Linux computer in safety manner.
SCP allows copy files from one computer to another computer
in safety manner.

1. check if ssh exists on the computer:
    $ ssh -V
     OpenSSH_8.9p1 Ubuntu-3ubuntu0.6, OpenSSL 3.0.2 15 Mar 2022

2. If not exists , install SSH server (daemon):
    $ sudo apt update
    $ sudo apt install openssh-server

3. run SSH server on both computers:
    $ sudo systemctl start ssh   #start ssh for this session
       or
    $ sudo systemctl enable ssh #start ssh from boot to computer
4. connect to computer :
    $ ssh user_name@computer_name
       or
    $ ssh user_name@computer_ip_address

    Example :
    $ ssh amit@192.168.1.12
     The authenticity of host '192.168.1.12 (192.168.1.12)'
     can't be established.
     ED25519 key fingerprint is SHA256:P5+OF1k8R/kSq4BSPgypkbda/
     o4XhXsdgHjVtvhuz5c.
     This key is not known by any other names
     Are you sure you want to continue connecting
     (yes/no/[fingerprint])? yes
     Warning: Permanently added '192.168.1.12' (ED25519)
     to the list of known hosts.
     amit@192.168.1.12's password:
     Last login: Sun Oct 29 06:17:17 2023 from 192.168.1.18

     amit@amit-vm-mint21:~$
     amit@amit-vm-mint21:~$
     amit@amit-vm-mint21:~$

     Note1 : $USER NAME is amit,
             $HOSTNAME is amit-vm-mint21

```
     Note2 : SSH uses computer fingerprint to recognize the remote
             computer.

There are two commands to operate SSH : systemctl and service
SSH operate commands:
   $ sudo systemctl status ssh  #check the status
   $ sudo systemctl stop ssh    #stop ssh for this session
   $ sudo systemctl start ssh   #start ssh for this session
   $ sudo systemctl disable ssh #stop ssh from boot to computer
   $ sudo systemctl enable ssh  #start ssh from boot to computer


$ sudo systemctl status ssh
   [sudo] password for amits:
   o ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; disabled;
    vendor preset: enabled)
   Active: inactive (dead)
   Docs: man:sshd(8)
         man:sshd_config(5)

 Virtual box settings to access external computer network :
 ----------------------------------------------------------
 To connect virtual machine to external network we must choose
 bridge adapter (it connects virtual machine to home network).
 Choose:
     Settings,
     Network,
     Attached to : Bridged adapter
     Name : MedaTek Wifi 6 M7921 Wireless LAN Card
 Note: you must be connected to Internet in outer computer !

 Check if network connection exists:
   $ ifconfig
   enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
   inet 192.168.1.12  netmask 255.255.255.0  broadcast 192.168.1.255
 ----------------------------------------------------------
 Note: NAT network in virtual box is internal network between
 different machines inside the virtual box. If we define NUT network for
 two machines, we can run ssh and scp commands on both virtual computers.
 May there is needed to run
     $ dhclient enp0s3 # in order to get IP address
     or

     $ ip address add
```

17.9 fail2ban : security intrusion prevention software framework

```
https://en.wikipedia.org/wiki/Fail2ban

Fail2ban is an intrusion prevention software framework.
Written in the Python programming language, it is designed to prevent
brute-force attacks.
```

17.10 who : display who is logged in

```
 - Display the username, line, and time of all currently
   logged-in sessions:
     $ who
   Example:
   $ who
     amit     tty7         2024-03-10 07:17 (:0)
     amit     pts/1        2024-03-10 17:37 (192.168.1.18)
```

17.11 SCP : OpenSSH secure file copy

```
scp - remote file copy program, copies files between hosts on
a network

SCP is Secure CoPy (copy to remote computer).
SCP allows copy files from one computer to another computer
in safety manner.

uses ssh for data transfer authentication and provides
the same security as SSH does.

Copies between two remote hosts are permitted also !

1. check if ssh exists on the computer:
    $ ssh -V
     OpenSSH_8.9p1 Ubuntu-3ubuntu0.6, OpenSSL 3.0.2 15 Mar 2022

2. If not exists , install SSH server (daemon):
    $ sudo apt update
    $ sudo apt install openssh-server

3. run SSH server on both computers:
    $ sudo systemctl start ssh   #start ssh for this session
       or
    $ sudo systemctl enable ssh  #start ssh from boot to computer
   4. Copy a local file to a remote host:
scp {{path/to/local_file}} {{remote_host}}:{{path/to/remote_file}}
      Copy a file from a remote host to a local directory:
```

```
scp {{remote_host}}:{{path/to/remote_file}} {{path/to/
                                                local_directory}}
       Recursively copy the contents of a directory from a remote
       host to a local directory:
scp -r {{remote_host}}:{{path/to/remote_directory}} {{path/to/
                                                local_directory}}
```

Example:

```
$ scp p_data.txt amit@192.168.1.12:/home/amit/Documents
  amit@192.168.1.12's password:
  p_data.txt              100%  648   126.7KB/s   00:00
```

Recursive copy of folders:

```
$ scp -r /home/amits/Documents/RTC
                  amit@192.168.1.12:/home/amit/Documents/RTC
```

Windows :

```
$ pscp.exe bennyc@192.168.42.38:/home/minicom.log c:
```

17.12 pscp : transfer files between Windows and linux

The open source PSCP utility makes it easy to transfer files and folders between Windows and Linux computers

Using PSCP

PSCP (PuTTY Secure Copy Protocol) is a command-line tool for transferring files and folders from a Windows computer to a Linux computer.

Download pscp.exe from its website.
https://www.chiark.greenend.org.uk/~sgtatham/putty/latest.html

Move pscp.exe to a folder in your PATH (for example, Desktop\App if you followed the PATH tutorial here on Opensource.com). If you haven't set a PATH variable for yourself, you can alternately move pscp.exe to the folder holding the files you're going to transfer.

Open Powershell on your Windows computer using the search bar in the Windows taskbar (type 'powershell' into the search bar.)

Type pscp -version to confirm that your computer can find the command.

Using PSCP

PSCP (PuTTY Secure Copy Protocol) is a command-line tool for transferring files and folders from a Windows computer to a Linux computer.

Download pscp.exe from its website.

Move pscp.exe to a folder in your PATH (for example, Desktop\App if you followed the PATH tutorial here on Opensource.com). If you haven't set a PATH variable for yourself, you can alternately move pscp.exe to the folder holding the files you're going to transfer.

Open Powershell on your Windows computer using the search bar in the Windows taskbar (type 'powershell' into the search bar.)

Type pscp -version to confirm that your computer can find the command.

17.13 ping : check connection

This is convinient way to check that the opposite side
is connected and supports ping protocol.

It sends packets to given machine and get ackonowlegment
packets in return.

If you can ping GW (router) - your network interface is OK
If you is able to ping an external IP address, your network settings
are correct.

ping also evaluates time and hops that takes to
the command to arrive the opposite side.

Ask ping to check google dns site 3 times only :

```
$ ping -c 3 8.8.8.8
  PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
  64 bytes from 8.8.8.8: icmp_seq=1 ttl=119 time=12.2 ms
  64 bytes from 8.8.8.8: icmp_seq=2 ttl=119 time=13.5 ms
  64 bytes from 8.8.8.8: icmp_seq=3 ttl=119 time=13.8 ms

--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
```

```
rtt min/avg/max/mdev = 12.215/13.174/13.791/0.687 ms
```

```
--- 8.8.8.8 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7008ms
rtt min/avg/max/mdev = 12.861/14.049/15.676/0.881 ms
```

TTL : how many hops it can made maximum !!!

TTL : time to leave

https://www.cloudflare.com/en-gb/learning/cdn/glossary
/time-to-live-ttl/
Time to live (TTL) refers to the amount of time or ‘‘hops’’
that a packet is SET to exist inside a network BEFORE being
DISCARDED BY A ROUTER.

Time To Leave:
How many jumps(hops) the packet maximum will pass
until the next router will throw it.

to set TTL:

```
$ ping rt-ed.co.il -t 50 # set TTL=50
```

In following picture : rectangular is computer and circle is
router. We want to send message from one computer to another.
Firstly the message arrives to 1-st router, it must to calculate
the best rout from it to second computer.
In the well case the data arrives the destination :

```
[]---> O ----> O ----> O ----> []
```

In case of fault the packet can enter the infinite loop between
routers. It can cause traffic jam. In order to prevent such
situation Internet uses TTL - time to live, how time the packet
can hop maximum between different routers.

```
[]---> O <---- O xxx O xxx []
       |       ^
       |       |
       v       |
       O-----> O
```

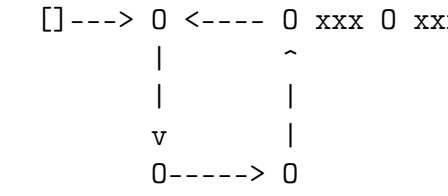
When some packet is in closed loop it can cause to conjesction it
the net, when other packets arrive.
To prevent conjesction , defined TTL.

17.14 iperf : measure quality and bandwidth of link

```
iperf is a tool to measure the bandwidth and the quality of a
network link.
The network link is delimited by two hosts running iperf.
The quality of a link can be tested as follows:
  - Latency (response time or RTT): can be measured with the Ping
     command.
  - Jitter (latency variation): can be measured with an
     iperf UDP test.
  - Datagram loss: can be measured with an iperf UDP test.

On the first computer (with IP 192.168.1.18) we run the server :
   $ iperf -s
------------------------------------------------------------
Server listening on TCP port 5001
TCP window size:  128 KByte (default)
------------------------------------------------------------
[  1] local 192.168.1.18 port 5001 connected with 192.168.1.12 port 38950
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-10.2928 sec  55.5 MBytes  45.2 Mbits/sec



On the second computer we run the client , enter the IP address of
the first computer :
   $ iperf -c 192.168.1.18
------------------------------------------------------------
Client connected to 192.168.1.18, TCP port 5001
TCP window size:  85 KByte (default)
------------------------------------------------------------
[  1] local 192.168.1.12 port 38950 connected with 192.168.1.18 port 5001
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-10.2928 sec  55.5 MBytes  45.2 Mbits/sec


Send ctrl+C signal to stop the iperf server.

To run the server in UDP mode :
  $ iperf -s -u

Change Network parameters and run iperf server :
  $ sysctl -q -w net.core.rmem_max=16777216
  $ sysctl -q -w net.core.rmem_default=16777216
  $ iperf -s -u

Run client in UDP aith defined size :
iperf -c 2.2.2.11 -u -b 2G
```

17.15 TTL : Time To Leave

In case of fault the packet can enter the infinite loop between routers. It can cause traffic jam. In order to prevent such situation Internet uses TTL - time to live, how time the packet can hop maximum between different routers.



When some packet is in closed loop it can cause to conjesction it the net, when other packets arrive.
To prevent conjesction , defined TTL.

Time To Leave:
How many jumps(hops) the packet maximum will pass
until the next router will throw it.

to set TTL:

```
$ ping rt-ed.co.il -t 50 # set TTL=50
```

https://www.cloudflare.com/en-gb/learning/cdn/glossary
/time-to-live-ttl/
Time to live (TTL) refers to the amount of time or ‘‘hops’’
that a packet is SET to exist inside a network BEFORE being
DISCARDED BY A ROUTER.

check Google dns server :

```
$ ping 8.8.8.8
  PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
  64 bytes from 8.8.8.8: icmp_seq=1 ttl=118 time=16.4 ms
  64 bytes from 8.8.8.8: icmp_seq=2 ttl=118 time=15.5 ms
  64 bytes from 8.8.8.8: icmp_seq=3 ttl=118 time=14.4 ms
  64 bytes from 8.8.8.8: icmp_seq=4 ttl=118 time=14.4 ms
```

17.16 Gate Way : Time To Leave

GW is routher/computer that is used to connect us out
of our local network.

17.17 traceroute : shows hopes(jumps) between devices

Print the route of packets trace to network host

```
$ traceroute 192.168.1.12

traceroute to 192.168.1.12 (192.168.1.12), 30 hops max, 60 byte packets
 1  192.168.1.12 (192.168.1.12)  11.090 ms  11.044 ms  11.284 ms

$ traceroute rt-ed.co.il

traceroute to rt-ed.co.il (212.199.114.183), 30 hops max, 60 byte packets
 1  InnboxG84 (192.168.1.1)  6.118 ms  6.123 ms  6.462 ms
 2  100.64.32.1 (100.64.32.1)  9.977 ms  11.990 ms  12.110 ms
 3  45.80.88.197 (45.80.88.197)  12.803 ms  12.791 ms  12.779 ms
 4  82.102.142.225 (82.102.142.225)  36.986 ms  37.342 ms  37.331 ms
 5  212.116.177.49.static.012.net.il (212.116.177.49)  25.972 ms  25.961 ms
    25.950 ms
 6  212.116.177.50.static.012.net.il (212.116.177.50)  20.347 ms  9.816 ms
    9.730 ms
 7  * * *
 8  * * *
 9  * * *
10  * * *

$ traceroute google.com

traceroute to google.com (216.239.38.120), 30 hops max, 60 byte packets
 1  InnboxG84 (192.168.1.1)  3.762 ms  3.687 ms  3.808 ms
 2  100.64.32.1 (100.64.32.1)  10.539 ms  14.998 ms  15.167 ms
 3  45.80.88.197 (45.80.88.197)  18.678 ms  18.653 ms  18.617 ms
 4  72.14.221.92 (72.14.221.92)  18.128 ms  18.109 ms  17.911 ms
 5  * * *
 6  any-in-2678.1e100.net (216.239.38.120)  17.687 ms  11.035 ms  10.995 ms
```

17.18 route : set/display the routing table

Use route cmd to set the route table
This command allows us to define the default Gate Way IP .

- Display the information of route table:
  ```
  $ route -n
  ```
  or
  ```
  $ route
  ```
- Add route rule:
  ```
  $ sudo route add -net {{ip_addr}} netmask {{netmask_addr}} gw {{gw_addr}}
  ```
- Delete route rule:
  ```
  $ sudo route del -net {{ip_addr}} netmask {{netmask_addr}} dev {{gw_addr}}
  ```

```
      $ route
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
default         InnboxG84       0.0.0.0         UG    600    0        0 wlp0s20f3
link-local      0.0.0.0         255.255.0.0     U     1000   0        0 wlp0s20f3
192.168.1.0     0.0.0.0         255.255.255.0   U     600    0        0 wlp0s20f3


      $ route -n
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0         192.168.1.1     0.0.0.0         UG    600    0        0 wlp0s20f3
169.254.0.0     0.0.0.0         255.255.0.0     U     1000   0        0 wlp0s20f3
192.168.1.0     0.0.0.0         255.255.255.0   U     600    0        0 wlp0s20f3


      To define default GW :

     $ sudo route add default gw 192.168.1.0
         SIOCADDRT: Network is unreachable
```

17.19 ip/ip route : ip command

```
        ip - show / manipulate routing, network devices,
             interfaces and tunnels

        ip [ OPTIONS ] OBJECT { COMMAND | help }

        OBJECT := { link | address | addrlabel | route | rule | neigh |
                    table | tunnel | tuntap | maddress | mroute
                    | mrule | monitor | xfrm | netns | l2tp | tcp_metrics |
                    token | macsec | vrf | mptcp | ioam }

        ------------------------------------------------------
         -Similar but newer than ifconfig command.
          display network status :

            $ ifconfig
           or
            $ ip address show #
            $ ip addr show    # short form

         - Make an interface up/down:
            $ ip link set enp0s3 up   # enable enp0s3 interface
            $ ip link set enp0s3 down # disable enp0s3 interface
```

```
    or
     $ ifconfig enp0s3 up   # enable enp0s3 interface
     $ ifconfig enp0s3 down # disable enp0s3 interface
-------------------------------------------------------------
  - Set IP address of eth0:

     $ ifconfig eth0 10.0.0.100
    or
     $ ip address add 10.0.0.100 dev eth0

  - Change IP of eth0 to 10.0.0.100
    and netmask to 255.255.255.0
     $ ifconfig eth0 10.0.0.100 netmask 255.255.255.0
    or
     $ ifconfig eth0 10.0.0.100/24
    or
     $ ip address add 10.0.0.100/24 dev eth0
-------------------------------------------------------------
  - Change default gateway to 10.0.0.1 for eth0 :
     $ route add default gw 10.0.0.1

     $ ip route add default via 10.0.0.1 dev eth0
-------------------------------------------------------------
 Show/manipulate routing, devices, policy routing and tunnels

  - List interfaces with detailed info:
      $ ip address

  1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN
     group default qlen 1000
     link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
     inet 127.0.0.1/8 scope host lo
     valid_lft forever preferred_lft forever
     inet6 ::1/128 scope host
     valid_lft forever preferred_lft forever
  2: enp1s0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc
     fq_codel state DOWN group default qlen 1000
     link/ether 04:bf:1b:72:b9:5d brd ff:ff:ff:ff:ff:ff
  3: wlp0s20f3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc
     noqueue state UP group default qlen 1000
     link/ether f4:6d:3f:8e:03:dc brd ff:ff:ff:ff:ff:ff
     inet 192.168.1.18/24 brd 192.168.1.255 scope global dynamic
     noprefixroute wlp0s20f3
     valid_lft 74418sec preferred_lft 74418sec
     inet6 fe80::dad7:1350:f3b3:451d/64 scope link noprefixroute
      valid_lft forever preferred_lft forever

  - Display the routing table:
      $ ip route
  - Show neighbors (ARP table):
```

```
$ ip neighbour
```

This command can be used to display or modify the existing IP routing table. We can add, delete, or modify specific static routes to specific hosts or networks using ip route command

- Display the routing table:
  ```
  $ ip route {{show|list}}
  ```

- Add a default route using gateway forwarding:
  ```
  $ sudo ip route add default via {{gateway_ip}}
  ```

- Add a default route using eth0:
  ```
  $ sudo ip route add default dev {{eth0}}
  ```

- Add a static route:
  ```
  $ sudo ip route add {{destination_ip}} via {{gateway_ip}} dev {{eth0}}
  ```

  Example :

  ```
  $ sudo ip route add default via 192.168.7.3 dev enp1s0
  ```

- Delete a static route:
  ```
  $ sudo ip route del {{destination_ip}} dev {{eth0}}
  ```

- Change or replace a static route:
  ```
  $ sudo ip route {{change|replace}} {{destination_ip}} via {{gateway_ip}} dev {
  ```

```
$ route -n
  Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref
Use Iface
0.0.0.0         192.168.1.1     0.0.0.0         UG    600    0
0  wlp0s20f3
169.254.0.0     0.0.0.0         255.255.0.0     U     1000   0
0  wlp0s20f3
192.168.1.0     0.0.0.0         255.255.255.0   U     600    0
0  wlp0s20f3
$ ip route
default via 192.168.1.1 dev wlp0s20f3 proto dhcp metric 600
169.254.0.0/16 dev wlp0s20f3 scope link metric 1000
192.168.1.0/24 dev wlp0s20f3 proto kernel scope link src 192.168.1.18
metric 600
```

17.20 DNS..server

https://www.cloudflare.com/en-gb/learning/dns
/what-is-a-dns-server/

Your programs need to know what IP address corresponds to a given
host name (ex.)
Domain Name Servers (DNS) take care of this.

You just have to specify the IP address of 1 or more DNS servers
in your /etc/resolv.conf file:
  nameserver 217.19.192.132
  nameserver 212.27.32.177
The changes takes effect immediately

DNS (Domain Name Server) server is used to translate internet
site names (host name) to IP address.
IP addresses of DNS servers are placed in /etc/resolv.conf file.

To add Google DNS server see following :
https://easyengine.io/tutorials/linux/google-public-dns/

The Domain Name System (DNS) is the phonebook of the Internet.
When users type domain names such as ‘google.com’ or
‘nytimes.com’ into web browsers, DNS is responsible for finding
the correct IP address for those sites.

Browsers then use those IP addresses to communicate with
origin servers or CDN edge servers to access website information.
This all happens thanks to DNS servers: machines dedicated to
answering DNS queries.

- Configure your network settings to use the IP addresses 8.8.8.8
  and 8.8.4.4 as your DNS servers.

check Google dns server :
```
  $ ping 8.8.8.8
   PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
   64 bytes from 8.8.8.8: icmp_seq=1 ttl=118 time=16.4 ms
   64 bytes from 8.8.8.8: icmp_seq=2 ttl=118 time=15.5 ms
   64 bytes from 8.8.8.8: icmp_seq=3 ttl=118 time=14.4 ms
   64 bytes from 8.8.8.8: icmp_seq=4 ttl=118 time=14.4 ms
```

17.21 arp : manage Address Resolution Protocol

The arp command in Linux is a network utility tool used to display,
add, and remove entries in the Address Resolution Protocol (ARP)
cache. This command is crucial for managing network communications in
a Linux system

```
        - Show the current ARP table (mac addresses of every machine that with current m
            arp -a

         display the ARP cache.
         The output shows the IP address (192.168.1.1),
         the corresponding MAC address (ab:cd:ef:gh:ij:kl),
         the network interface (eth0) on which the ARP entry is located.


        - Delete a specific entry:
            arp -d {{address}}

        - Create an entry in the ARP table:
            arp -s {{address}} {{mac_address}}
```

```
-a Show all entries in the ARP cache.     arp -a
-d Delete an entry in the ARP cache.         arp -d 192.168.1.1
-s Add a new entry to the ARP cache.        arp -s 192.168.1.1 ab:cd:ef:gh:ij:kl
-v Enable verbose mode.                      arp -v -a
-n Do not resolve names.                   arp -n -a
-i Specify the network interface.           arp -i eth0 -a
-D Read hardware address from given device. arp -D eth0 -s 192.168.1.1
-e Display in default (Linux) style.          arp -e -a
-H Specify the hardware type.               arp -H ether -a
-p Publish the entry.                  arp -s 192.168.1.1 ab:cd:ef:gh:ij:kl -p
-r Read new entries from file or standard input.       arp -r < arp_entries.txt
-t Specify the hardware type.              arp -t ether -a
```

17.22 ifconfig : InerFace CONFIGurator

```
https://www.baeldung.com/linux/ifconfig-command


ifconfig can be installed with command :
$ sudo apt install net-tools

Connection names in Linux:
lo            ---> loopback

(w)lps20 wifi0  ---> Wireless connection

(e)nps20 eth0   ---> Ethernet cable connection
(e)nps21 eth1

Prints details about all the network interfaces available
on your system.
```

It is used to view and change the configuration of the network
interfaces on your system.

Display information about all network interfaces currently in operation:
```
$ ifconfig
```

To ENABLE an inactive interface, provide ifconfig with the interface
name followed by the keyword up:
```
$ sudo ifconfig eth1 up
```

You can DISABLE an active network interface using the down keyword.
```
$ sudo ifconfig eth1 down
```

Configuring an interface
---
ifconfig can be used at the command line to configure
(or re-configure) a network interface. This is often unnecessary
since this configuration is often handled by a script when you
boot the system. If you'd like to do so manually, you need superuser
privileges, so we'll use sudo again when running these commands.

To assign a static IP address to an interface, specify the interface
name and the IP address.
For example, to assign the IP address 192.168.7.100 to the interface
enp1s0, use the command:

```
$ sudo ifconfig enp1s0 192.168.7.100
```

To assign a network mask to an interface, use the keyword "netmask" and
the netmask address. For instance, to configure the interface enp1s0 to
use a network mask of 255.255.255.0, the command would be:

```
$ sudo ifconfig enp1s0 netmask 255.255.255.0
$ sudo ifconfig enp1s0 10.0.0.100/16     # netmask
                                         # of 16 bit for subnet
```

These configurations can combined in a single command. For instance, to
configure interface enp1s0 to use the static IP address 192.168.7.100,
and the network mask 255.255.255.0:

```
$ sudo ifconfig enp1s0 192.168.7.100 netmask 255.255.255.0
```

mtu N
This parameter sets the MTU (maximum transfer unit) of an interface.
This setting is used to limit the maximum packet size transferred by the
interface. If you're unsure about it, you can safely leave this setting
alone.

netmask address
Set the IP network mask for this interface. This value defaults to the

```
usual class A, B or C network mask (as derived from the interface IP
address), but it can be set to any value.
-------------------------------------------------------------

What about DHCP (Dynamic Host Configuration Protocol)?
ifconfig can only assign a static IP address to a network interface. To
assign a dynamic IP address using DHCP, use the "dhclient" command.

Dynamic Host Configuration Protocol (DHCP) is a client/server protocol
that automatically provides an Internet Protocol (IP) host with its IP
address and other related configuration information such as the subnet
mask and default gateway. RFCs 2131 and 2132 define DHCP as an Internet
Engineering Task Force (IETF) standard based on Bootstrap Protocol
(BOOTP), a protocol with which DHCP shares many implementation details.
DHCP allows hosts to obtain required TCP/IP configuration information
from a DHCP server.

Examples :

Display details of all interfaces, including disabled interfaces:
We can see whether an interface is enabled or disabled by looking
at the flags, in particular the UP and RUNNING keywords.
      $ ifconfig -a

For a quick summary of the interfaces, with some brief network stats,
we can use the -s option:
      $ ifconfig -s
View network settings of an Ethernet adapter enp0s3:
      $ ifconfig enp0s3

Enable enp0s3 interface:
      $ ifconfig enp0s3 up

Disable enp0s3 interface:
      $ ifconfig enp0s3 down
```

17.23 ping : ping command

```
$ ping rt-ed.co.il # We can see that IP of the site is 212.199.114.183

  PING rt-ed.co.il (212.199.114.183) 56(84) bytes of data.
  64 bytes from 212.199.114.183.static.012.net.il (212.199.114.183):
     icmp_seq=2 ttl=57 time=13.3 ms
  64 bytes from 212.199.114.183.static.012.net.il (212.199.114.183):
     icmp_seq=3 ttl=57 time=36.1 ms
```

```
$ ping 192.168.1.12

 PING 192.168.1.12 (192.168.1.12) 56(84) bytes of data.
 64 bytes from 192.168.1.12: icmp_seq=1 ttl=64 time=129 ms
 64 bytes from 192.168.1.12: icmp_seq=2 ttl=64 time=4.95 ms
 64 bytes from 192.168.1.12: icmp_seq=3 ttl=64 time=4.64 ms



Send ICMP ECHO_REQUEST packets to network hosts.
- Ping host:
    ping {{host}}

- Ping a host only a specific number of times:
    ping -c {{count}} {{host}}

- Ping host, specifying the interval in seconds between
  requests (default is 1 second):
    ping -i {{seconds}} {{host}}

- Ping host without trying to lookup symbolic names for
  addresses:
    ping -n {{host}}

- Ping host and ring the bell when a packet is received
  (if your terminal supports it):
    ping -a {{host}}

- Also display a message if no response was received:
    ping -O {{host}}
```

17.24 hostname : system's hostname

```
hostname command in Linux is used to obtain the DNS (Domain Name System)
name and set the system's hostname or NIS (Network Information System)
domain name.
A hostname is a name given to a computer and attached to the network.
Its main purpose is to uniquely identify over a network.
    $ hostname
        amit-vm-mint21
    $ hostname -i
        127.0.1.1
```

17.25 wget : network downloader

```
  GNU Wget is a free utility for non-interactive download of
  files from the Web.
```

```
  It supports HTTP, HTTPS, and FTP protocols, as well as
  retrieval through HTTP proxies.
  Wget is non-interactive, meaning that it can work in the
  background, while the user is not logged on.
  This allows you to start a retrieval and disconnect from
  the system, letting Wget finish the work.  By contrast,
  most of the Web browsers require constant user's presence,
  which can be a great hindrance when transferring a lot of
  data.

 Wget provides a number of options allowing you to download
 multiple files, resume downloads, limit the bandwidth,
 recursive downloads, download in the background, mirror a
 website, and much more.

 Mirror a site:
     wget -m http://lwn.net

 Mirror a site (-r : recusive):
     wget -r -np http://www.xml.com/ldd/chapter/book/

 #recursively download an online book
 #for offline access. -np : no parent
```

18 TextToSpeech

18.1 TextToSpeech..apps

```
say : converts text to audible speech using the GNUstep speech engine.
        sudo apt-get install gnustep-gui-runtime
        say "hello"

festival : General multi-lingual speech synthesis system.
        sudo apt-get install festival
        echo "hello" | festival --tts

spd-say : sends text-to-speech output request to speech-dispatcher
        sudo apt-get install speech-dispatcher
        spd-say "hello"

espeak : is a multi-lingual software speech synthesizer.
        sudo apt-get install espeak
        espeak "hello"

  Example to see voices :
     espeak --voices=variant  # to see avalable languages
     espeak -compile=en-scottish
```

18.2 spd-say : text-to-speech output

```
See: https://askubuntu.com/questions/501910/
how-to-text-to-speech-output-using-command-line

Install :
sudo apt-get install speech-dispatcher
spd-say "hello"


See spd-say

-r, --rate
       Set the rate of the speech (between -100 and +100, default: 0)

-p, --pitch
       Set the pitch of the speech (between -100 and +100, default: 0)

-i, --volume
       Set the volume (intensity) of the speech (between -100 and +100,
       default: 0)

-l, --language
       Set the language (ISO code)

-t, --voice-type
       Set  the preferred voice type (male1, male2, male3, female1, fe-
       male2 female3, child_male, child_female)

-L, --list-synthesis-voices
       Get the list of synthesis voices

 spd-say -t Robert "Hello, how do you do "
 spd-say -t Pablo "Hello, how do you do "
 spd-say -t female2 "Hello, how do you do "
```

19 Appendix.

19.1 LaTeX : editor

```
 Bookmarks addition was based on following explanation :
      https://michaelgoerz.net/notes/pdf-bookmarks-with-latex.html

sudo apt install texlive-full # install full LaTex editor

 To add bookmarks do following steps:
   1. sudo apt install texlive-full # install LaTex editor
   2. cd ~/Documents/docs/c_for_embeded_pdf
```

```
  3. touch c_for_embed.tex
  4. geany & c_for_embed.tex     # prepare LaTex document with bookmarks
     definitions
  5. pdflatex c_for_embed.tex    # this command generates linux_admin.pdf with
     desired bookmarks :)
Note : the latex document can be edited with texmaker


 see : https://artofproblemsolving.com/wiki/index.php/LaTeX

 To enlarge interface font run following command
 from terminal window :
     $ texmaker -dpiscale 1
```

19.2 Shell..built..in..commands : shell built in commands

```
See :
https://www.gnu.org/software/bash/manual/html_node/Shell-Builtin-Commands.html

Shell Builtin Commands
Builtin commands are contained within the shell itself.
When the name of a builtin command is used as the first word
of a simple command (see Simple Commands), the shell executes the command
directly, without invoking another program.
Builtin commands are necessary to implement functionality impossible or
inconvenient to obtain with separate utilities.

Example:
  $ type -a time
     time is a shell keyword # time is built in inside the shell
     time is /usr/bin/time   # also time is in this location
     time is /bin/time       # also time is in this location
```

19.3 shift+PgUp/PgDn : page scroll in terminal

```
Page scroll in terminal
```

19.4 niceness : process priority

```
To my knowledge, the Linux kernel does not change the niceness
of a process, and I fail to see why it would since it doesn't have
to to lower the priority of a process.
The niceness is an information given to the kernel, telling it how
nice that process is willing to be.
The kernel scheduler is free to take this information into account
the way it wants in order to change the priority of a process,
it doesn't need to change its value.
```

On the other hand, in user land, there are daemons like AND whose
task is to renice processes according to rules set up by the admin.
Do you have such a daemon installed on your server?

Niceness values range from -20 (highest priority - most favorable to the
process) to 19 (lowest priority - least favorable to the process).

Niceness can be changed with command renice for the running process.

The NI field in ps command shows the nice value of the process.
The nice value determines the priority of the process. The higher the value,
the lower the priority--the "nicer" the process is to other processes.
The default nice value is 0 on Linux workstations.

19.5 VT : virtual terminal

Linux systems come with a bunch of virtual terminals,
or VTs. Your graphical user interface on Ubuntu runs on VT2,
and VT3 to VT6 allow you to login via command line.

When the graphics is stuck, you can still access
the console window.

To access consoles: [Ctrl][Alt][F3]/[F4]/[F5]

It can be used to work with terminal tty3/tty4/tty5 only.
In the text console, you can try to kill the guilty application

Once this is done, you can go back to the graphic session
by pressing [Ctrl][Alt][F2] (depends on distribution)

19.6 ctrl+shift+arrowup/down : scroll in terminal

Scroll in terminal window

19.7 Additional...shells

Different shells :
sh #bourne shell
#bourne again shell : bash
sch # c shell , less recommended
zsh # another shell

19.8 Special...files

```
Runs automation scripts every time the terminal runs:
   .bashrc - remote control
Runs automation scripts every time before the program run:
   .profile
```

19.9 inode : Create files or set access/modif. times

https://en.wikipedia.org/wiki/Inode

- The inode (index node) is a data structure in a Unix-style file system that describes a file-system object such as a file or a directory.
- Each inode stores the attributes and disk block locations of the object's data. File-system object attributes may include metadata (times of last change,access, modification), as well as owner and permission data

19.10 Disc..Partitions

```
/       root directory

/bin/   Basic, essential system commands

/boot/  Bootloader, Kernel images, initrd (initial RAM disk)
        and configuration files

/dev/   Files representing devices

/etc/   System configuration files(as registry)

/home/  User directories

/lib/   Basic system shared libraries

/lost+found Corrupt files the system tried to recover

/media  Mount points for removable media
        (/media/usbdisk, /media/cdrom)
/mnt/   Mount points for temporarily mounted filesystems

/opt/   Specific tools installed by the sysadmin
        (for all users)
/proc/  Access to system information (list of processes)

/root/  root user home directory
```

```
/sbin/  superuser only commands

/sys/   System and device controls (cpu frequency,
        device power, etc.

/tmp/   temporary files

/usr/   Regular user tools (not essential to the system)
        /usr/bin/, /usr/lib/, /usr/sbin...

/usr/local/  Specific software installed by the sys-admin
             (often preferred to /opt/

/var/   Data used by the system or system servers
        /var/log/, /var/spool/mail (incoming mail),
        /var/spool/lpd (print jobs)...
```

19.11 Globbing

```
Expansion :
 ~  : home directory
 [], ? , *
 !
 $

 ‘‘
 ‘


 *  : any characters
 ?  : exactly one character

          echo ?.txt     # ? replace exactly one character
          echo ??.txt

          ^ is fot not :
          $ echo [^0-9].txt
              a.txt b.txt g.txt i.txt
```